Data Modeling in a Cross-domain Ontology for Cyber Intelligence in Smart-Grids Using Reinforcement Learning

Vincenzo Grimaldi
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Master Thesis

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RWTH Aachen University
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Institute for Automation of Complex Power Systems
Univ.-Prof. Antonello Monti, Ph. D.

Supervisor: Charles Emehel, M. Sc.

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1. Prüfer

Abstract

Diese Arbeit entwickelt eine Integrationsmethodik, die beide Domänen durch ontologische Rahmenwerke verbindet. Die Forschung etabliert einen systematischen Ansatz zur Verknüpfung des Common Information Model (CIM) mit dem ThreMA-Cybersicherheits-Framework. Dadurch entstehen einheitliche semantische Repräsentationen, die physische Stromkomponenten mit Sicherheitskonzepten wie Schwachstellen und Schutzmaßnahmen verbinden. Die Methodik schafft formale Zuordnungen zwischen CIM-Stromklassen und ThreMA-Sicherheitskonzepten. Die Validierung erfolgt anhand eines erweiterten IEEE-9-Bus-Testsystems mit realistischen Netzwerkstrukturen und dokumentierten Angriffsszenarien. Die experimentellen Ergebnisse zeigen verbesserte Leistung gegenüber konventionellen Sicherheitsansätzen, insbesondere bei komplexen Angriffen auf cyber-physische Systeme.

Schlüsselwörter: Smart-Grid-Sicherheit; Ontologie-Integration; CIM; ThreMA; Anomalieerkennung; Machine Learning.

Abstract

This thesis presents an integration methodology that bridges electrical engineering and cybersecurity domains through ontological frameworks. The research develops a systematic approach linking the Common Information Model (CIM) with the ThreMA cybersecurity framework, creating unified semantic representations that connect physical power components with security concepts including vulnerabilities and protective measures. The methodology establishes formal mappings between CIM power system classes and ThreMA security concepts. Validation employs an enhanced IEEE 9-Bus test system incorporating realistic network infrastructure and documented attack scenarios. Experimental results demonstrate improved performance compared to conventional security approaches, particularly for complex attacks targeting cyber and physical systems.

Keywords: Smart-grid security; ontology integration; CIM; ThreMA; anomaly detection; machine learning.

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Chapter 1

Introduction

1.1 Context: The Cyber-Physical Transformation of Power Systems

Cyber intrusions in power systems demonstrate the potential for cascading failures from digital attacks to physical equipment damage [1]. Addressing these challenges requires integrated security methodologies that bridge power engineering and cybersecurity domains. Ontological frameworks like [2] provide formal semantic structures for representing shared concepts and data models across these traditionally separate fields, enabling systematic knowledge integration.

1.2 Motivation and Problem Statement

The increasing digitalization of power systems creates an urgent need for integrated security approaches that bridge the knowledge gap between power engineering and cybersecurity domains. While both fields have developed sophisticated tools and methodologies independently, their integration remains challenging due to different terminologies, priorities, and conceptual frameworks.

As emphasized in a survey [3], the security challenges in Smart-Grids require interdisciplinary approaches that bridge power engineering and cybersecurity domains, particularly in developing unified semantic frameworks for threat modeling. Figure 1.1 shows that risk management is usually at the intersection of electric grid asset, securoty cyber threat and system device vulnerabilities. Grid assets are critical and valuable assets that needs protection from faults, attacks and renewable disturbances. Cyber threats are events that are potentially harmful to the grid assets while vulnerabilities are lapses or weaknesses that can be exploited by malicious attack vectors. [4]

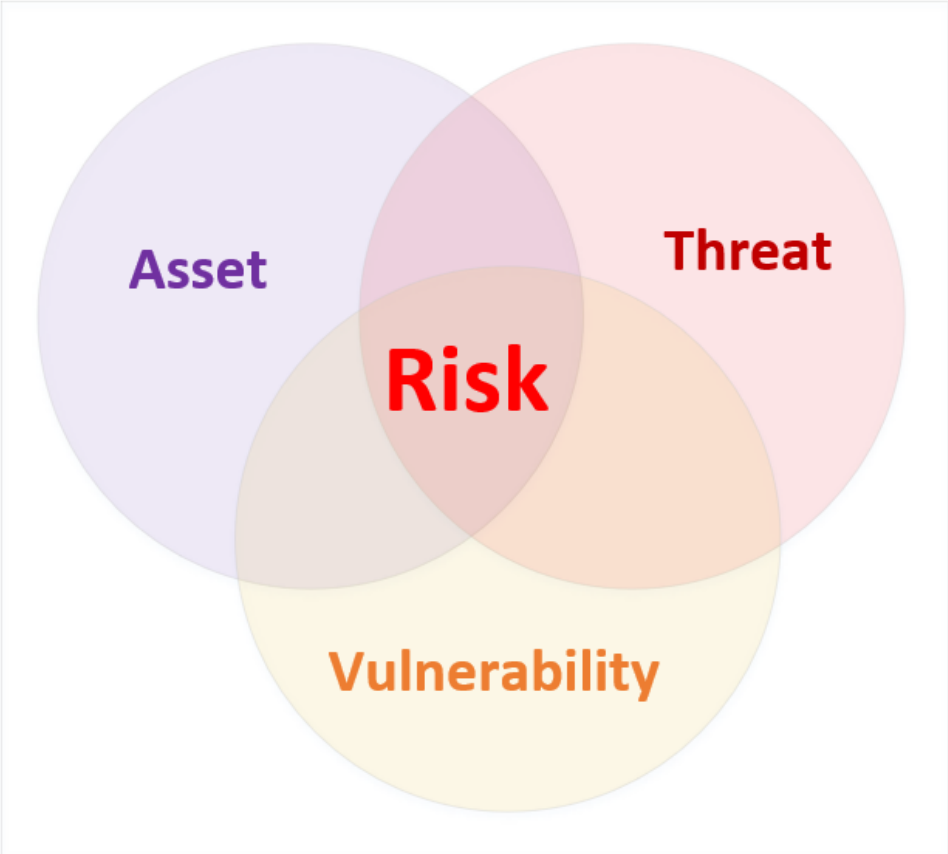

Figure 1.1: Risk as an intersection point between grid assets, cyber threats and system vulnerabilities.

1.3 The Role of Ontologies: CIM and Cybersecurity Frameworks

Within the power systems, the Common Information Model (CIM) [15, 16], standardized by the IEC (IEC 61970/61968), has become a widely used standard for data exchange. Further analysis of CIM's structure reveals it had a significant blind spot: cybersecurity.

Cybersecurity experts have their own set of ontologies and structured knowledge frameworks [1, 35]. Examples include the Threat Management (ThreMA) ontology [51], STIX (Structured Threat Information eXpression) for threat intelligence, CAPEC (Common Attack Pattern Enumeration and Classification) for attack patterns, and CVE/CWE (Common Vulnerabilities and Exposures/Common Weakness Enumeration) for vulnerabilities [1, 8, 3]. These frameworks provide rich vocabularies for describing security concepts but typically lack the granular detail required to represent the specific components, operational constraints, and physical dynamics of high-voltage power systems.

The approach requires a system that can eventually translate between these two worlds. By connecting CIM (the power engineering standard) with ThreMA (the cybersecurity framework), I attempt to create a unified knowledge system that enables:

- Cyber-physical threat modeling.
- Situational awareness correlating cyber events with physical impacts.

- Improved risk assessment considering operational consequences.
- More effective and context-aware decision support for security operations.

This research addresses the integration challenge through a systematic approach that bridges **CIM** and **ThreMA** domains via a unified CrossDomainOntology framework. The integrated knowledge system enables machine learning algorithms to process both electrical and security aspects of attacks simultaneously, improving threat detection capabilities through enhanced contextual understanding. Existing ontological frameworks, including SAREF and SARGON from European smart energy initiatives [9, 10], provide valuable semantic modeling approaches. However, these frameworks lack specific provisions for cybersecurity integration in power system contexts.

1.3.1 Extending the Interoperability: SAREF and SARGON

The literature review also examined the SAREF and SARGON ontologies used in European smart energy projects [26, 49]. Although these frameworks offer insights, they did not address the specific cybersecurity integration challenges I was trying to solve.

As we look beyond SAREF, SARGON is able to extend these capabilities specifically for the smart energy domain. SARGON (SmArt eneRGy dOmain oNtology) provides semantic descriptions of smart assets in building automation and smart grid environments, along with the relationships between them [2, 49]. Figure 1.2 illustrates this integration, showing how SARGON extends SAREF while providing mappings to power system components represented in CIM. This multiontology approach enables a semantic modeling across the cyber-physical boundary: physical devices can be represented through their energy domain characteristics (via SARGON and CIM) while simultaneously maintaining their security-relevant attributes (via SAREF's connections to security frameworks like ThreMA) [26, 49, 51].

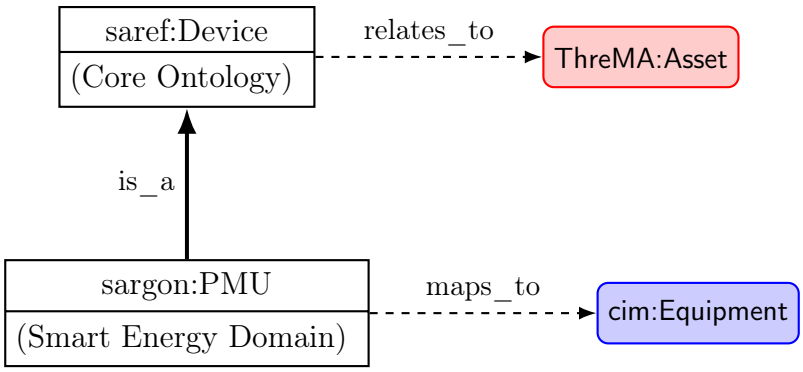

Figure 1.2: Integration of SARGON and SAREF ontologies within the cross-domain semantic framework. SARGON extends SAREF for smart energy domain applications, providing semantic mappings to CIM power system components while maintaining compatibility with security concepts in ThreMA. Please note: *All labeled or unlabeled **bold** and **black arrows** between nodes or boxes in all subsequent figures represents inheritance with the "is_a" or "subClassOf" term in standard ontology vocabularies.*

In our approach, we draw inspiration from this SAREF-SARGON relationship to inform our CIM–ThreMA integration methodology. Specifically, we adopt the pattern of creating explicit semantic bridges between domain ontologies while preserving their

individual strengths and specialized vocabularies. For example, just as SARGON maps its energy-specific device concepts (e.g., `sargon:PMU`) to both SAREF's general device model and CIM's equipment representations, our integration layer creates mappings between ThreMA security concepts and CIM power system components [9, 10].

Combining multiple knowledge systems like this gives us several important benefits:

- **Domain Integrity:** Each domain ontology (CIM, ThreMA, SAREF, SARGON) maintains its specialized vocabulary and conceptual integrity.
- **Flexible Mapping:** Explicit semantic bridges between domains can be defined at various levels of granularity as needed.
- **Evolutionary Development:** Each knowledge system can develop independently while maintaining interoperability through updated mapping layers.
- **Cross-Domain Reasoning:** Security analytics can traverse semantic connections across domains to identify complex attack vectors and cascading impacts.

1.4 Research Contributions

This research builds upon Smart-Grid cybersecurity through ontology-based knowledge representation for high-voltage systems. The primary contributions include an integrated cross-domain ontology framework that unifies the CIM and ThreMA standards.

1. **Integrated Cross-Domain Ontology Framework:** The research develops an ontology that brings together CIM and ThreMA, creating formal semantic connections between power system components (like substations, breakers, and protection relays) and cybersecurity concepts (such as threats, vulnerabilities, malware, and security controls). The proposed framework builds upon existing standards (IEC 61970/61968, STIX) to specifically address the unique cyber-physical security needs of Smart-Grids [2].
2. **Systematic Cross-Domain Mapping Methodology:** Contribution of a structured methodology for aligning concepts, properties, and relationships between the power system and cybersecurity domains, employing ontology alignment techniques and defining reusable semantic bridge patterns, validated for consistency and completeness.
3. **Application and Validation on Cyber-Physical Benchmark:** Validation of the ontology's practical utility by modeling the IEEE 9-Bus benchmark system augmented with cybersecurity components (e.g., firewalls, communication links, SCADA servers). This includes mapping physical assets to cyber counterparts and modeling cross-domain vulnerabilities and attack scenarios.
4. **Synergistic Integration with AI for Cyber Intelligence:** A key innovation involving the integration of the ontology with machine learning (neural networks for anomaly detection) and reinforcement learning (for response optimization) [32, 31, 43, 47]. This includes developing ontology-based feature engineering techniques and illustrating how semantic context improves AI model performance, accuracy, and interpretability in detecting and responding to cyber-physical threats.

5. **Evaluation Framework and Empirical Validation:** Establishment of an evaluation framework and presentation of empirical results quantifying the benefits of the cross-domain approach for threat detection, impact assessment, and response optimization, compared against baseline methods.

1.5 Thesis Structure

This thesis comprises five chapters Chapter 2 reviews the existing literature on power system ontologies and cybersecurity frameworks that influenced my approach. Chapter 3 describes the proposed methodology for integrating these different knowledge domains. Chapter 4 presents experimental results using the IEEE 9-Bus test system. Finally, Chapter 5 discusses the findings and suggests directions for future work.
Throughout these chapters, we maintain a consistent focus on the semantic integration of power system and cybersecurity domains, demonstrating how this integration enhances both human understanding and machine learning capabilities for Smart-Grid security.

Chapter 2

Literature Review

2.1 The Common Information Model (CIM) in Power Systems

2.1.1 Standardization Context

The Common Information Model (CIM) emerged from the Electric Power Research Institute's initiative to address semantic interoperability challenges in power system data exchange. Through its object-oriented modeling approach, CIM establishes a unified vocabulary that enables heterogeneous energy management platforms to interpret and process power system information consistently, thereby reducing integration complexity in multi-vendor environments.

Prior work on interoperability in smart-grid cybersecurity often stops at the ontology level, leaving the data-exchange layer underspecified. FIWARE's Smart Data Models initiative, originally conceived for smart-city deployments, has recently been extended to the energy vertical. These models adopt NGSI-LD's graph-oriented information structure, providing native support for relationships, temporal properties, and JSON-LD @context declarations—features that map naturally to ontology triples. Comparable efforts, such as the IEC Common Semantic Model (CSM) and the NIST IoT Knowledge Framework, acknowledge the need for harmonised data models but lack the ready-made RESTful interfaces and mature open-source tooling offered by FIWARE [11].

Recent advancements in smart grid interoperability have demonstrated the significant benefits of integrating CIM with NGSI-LD frameworks. Haghgoo et al. [12] introduced a cloud-based platform for service restoration in active distribution networks that leverages the SARGON ontology—a semantic model based on IEC 61850 and CIM standards—to enable real-time monitoring and control of electrical grids via Intelligent Electronic Devices (IEDs). This integration represents a crucial step toward bridging operational technology (OT) and information technology (IT) domains in smart grid environments. The European Telecommunications Standards Institute (ETSI) has further formalized this convergence in their 2025 Group Report on Context Information Management [6], which explicitly positions NGSI-LD as an enabling technology for data spaces in energy systems, providing "a common data model and interface for representing and sharing data, enabling greater interoperability between different systems and domains." This standards-based approach addresses what Coppolino et al. [6] identify as the "semantic gap" in cybersecurity digital twins for critical infrastructure, where the integration of CIM's smart data model with NGSI-LD "facilitates a detailled and standardized repre-

sentation" that supports both operational efficiency and security monitoring. The Risk Management data models from the Smart Data Models initiative exemplify this integration, offering a standardized approach to representing security measures and vulnerabilities that can be seamlessly exchanged across domain boundaries while maintaining their semantic richness and relationships to physical grid assets.

2.1.2 Core Structure: An Object-Oriented Framework

The architectural foundation of CIM leverages object-oriented design principles, utilizing UML-based class hierarchies to represent power system entities. This hierarchical organization supports inheritance of common attributes from abstract parent classes to specialized equipment representations, facilitating both semantic consistency and extensibility across diverse power system components.

The root class, `IdentifiedObject`, provides essential attributes common to most elements, such as a unique identifier (`mRID`), name, and description. Key branches of the hierarchy relevant to high-voltage engineering and cyberphysical modeling include:

- `PowerSystemResource`: Represents all physical and logical resources within the power system, encompassing attributes related to operational status, ownership, and geographical location.
- `Equipment`: A subclass of `PowerSystemResource`, representing tangible devices like transformers, generators, switches, and protection relays. It includes attributes like `aggregate` (indicating if it represents multiple physical devices) [11].
- `ConductingEquipment`: Specializes `Equipment` for components designed to carry electrical current, such as transmission lines, busbars, and machine windings. Includes attributes like `phases`.
- `Terminal` and `ConnectivityNode`: Essential for defining the electrical network topology. Terminals represent connection points on conducting equipment, while Connectivity Nodes represent the points where terminals connect (electrical buses).
- Specialized Equipment Classes: Granular representations like `SynchronousMachine`, `PowerTransformer`, `ACLineSegment`, `Breaker`, `Disconnector`, providing detailed parameters for specific equipment types.
- Operational and Measurement Classes [2, 11]: Hierarchies for representing `Measurement` values (e.g., `Analog`, `Discrete`), `Control` actions, and `Protection` equipment.

2.1.3 Core Class Structure and Associations

The Common Information Model (CIM) defines both a rich inheritance hierarchy and explicit associations that together capture the full cyber-physical topology of a power system. Figure 2.1 presents a high-level view of the core classes, their key attributes, and the two primary association types—"has" (linking measurements to assets) and "isControlledBy" (linking control actions to equipment).

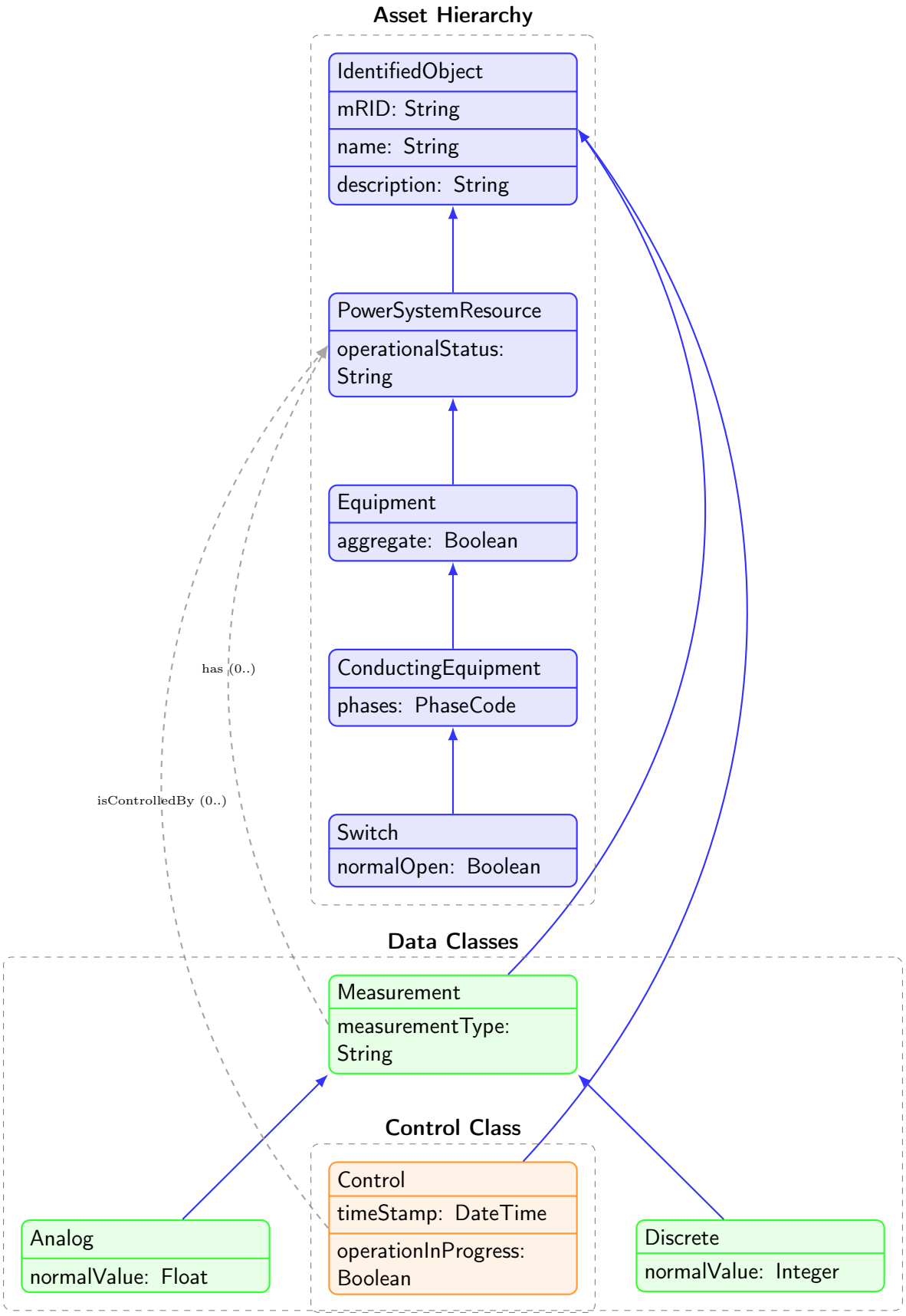

Figure 2.1: High-level CIM core model using ‘rectangle split‘. **Asset Hierarchy:** from `IdentifiedObject` down to concrete equipment. **Data Classes:** `Measurement` and its subtypes. **Control Class:** control actions. Dashed associations (‘has‘, ‘isControlledBy‘) tie measurements and controls back to power-system resources.

2.1.4 Class Properties: The `Equipment` Example

In CIM, each class encapsulates a precise set of typed properties that define its semantics and behavior within the power-system model. For example, the `Equipment` class not only inherits the unique identifier from `IdentifiedObject` but also declares key operational flags—such as `aggregate`, `networkAnalysisEnable`, and `normallyInService`—alongside status indicators like `operationalStatus`. Figure 2.2 provides a compact schematic of these attributes.

Integrating domain-specific ontologies like CIM and cybersecurity frameworks presents significant challenges. For example, mapping the CIM `Equipment` class to ThreMA's `Asset` hierarchy requires careful consideration of property alignment. The CIM `operationalStatus` property must be semantically linked to ThreMA's asset state model, while ensuring that the inheritance hierarchies from both domains remain consistent. This involves creating explicit OWL axioms such as:

```
EquivalentClasses(
  cim:Equipment
  ObjectIntersectionOf(
    threma:Asset
    ObjectSomeValuesFrom(rdf:type PowerSystemAsset)
  )
)
```

Such formal mappings enable cross-domain reasoning while preserving the semantic integrity of both original ontologies.

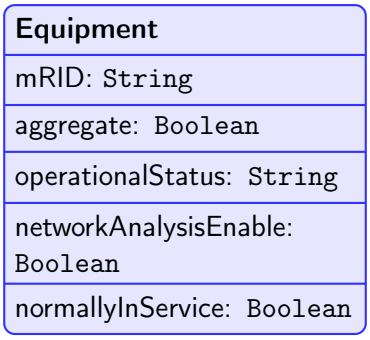

Figure 2.2: Core properties of the CIM `Equipment` class. **Part 1:** class name; **Part 2:** inherited identifier; **Parts 3–6:** key operational attributes and flags.

2.1.5 Applications in Power System Engineering

Recent work by Varma et al. [13] shows the practical application of CIM in coordinating Solar PV and STATCOM for voltage stability specifically in the IEEE 9-Bus system, which serves as a foundation for our experimental validation framework. CIM serves as a foundational element for numerous power system engineering applications:

- **Interoperable Data Exchange:** Standardized XML/RDF serializations (defined in IEC 61970-552, IEC 61968-100, etc.) facilitate seamless data sharing between diverse utility applications (EMS, DMS, Outage Management Systems (OMS), Geographic Information Systems (GIS)), replacing cumbersome proprietary formats.

- **Network Analysis and Simulation:** The detailed representation of components, their electrical parameters, and network topology enables sophisticated analyses, including power flow calculations, contingency analysis, short-circuit studies, and dynamic stability simulations.

- **Asset Management:** CIM provides a structured inventory of power system assets, supporting maintenance scheduling, lifecycle management, regulatory compliance tracking, and performance analysis by capturing operational context and historical data.

- **Distributed Energy Resource (DER) Integration:** Extensions to CIM accommodate the modeling of solar PV, wind generation, energy storage systems, and microgrids, essential for managing bidirectional power flows and the complexities of active distribution networks.

- **Smart-Grid Initiatives:** CIM gives us various Smart-Grid functionaly structures s, including Advanced Metering Infrastructure (AMI), demand response programs, Volt/VAR optimization, and grid modernization efforts [5, 11].

2.1.6 Limitations of CIM for Cybersecurity Modeling

Syed et al. [5] proposed OntoPowSys, a power system ontology for cross-domain interactions, which addresses some of these limitations but still lacks integration with cybersecurity frameworks [5]. As I worked with CIM in my early experiments, I kept running into the same problem: it describes electrical equipment but ignores/avoids security aspects [1, 5].

1. **Limited Cyber Component Representation**: CIM primarily focuses on physical power system components and their electrical characteristics. It lacks detailed representations of cyber components such as firewalls, intrusion detection systems, and communication protocols that are essential for security modeling.

2. **Absence of Threat Concepts**: CIM does not include concepts for representing threats, vulnerabilities, attacks, or security controls. These are fundamental elements needed for cybersecurity risk assessment and management.

3. **Insufficient Behavioral Modeling**: While CIM can represent the static structure of power systems, it has limited capabilities for modeling dynamic behaviors, especially those related to cyber attacks and their propagation through the system.

4. **Inadequate Security Property Representation**: Security properties such as confidentiality, integrity, and availability are not explicitly represented in CIM, making it difficult to model security requirements and violations.

These limitations highlight the need for integrating CIM with dedicated cybersecurity ontologies like ThreMA to create a complete system for Smart-Grid security modeling. This integration forms a core contribution of this thesis.

2.2 CIM Ontology

Building on the common information model in section 2.1, the CIM ontology is an ontology of the electric grid that aims to captures all the entities, properties and relations of the power systems. It defines the RDF, RDFS and OWL specifications as standardized by the w3C's standards classes [16].

According to the work of [15]. future electric systems will be self-managed due to the complexity of the grid infrastructure. This will receive data and information from different distribution system as well as applications. The CIM ontology will be a common model for the electric grid. The CIM User working group developed an owl (web ontology language) and ttl (Turtle Triple Language) ontology which is hosted in a github namespace [16]. This owl and ttl based ontology can be re-used to build a self-managed grid that will support the grid operator on monitoring and operation tasks for a resilient grid.

The CIM Ontology could be anchored on a top level ontology like the ISO-IEC 21838-2:2021 standard for interoperability with other ontology for the purpose of faster convergence in decision support that cut across heterogeneous systems or multiple domains that transcend the grid and the cyber domain [17]. The work of [18] demonstrated an ontology matching approach developed for the semantic web was used in the harmonization of CIM and IEC61850 data model standard. This study helped in solving the mismatches that happens among datasets that where used in building intelligent systems. *L. Wang et al* in [19] focused on the transformation of the CIM and UML to the IEC62656 by building a CIM Ontology management system. This study tried to create a continuous CIM ontology management. The work of *T. Andreas et al* in [20] developed a CIM ontology architecture for managing IT system. This is reproduced in figure 2.3 below.

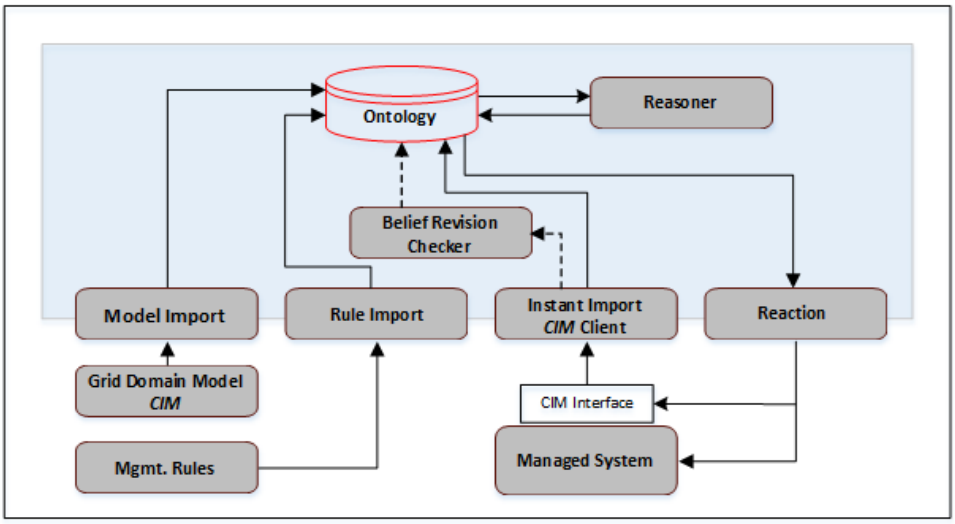

Figure 2.3: CIM Ontology based Management System Architecture.

Figure 2.4 drills into the pure inheritance relationships of the electric grid CIM ontology. Here we see how general CIM concepts (e.g. `IdentifiedObject`) specialize into domain-specific equipment classes (e.g. `ConductingEquipment`, `EnergyConsumer`) alongside the measurement and control branches. This focused view highlights the parallel structures that underpin asset modeling and the separation of operational data classes [1, 11].

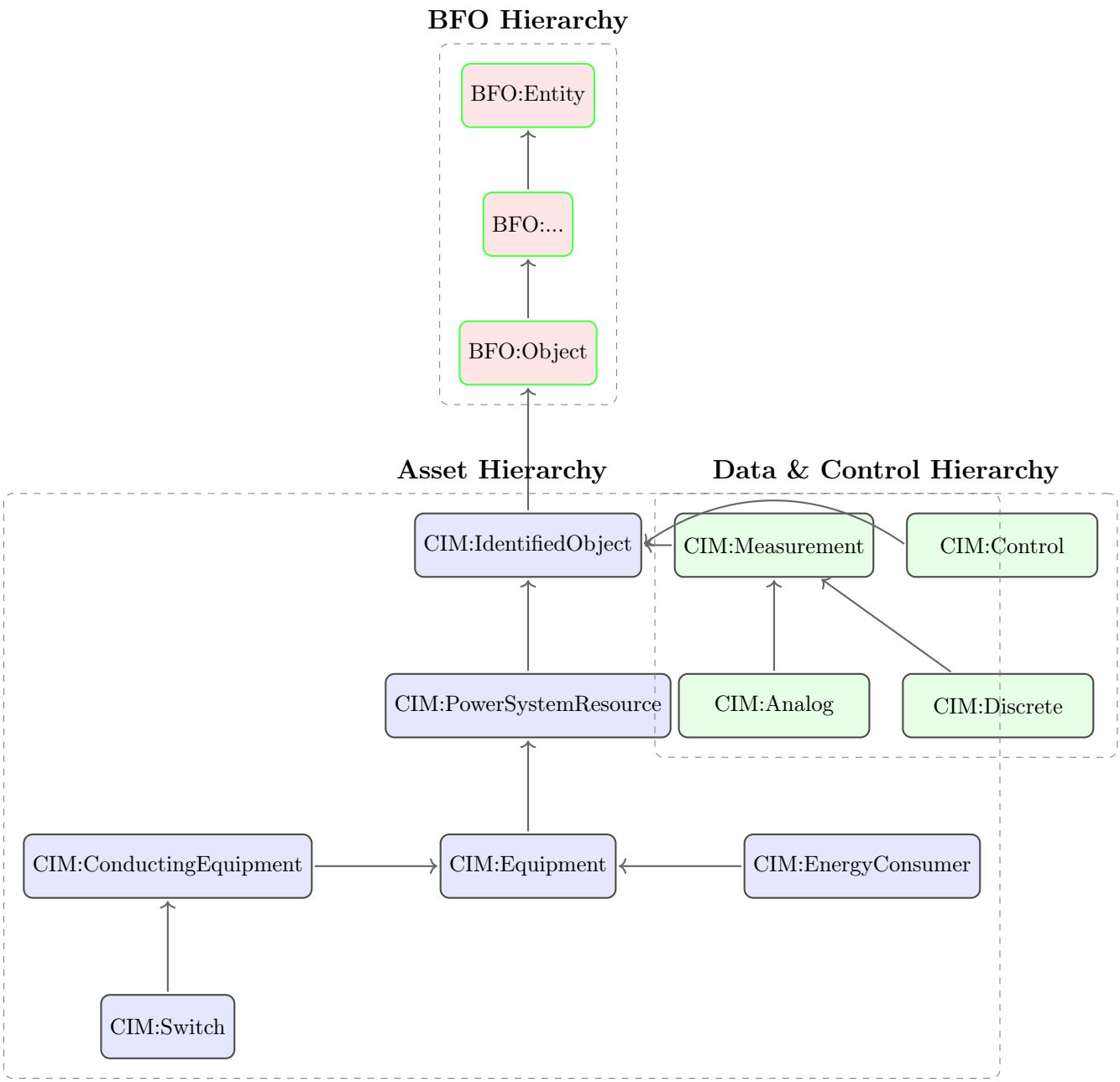

Figure 2.4: Core CIM inheritance: on the left, how `IdentifiedObject` specializes through `PowerSystemResource` and `Equipment` into physical asset classes (`ConductingEquipment`, `EnergyConsumer`, `Switch`); on the right, the parallel branch for operational data classes (`Measurement` → `Analog`, `Discrete`) and `Control`.

Together, these figures provide both the broad architecture of CIM's core schema and the focused class-specialization pathways that underpin precise modeling of assets, measurements, and control actions in a modern Smart-Grid.

The CIM ontology namespace developed by TNO (Netherlands Organisation for Applied Scientific Research - Nederlandse Organisatie voor Toegepast Natuurwetenschappelijk Onderzoek) in [52] gives a comrehensive vocabuilary of the grid objects, classes and their various relationships. It has extension for SAREF4ENERGY amongst others.

2.3 ThreMA Ontology

The ThreMA cybersecurity ontology defines a structured framework [7, 4] where instances are characterized by literal-valued attributes through data properties. Each property is visually identified with a red diamond prefix (♦) and utilizes OWL-compatible XML Schema datatypes to ensure standardization and interoperability [2].

Data Properties

As defined in the ThreMA documentation [7, 4] and extended in this work, the following data properties characterize the core security concepts:

- ♦::Object-Property<>`ThreMA:assetName` **Range:** `^^xsd:string`
  The human-readable name of an asset within the system.
- ♦::Object-Property<>`ThreMA:assetType` **Range:** `^^xsd:string`
  The category or classification of the asset within the infrastructure.
- ♦::Object-Property<>`ThreMA:cveID` **Range:** `^^xsd:string`
  The Common Vulnerabilities and Exposures (CVE) identifier associated with a specific vulnerability.
- ♦::Object-Property<>`ThreMA:severity` **Range:** `^^xsd:float`
  A numeric severity score for a vulnerability (e.g., Common Vulnerability Scoring System - CVSS).
- ♦::Object-Property<>`rdfs:comment` **Range:** `^^xsd:string`
  A free-text annotation or note providing additional context or information.
- ♦::Object-Property<>`ThreMA:capecID` **Range:** `^^xsd:string`
  The Common Attack Pattern Enumeration and Classification (CAPEC) identifier for a specific attack pattern.
- ♦::Object-Property<>`ThreMA:controlID` **Range:** `^^xsd:string`
  Unique identifier for a security control implemented within the system.
- ♦::Object-Property<>`ThreMA:effectiveness` **Range:** `^^xsd:float`
  A numeric rating quantifying a control's effectiveness in mitigating vulnerabilities.
- ♦::Object-Property<>`ThreMA:impactType` **Range:** `^^xsd:string`
  The category of impact (e.g., confidentiality, integrity, availability) resulting from a security incident.
- ♦::Object-Property<>`ThreMA:impactSeverity` **Range:** `^^xsd:float`
  A numeric score quantifying the severity of an impact on the system or organization.

Further Object Properties

Having established the data properties that characterize individual entities, we now turn to the relationships between them. The core classes in the ThreMA ontology are interconnected through the following object properties, which define the semantic relationships

crucial for threat modeling and security analysis. Each entry displays its IRI (Internationalized Resource Identifier), domain, range, and a concise description of its semantic meaning within the model. Understanding that the "Basic Formal Ontology" [21], BFO, works as a top-level ontology designed to support information integration, retrieval, and analysis across different domains. In the context of the ThreMA cybersecurity ontology and CIM ONtology: BFO provides foundational classes and relations that serve as building blocks for domain-specific ontologies like ThreMA and CIM [21].

- `bfo:affects` Domain: `ThreMA:Vulnerability` Range: `ThreMA:Asset`
  Establishes a relationship indicating that a specific vulnerability affects a particular asset within the system. This follows the Basic Formal Ontology (BFO) relation pattern [21].
- `bfo:targets` Domain: `ThreMA:Threat` Range: `ThreMA:Asset`
  Defines which asset is the intended target of a specific threat in the security model.
- `bfo:exploits` Domain: `ThreMA:Threat` Range: `ThreMA:Vulnerability`
  Links a threat to the specific vulnerability it exploits to compromise system security.
- `bfo:realizes` Domain: `ThreMA:AttackPattern` Range: `ThreMA:Threat`
  Associates a concrete attack pattern with the abstract threat concept it implements or realizes.
- `bfo:mitigates` Domain: `ThreMA:SecurityControl` Range: `ThreMA:Vulnerability`
  Specifies that a security control reduces or eliminates the risk posed by a particular vulnerability.
- `bfo:protects` Domain: `ThreMA:SecurityControl` Range: `ThreMA:Asset`
  Indicates that a security control safeguards an asset from potential threats and attacks.
- `bfo:resultsIn` Domain: `ThreMA:AttackPattern` Range: `ThreMA:Impact`
  Shows that the execution of a specific attack pattern leads to a measurable security impact.

Figure 2.5 shows a simplified example of asset inheritance within a cybersecurity context. This hierarchical structure is fundamental to understanding how different types of assets—hardware, software, network components, and data—inherit common properties while maintaining their domain-specific attributes. The inheritance relationships enable efficient modeling of security properties and vulnerabilities across diverse asset types in the Smart-Grid environment.

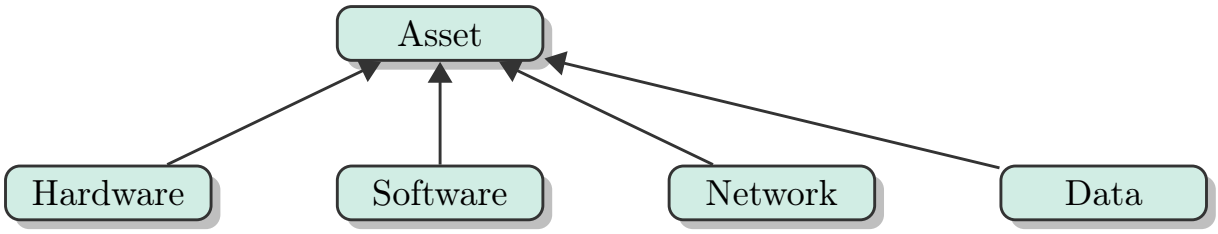

Figure 2.6: Inheritance hierarchy of cybersecurity asset types, using the `ontologyAsset` color palette.

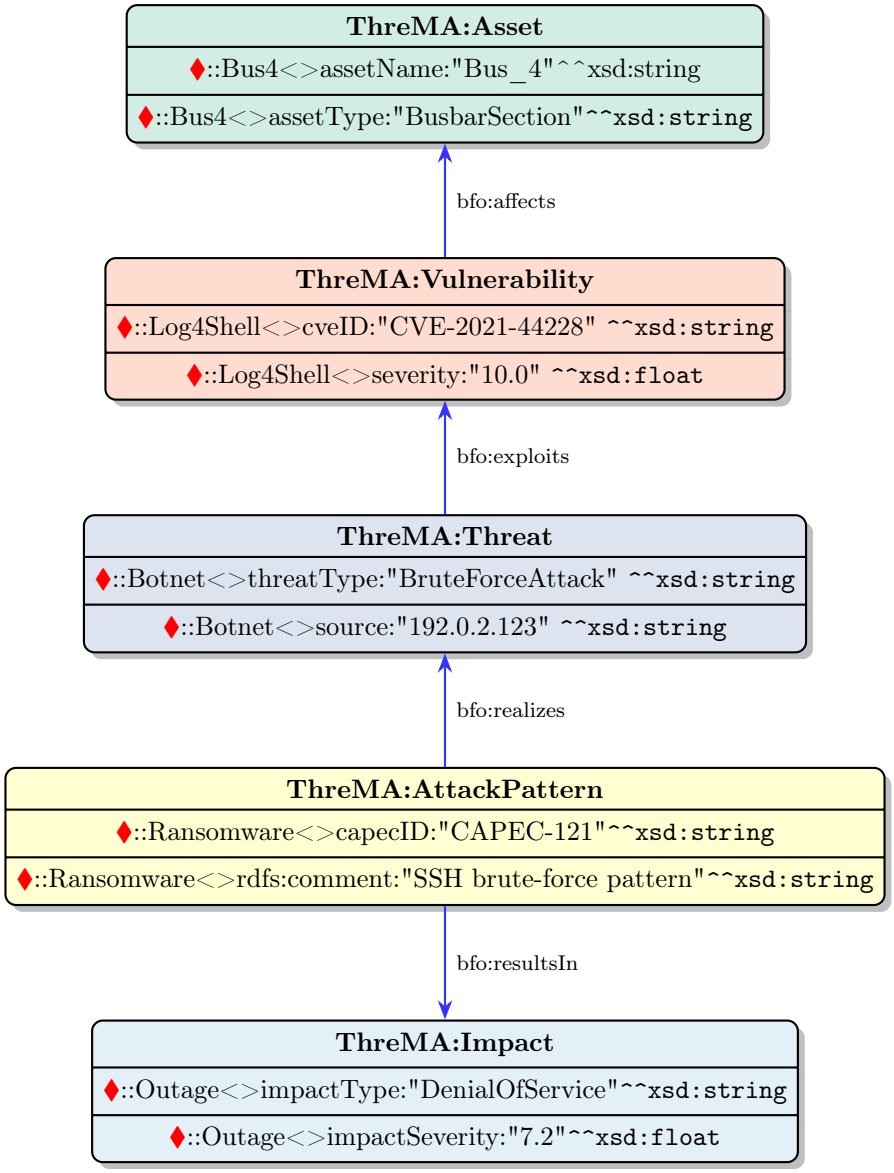

Figure 2.5: Vertical-stacked layout of the ThreMA cybersecurity ontology with example data properties (♦) and object-property links.

2.3.1 The ThreMA Framework: A Focus on Threat Management

Choosing ThreMA as my cybersecurity framework wasn't obvious at first. I evaluated several options including STIX and CAPEC, but ThreMA stood out for a specific reason: it was designed with formal reasoning in mind [7, 4]. This mattered because I needed a framework that could logically connect threats to assets—something that would be crucial for my cross-domain integration. After testing ThreMA's reasoning capabilities with simple power system scenarios, I was convinced it was the right choice, such involve:

- **ICT Sub-ontology:** Models the Information and Communication Technology assets, including hardware, software, network resources, and security mechanisms. This provides the basis for representing the cyber components of the Smart-Grid. Figure 2.6 depicts its core structure.

- **DFD Sub-ontology:** Represents Data Flow Diagrams, capturing data flows, trust

boundaries, and user interactions, crucial for understanding information exchange and potential interception points.

- **Threat Sub-ontology:** Defines threat categories, often linked to CAPEC patterns, and maps them to threat modeling methodologies like STRIDE (Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, Elevation of Privilege). This allows for systematic threat identification.

As an aside note, the economic aspects of cybersecurity investment decisions have been modeled using Stackelberg game theory, providing a quantitative approach to optimizing security resource allocation in grid communication networks.

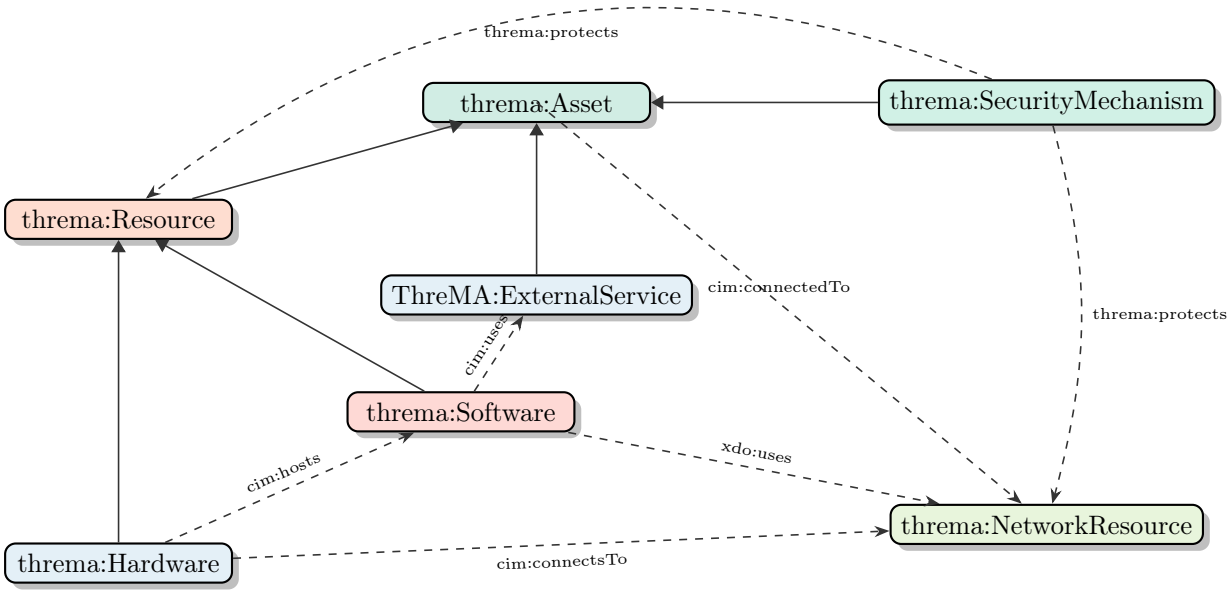

Figure 2.7: Core structure of the ThreMA ICT sub-ontology, modeling `Asset`, `Resource`, `Hardware`, `Software`, `NetworkResource`, `SecurityMechanism`, and `ExternalService` with a consistent color scheme.

Figure below illustrates the concepts within the ThreMA DFD sub-ontology.

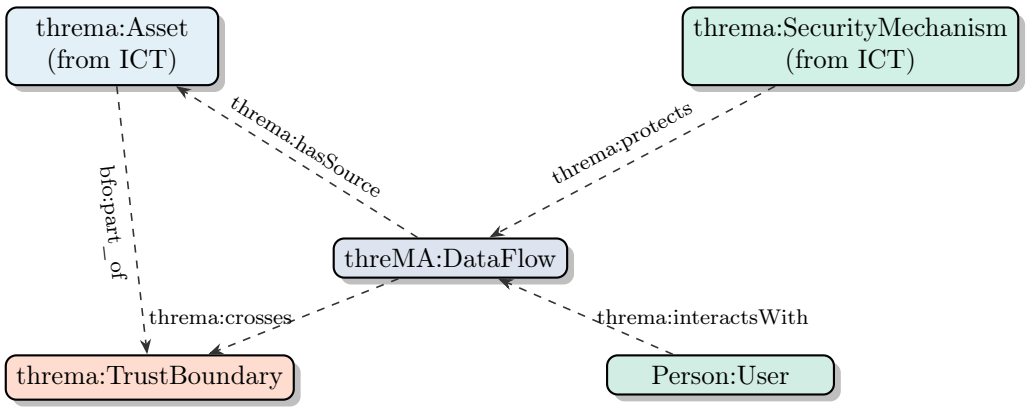

Figure 2.8: Core structure of the ThreMA DFD sub-ontology, depicting `DataFlow`, `TrustBoundary`, `User`, and re-used ICT concepts with cardinality and semantic annotations.

Furthermore, the ThreMA Threat sub-ontology provides structured categories for threats, linking them to established classifications like CAPEC and STRIDE.

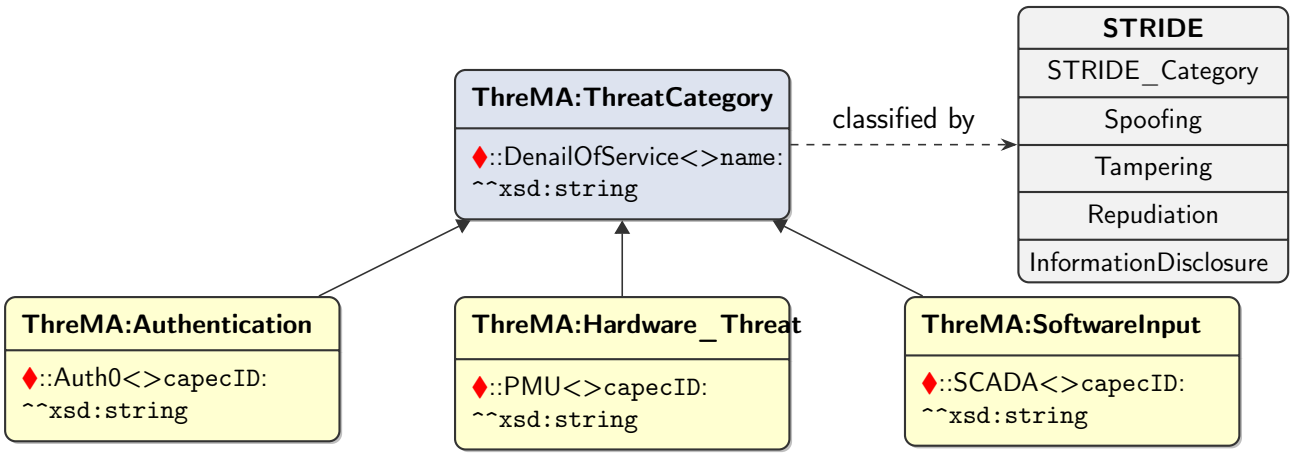

Figure 2.9: CAPEC-based subcategories of `ThreatCategory` with their data properties (♦) and the STRIDE enumeration.

Phasor Measurement Units (PMUs) Supervisory Control and Data Acquisition (SCADA)

ThreMA also provides detailed hierarchies for assets and security mechanisms, crucial for granular modeling, as depicted in Figures 2.8 and 2.9. These hierarchical structures form the foundation for our cross-domain integration approach, allowing precise mapping between power system components and security concepts.

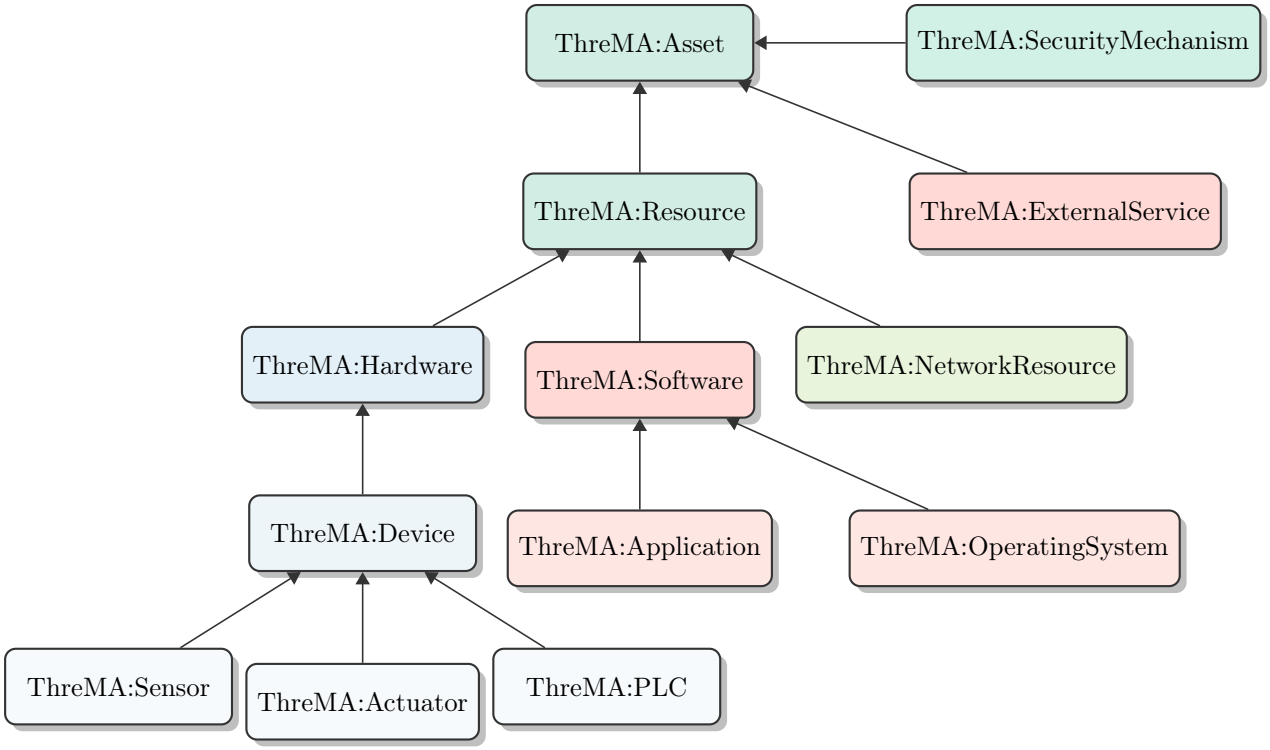

Figure 2.10: Detailed asset specialization hierarchy in ThreMA, illustrating how `Resource` splits into `Hardware`, `Software`, and `NetworkResource`, and further into concrete device and software types.

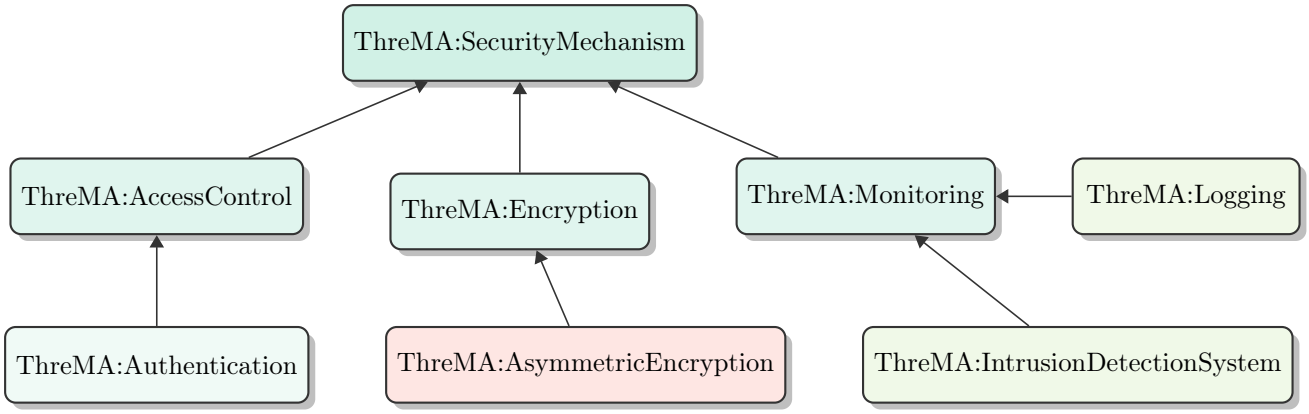

Figure 2.11: Hierarchy of security mechanisms in ThreMA, showing core categories—`AccessControl`, `Encryption`, `Monitoring`—and their specialized controls.

ThreMA provides a strong foundation for the cybersecurity aspects of our cross-domain ontology, offering a structured way to model threats and countermeasures relevant to the IT/OT components supporting the power grid.

2.4 Cross-Domain Ontology Integration Efforts

2.4.1 Challenges in Bridging Power Systems and Cybersecurity

Integrating domain-specific ontologies like CIM and cybersecurity frameworks presents significant challenges:

- **Semantic Heterogeneity:** Different terminologies, modeling paradigms (e.g., UML vs. OWL), and levels of abstraction exist between domains.
- **Scope Mismatch:** CIM focuses on power system function, while cybersecurity models focus on threats and defenses, often lacking detailed physical context.
- **Complexity Management:** Combining large, complex ontologies can lead to unwieldy models that are difficult to manage, query, and reason over.
- **Mapping Identification:** Identifying meaningful semantic correspondences between concepts across domains (e.g., linking a specific PMU modeled in CIM to its associated vulnerabilities and potential threats in a cybersecurity model) requires deep domain expertise.
- **Maintaining Consistency:** Ensuring consistency and avoiding contradictions when merging or aligning ontologies is crucial for reliable reasoning [22].

2.4.2 Existing Approaches and Their Limitations

Several research efforts have explored integrating power system and cybersecurity knowledge:

- **CIM Extensions:** Some approaches attempt to extend CIM directly by adding cybersecurity-related classes and attributes. While leveraging the existing CIM structure, this can lead to bloating the standard and may not fully capture the richness of dedicated cybersecurity models.

- **Manual Mapping:** Experts manually define mappings between CIM elements and cybersecurity concepts. This is labor-intensive, error-prone, and difficult to maintain as standards evolve.
- **Upper Ontology Alignment:** Aligning both CIM and cybersecurity ontologies to a common upper ontology (e.g., BFO, DOLCE) can provide a formal basis for integration but adds complexity and requires significant re-engineering effort.
- **Federated Ontologies:** Maintaining separate domain ontologies while defining explicit mapping relationships between them. This preserves domain specificity but requires robust mechanisms for cross-ontology querying and reasoning.
- **Knowledge Graph Approaches:** Building integrated knowledge graphs by extracting entities and relationships from both domains and linking them. While flexible, this may lack the formal semantics and reasoning capabilities of OWL ontologies [22].

Existing approaches often struggle to achieve a balance between expressiveness, scalability, maintainability, and practical usability, particularly in the context of complex, real-time Smart-Grid operations. There remains a need for a systematic, semi-automated methodology for integrating CIM with cybersecurity frameworks like ThreMA, specifically tailored for Smart-Grid cyber-physical security analysis.

While these existing approaches provide valuable insights, they fall short of addressing the integration needs identified in our research questions. Specifically, none of the reviewed works provides: (1) a systematic methodology for mapping between `cim:` and `threma:` domains that preserves the semantic richness of both ontologies; (2) a formal definition of cross-domain relationships through an explicit `xdo:` namespace; and (3) a framework for leveraging these semantic connections to enhance machine learning and reinforcement learning for security applications. Our methodology, presented in Chapter 3, addresses these gaps through a connection cross-domain ontology approach that affects a more effective detection and response to cyber-physical threats in Smart-Grid environments.

Having established the theoretical foundations and identified gaps in existing approaches, we now present our methodology for developing a Cross-Domain Ontology that addresses these limitations. Chapter 3 details our systematic approach to integrating and domains, along with the technical implementation and validation framework.

2.5 Data Model for Grid and Cyber Interoperability

Achieving seamless interoperability between the electric grid operations and the cyber layer of the grid operator ICT (Information and Communication Technology) network requires a common data model that is both Power system-aware and cybersecurity enriched.

The cross-domain ontology integration efforts mentioned in section 2.4 is to be able to achieve a common data model across the integration of datasets from these two domains during prediction or classification task in distinguishing between fault and attacks on the grid while retaining their semantic meaning within a unified data structure.

The electrical grid has a standard data model and communication protocol as specified in the IEC-61850 and IEC61400-25. These standard has five hierarchical levels that describes the composition of data models in grid equipments, devices as well as functions and sub-functions as determined by the use cases. This hierarchical model from the physical grid server or substation equipment to its logical devices, logical nodes, data objects and data attributes as illustrated by [53] and showned in figure 2.12. This helps in the interoperabilty acroos layers, sub domains and zones in the grid. This gives rise to the Smart Grid Architectural model (SGAM) as a 3-dimensional reference standard [54].

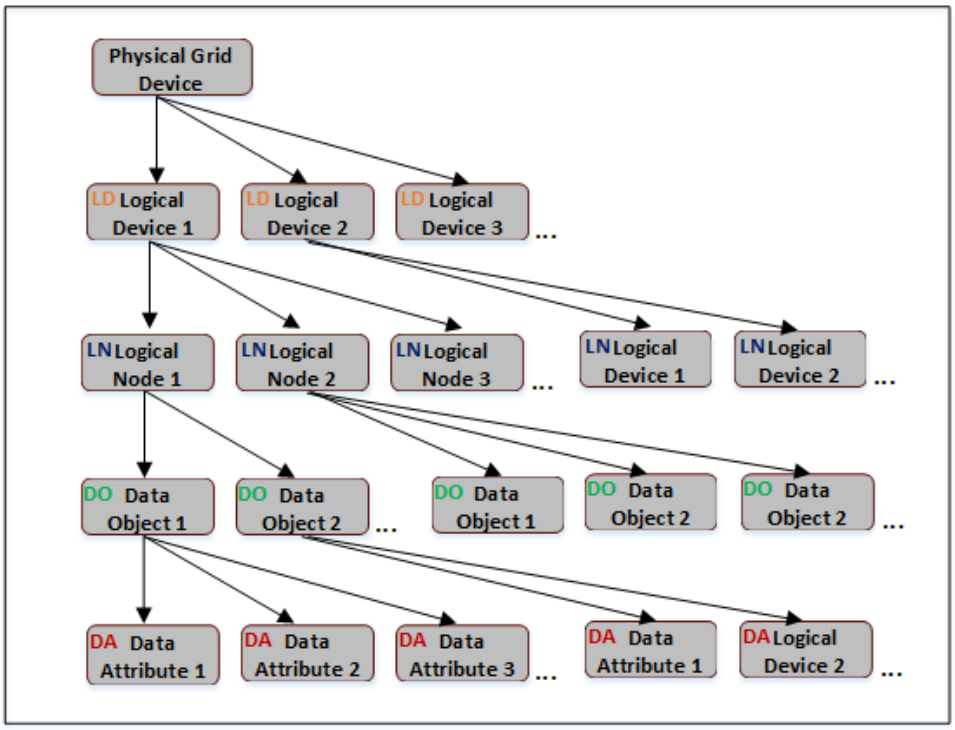

Figure 2.12: IEC-61850 Data Model

Similar standard is difficult for the cyber domain and hence the choice of adopting the Fiware smart data model by curating these standards to fit into a common data model for our use case. However, a comprehensive survey of smart grid standards shows that adding CIM entities with security attributes (e.g., confidentiality labels, provenance, threat indicators) closes critical gaps between grid-codes compliance and IEC-62351/ISO 27000 control requirements. [55] recent work goes further, proposing an authorization framework that aatches role-and-context metadata direcly to the CIM instances, thereby neutralizing the insider threats without breaking interoperability standards.

The work of [56] also motivates this thesis to make an an attemp to create a common data model for the Grid and Cyber domain and integrating reinforcement learning and react native application and neo4j database for dynamic monitoring purpose for the grid operator.

Chapter 3

Methodology

This chapter presents our systematic approach to developing and validating the Cross-Domain Ontology (`xdo:`) that bridges power engineering (`cim:`) and cybersecurity (`threma:`) domains. We begin by outlining the overall methodology for ontology development, followed by our structured data representation approach that enables AI-driven grid modeling. We then detail the semantic mapping process that creates meaningful connections between domains, the toolchain used for implementation, and finally, the integration with reinforcement learning for security intelligence.

My approach would tackle the research questions established in Chapter 1 by providing: (1) a systematic integration of domain knowledge from `cim:` and `threma:` ontologies; (2) formal mapping techniques for connecting physical power system components with cybersecurity concepts; (3) mechanisms for enhancing AI-based anomaly detection through ontological context; and (4) approaches for ontology-driven feature engineering to improve machine learning performance.

3.1 Systematic Approach to Cross-Domain Ontology Development

Developing a cross-domain ontology for Smart-Grids meant wearing two hats at once: that of an electrical engineer and that of a cybersecurity specialist. My goal was to bridge these two worlds—each with its own terminology, assumptions, and priorities—while keeping every technical detail accurate and useful.

1. **Understanding Each Domain and Gathering Requirements.** First, I immersed myself in how Smart-Grids operate at both the transmission and distribution levels. I reviewed the Common Information Model (CIM) to see how power systems are formally described and dug into cybersecurity references—especially IEC 62351 for securing power data and NIST guidelines for IT/OT convergence. At the same time, I sketched concrete use cases where cyber intelligence matters, such as detecting cascading failures, assessing how a vulnerability might propagate, or spotting coordinated attacks. Those scenarios drove my list of what the ontology had to capture.

2. **Choosing and Analyzing Core Ontologies.** Next, I examined the CIM standards (IEC 61970/61968) to model power-system components, and I looked at existing security ontologies (for example, ThreMA, UCO, or STIX-based schemas) to represent threats and controls.

3. **Aligning Concepts Across Domains.** Merging two ontologies isn't just a matter of renaming classes. I applied automated techniques (such as lexical comparison and structural matching) to propose initial alignments, then verified each suggestion manually. For example, I matched CIM's `Breaker` or `ProtectionRelay` to security-related notions like `Asset` or `Control`. Whenever possible, I expressed these correspondences with OWL axioms—using `owl:equivalentClass`, `owl:subClassOf`, and property chains—so a reasoner could understand that, say, a "protected breaker" in the power grid context is also an "asset under control" in the security context.

4. **Building the Integrated Ontology.** I organized the ontology into modules: one for core CIM concepts, one for core cybersecurity concepts, one purely for mapping rules, and an extension module tailored to Smart-Grid specifics.

5. **Populating with Real-World Data.** To make the ontology useful, I injected instances from well-known benchmarks—like the IEEE 9-Bus system—augmented with realistic IT/OT labels. I also imported vulnerability records from the National Vulnerability Database and mapped in attack patterns from CAPEC. This step turned abstract classes into concrete examples: a particular substation relay, its firmware version, a CVE entry affecting it, and so on.

6. **Designing for AI/ML Integration.** With the ontology in place, I sketched how it could feed AI or machine-learning pipelines. For instance, ontology-based feature engineering can transform raw grid measurements or log files into semantically enriched inputs—while the same ontology helps interpret model outputs (e.g., "this anomaly corresponds to a misconfigured relay seen as a high-risk asset").

7. **Validating and Refining Iteratively.** Finally, I tested the ontology against the original use cases by writing competency questions like "Which assets, if compromised, could cascade to a regional blackout?" or "What's the impact range if a relay vulnerability is exploited?" I ran logical consistency checks with reasoners such as HermiT and Pellet and simulated various cyber-physical scenarios to see if any contradictions arose.

3.2 Structured Data Representation for AI-Driven Grid Modeling

A well-organized data representation and data flow is fundamental for an AI IEEE 9-Bus system, as it highlights accurate simulation, predictive modeling data patterns, and asynchronous response on decision-making. In this work, we develop a modular data architecture that integrates power system parameters—including short-circuit capacities (SCC), transformer ratings, and time-varying load profiles—with advanced AI/ML processing. The methodology facilitates data exchange among API's and data flow between the simulation backend and the interactive visualization, thereby enabling system analysis and control.

3.2.1 Modular Data Representation for Cross-Domain Integration

Here, I attempt to designed what I call "domain modules"—think of them as specialized translators. Each module speaks its native language (electrical or security) but can communicate with other modules through standardized interfaces. This approach emerged from my frustration with earlier attempts that created confusing hybrid concepts that satisfied neither domain.

This method provides several benefits for cross-domain ontology development:

1. **Domain Independence**: Each module encapsulates domain-specific knowledge without requiring changes to other modules.
2. **Flexible Integration**: Modules can be connected through clearly defined relationships, allowing for dynamic reconfiguration as understanding evolves.
3. **Incremental Development**: The ontology can be extended by adding new modules without disrupting existing structures.
4. **Validation Simplification**: Individual modules can be validated by domain experts before integration into the complete ontology.

This modular representation forms the foundation for our semantic mapping methodology described in Section 3.4.

3.2.2 Real-World Cybersecurity Threats in Smart-Grids

Building on the data representation framework described above, it is essential to consider how this structure addresses real-world security challenges. Modern power grids are increasingly vulnerable to cyber risks due to their online integration, with false data injection (FDI) attacks becoming eminent and advanced persistent threats (APT) [23, 3] more commonly experienced.

3.3 Ontology Engineering and Knowledge Acquisition Methods

Our methodology leverages a combination of established and tailored techniques for knowledge acquisition and ontology construction. The approach that follows, builds upon the integrated risk assessment framework proposed by Phadke et al. [22], which provides a quantitative methodology for evaluating cyber-physical risks in Smart-Grid environments.

3.3.1 Ontology Engineering Principles

We adopt principles from recognized methodologies like NeOn, emphasizing ontology reuse, modularity, and alignment. Key techniques include:

- **Modular Design:** Partitioning the ontology into coherent modules (CIM-core, ThreMA-core, Mappings, SmartGrid-Extensions) facilitates maintenance, scalability, and independent development.

- **Ontology Alignment Techniques:** Utilizing tools and methods from the Ontology Alignment Evaluation Initiative (OAEI) community, adapted for the specific challenges of CIM-Cybersecurity integration.
- **Axiom Pattern Utilization:** Employing common OWL axiom patterns for representing part-whole relationships, dependencies, and causal links relevant to cyber-physical systems.

Kasaei et al. [23] established foundational principles for integrated quantitative modeling for risk management in power systems that inform our ontology development process.

3.3.2 Knowledge Acquisition Strategy

A multi-faceted strategy ensures knowledge capture:

- **Standards Analysis:** Detailed review of relevant IEC standards (61970, 61968, 61850, 62351), cybersecurity frameworks (NIST SP 800 series, ISA/IEC 62443), and knowledge bases (CAPEC, CWE, CVE, ATT&CK) [25].
- **Use Case Driven Modeling:** Defining concrete cyber-physical scenarios (e.g., false data injection targeting state estimation, coordinated relay tripping attacks) to guide the modeling of specific entities and relationships.

3.4 Semantic Mapping and Cross-Domain Integration

Bridging the semantic gap between CIM and cybersecurity ontologies is central to this work. We focus on establishing meaningful links that enable reasoning across domains.

3.4.1 Mapping Methodology

A semi-automated approach is employed:

- **Mapping:** Automated tools (e.g., LogMap, AML) generate potential correspondences based on lexical and structural similarities.
- **Validation and Refinement:** Domain experts review candidate mappings, resolving ambiguities, correcting errors, and defining complex mappings that automated tools miss. For instance, mapping a CIM `ProtectionRelay` to a ThreMA `Hardware` asset requires understanding its function and potential vulnerabilities.
- **Formal Representation:** Mappings are formalized using OWL axioms, primarily `owl:equivalentClass` and `owl:subClassOf`, potentially augmented with SWRL rules for more complex conditional mappings.

Figure 3.1 provides a high-level conceptual view of the core mappings established between CIM and ThreMA entities.

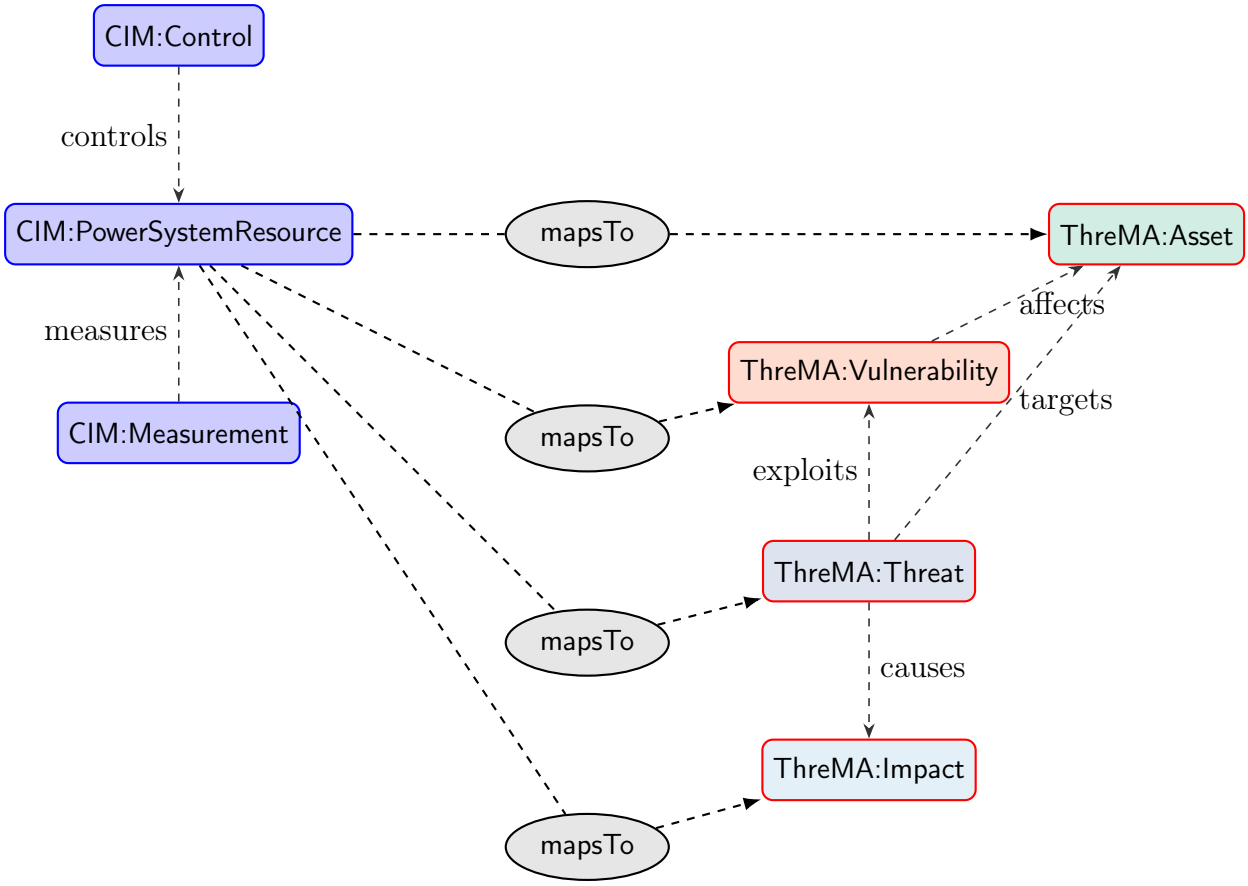

Figure 3.1: Cross-domain mappings: blue nodes (CIM classes), gray split blocks (generic `mapsTo` alignments), and ThreMA threat nodes colored along a risk spectrum—light green for Assets (secure), light orange for Vulnerabilities (caution), light red for Threats (danger), and light purple for Impacts (outcome). Dashed arrows show CIM→ThreMA alignments.

3.5 Toolchain and Representation Languages

Effective management of power system ontologies and analytical frameworks demands a cohesive toolchain and formal representation languages. We leverage ontology editors such as **Protégé** for OWL modeling and visualization [5, 26], version control systems like `Git` for collaborative development, and optional repositories (e.g., BioPortal) for sharing modular knowledge components. Together, these tools maintain the integrity and evolution of domain ontologies.

Underlying semantic representations are defined using standards designed for expressivity and reasoning: the **OWL 2 DL** profile for decidable ontology semantics, the triple-based **RDF** data model, **SPARQL** for graph querying, and **SWRL** for complex rule formulations [9, 10]. Reasoning engines (HermiT, Pellet, FaCT++), SPARQL endpoints/triple stores (Apache Jena Fuseki, GraphDB), and AI/ML libraries in Python (RDFLib, OWLReady2, spaCy, NetworkX) integrate seamlessly with machine learning frameworks (TensorFlow, PyTorch, scikit-learn) [25], graph-embedding toolkits (PyKEEN, AmpliGraph, Node2Vec) and reinforcement learning suites (Stable Baselines3, RLlib) [27].

Building on this foundation, the next section examines the *Critical Role of Short-*

Circuit and Voltage Drop Ratings, covering short-circuit capacity, relay coordination, and transient stability, followed by a formal fault-current analysis. We then introduce *Transformer Ratings and Adaptive Control Strategies*, including reinforcement learning–driven tap-changing mechanisms, and conclude with *Integration with AI-Based Fault and Load Prediction Models*, demonstrating how polynomial regression, neural networks, and Q-learning enhance system resilience and operational efficiency.

3.5.1 Ontology Validation and Verification

Here we ensure the ontology's quality, consistency, and fitness for purpose:

- **Automated Reasoning:** Employing OWL reasoners (e.g., HermiT, Pellet, FaCT++) to check for logical inconsistencies (unsatisfiable classes, contradictory axioms) introduced during modeling and integration.
- **Instance-Based Validation:** Checking the consistency and plausibility of instance data populated from benchmark systems or real-world data sources.

3.5.2 Cross-Domain Ontology Modeling with Protégé

The implementation of our cross-domain ontology leverages Protégé as the development environment and the HermiT reasoner for logical inference. This combination enables the semantic integration of power system concepts (CIM) with cybersecurity frameworks (ThreMA) through our bridging ontology layer.

HermiT Reasoner Selection Rationale

While several reasoners are available for ontology processing (Pellet, FaCT++, ELK), we selected HermiT specifically for our smart grid cyber-physical security use case for the following reasons:

- **Complete OWL 2 DL Support**: HermiT processes all OWL 2 DL constructs, essential for our complex cross-domain relationships, unlike ELK which only supports the OWL 2 EL profile.
- **Hypertableau Algorithm**: Offers significant performance advantages over tableau algorithms when processing intricate inheritance relationships between CIM and ThreMA hierarchies.
- **Rule Support**: HermiT's SWRL rule processing capability enables implementation of cross-domain impact propagation logic, critical for modeling how cyber attacks affect physical power components.
- **Explanations for Inconsistencies**: Provides detailed justifications for logical inconsistencies, accelerating the debugging of complex cross-domain mappings.

Data Model Integration for Standardized Risk Representation

As part of the ontology validation and verification process, my work incorporates standardized data models to ensure interoperability and alignment with industry best practices. The Smart Data Models initiative, which provides harmonized data models across various domains, offers particularly valuable resources for representing cybersecurity and risk management concepts in a standardized format [28].

For this thesis, the `Measure` data model from the Smart Data Models Risk Management suite was selected as the most appropriate representation for actionable security measures derived from our ontological framework. This model provides a structured approach to representing specific mitigation actions that can be executed within cyber-physical systems, aligning perfectly with the cross-domain security focus of our research.

The `Measure` entity enables the translation of abstract security requirements and risk assessments into concrete, implementable actions. This capability is essential for bridging the gap between theoretical ontology-based security analysis and practical security operations in smart grid environments. By adopting this standardized data model, we ensure that the outputs of our ontological reasoning can be seamlessly integrated with existing security management systems and processes.

Listing 1 presents a JSON-LD example of the `Measure` data model populated with values relevant to our ontology-based security framework for smart grids. This example proves how a specific security measure—isolating a compromised PMU to prevent attack propagation—can be represented in a standardized, machine-readable format.

Listing 1 JSON-LD representation of a security measure using the Smart Data Models `Measure` entity

```
{
  "id": "urn:ngsi-ld:Measure:PMU4_Isolation",
  "type": "Measure",
  "action": [
    {
      "id": "urn:ngsi-ld:xdo:PMU_4",
      "value": "ISOLATE_NETWORK"
    },
    {
      "id": "urn:ngsi-ld:xdo:FirewallRule_PMU4",
      "value": "BLOCK_PORT_8080"
    }
  ],
  "description": "Isolate PMU_4 from the network to prevent DoS attack propagation",
  "name": "PMU4 Network Isolation Measure",
  "severity": "high",
  "category": "preventive",
  "validFrom": "2025-05-31T00:00:00Z",
  "validTo": "2025-06-30T23:59:59Z",
  "source": "threma:AttackPattern_DoS",
  "refAsset": "urn:ngsi-ld:xdo:NetworkPort_PMU4",
  "@context": [
    "https://raw.githubusercontent.com/smart-data-models/dataModel.RiskManagement/master/context.jsonld"
  ]
}
...
```

The example in Listing 1 illustrates several key aspects of our approach:

- **Cross-domain integration**: The measure connects cyber components (`NetworkPort_PMU4`, `FirewallRule_PMU4`) with physical assets (`PMU_4`), demonstrating the cross-domain capabilities of our ontology.

- **Actionable security**: The `action` array contains specific, executable commands that can be implemented by security systems, translating abstract security requirements into concrete operations.
- **Threat context**: The `source` field links the measure to a specific attack pattern (`threma:AttackPattern_DoS`), providing context for why the measure is necessary.
- **Temporal validity**: The `validFrom` and `validTo` fields define the time period during which the measure should be applied, supporting dynamic security management.

By integrating standardized data models like the `Measure` entity with our ontological framework, we enhance both the theoretical rigor and practical applicability of our approach. This integration provides how semantic technologies can be leveraged to improve cybersecurity in smart grid environments through standardized, interoperable representations of security knowledge and actions.

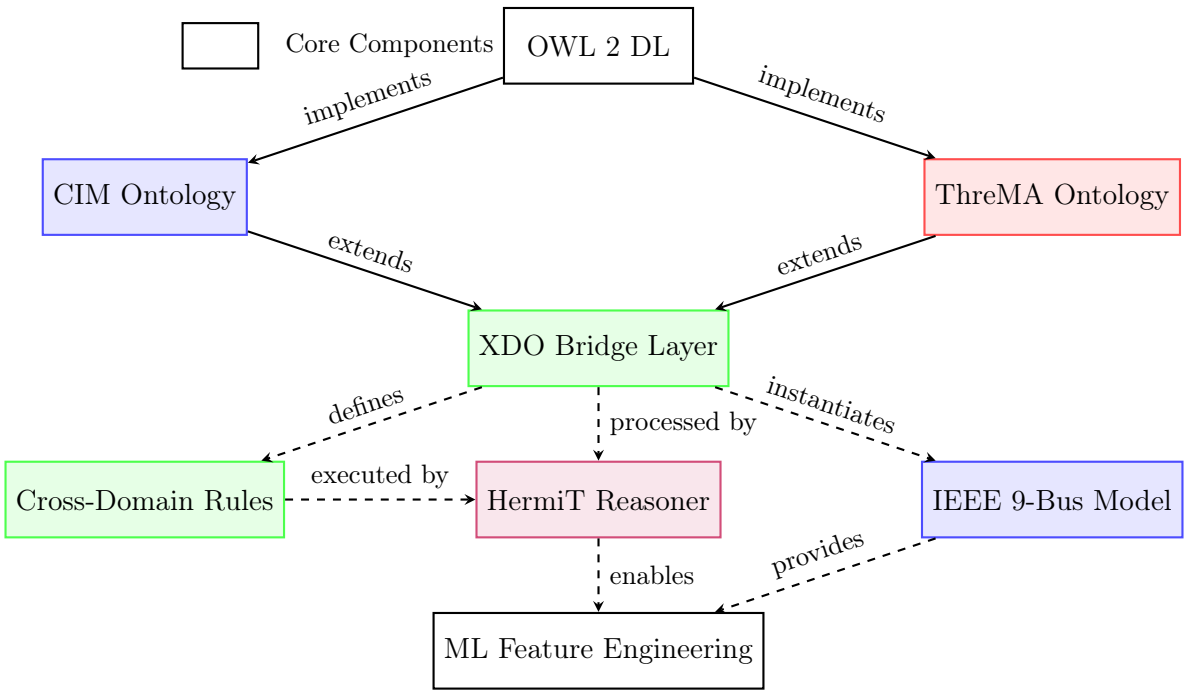

Figure 3.2: Cross-Domain Ontology Architecture showing core building blocks (solid) and extensions (dashed) for smart grid cyber-physical security modeling. The HermiT reasoner processes the integrated ontology to enable cross-domain inference and feature engineering for machine learning.

Ontological Foundation for Security Measures

While the JSON-LD representation in Listing 1 provides a standardized format for expressing security measures, it is built upon a formal ontological foundation that defines the semantic relationships between entities in our cross-domain security model. To illustrate this foundation, Listing 2 presents a small section of the RDF/OWL ontology that underpins our approach.

The RDF/OWL snippet in Listing 2 defines several critical object properties that form the semantic backbone of our cross-domain security ontology. These properties establish the relationships that enable sophisticated reasoning about security measures:

Listing 2 RDF/OWL representation of key object properties in the Cross-Domain Ontology

```
<!-- Object Properties -->
<owl:ObjectProperty rdf:about="http://www.semanticweb.org/vincenzovceccarellig/ontologies/2025/4/untitled-ontology-7/xdo:affects">
    <rdfs:domain rdf:resource="http://www.semanticweb.org/vincenzovceccarellig/ontologies/2025/4/untitled-ontology-7/xdo:Vulnerability"/>
    <rdfs:range rdf:resource="http://www.semanticweb.org/vincenzovceccarellig/ontologies/2025/4/untitled-ontology-7/xdo:Asset"/>
    <rdfs:comment>Relates a vulnerability to the asset it affects</rdfs:comment>
</owl:ObjectProperty>

<owl:ObjectProperty rdf:about="http://www.semanticweb.org/vincenzovceccarellig/ontologies/2025/4/untitled-ontology-7/xdo:controls">
    <rdfs:domain rdf:resource="http://www.semanticweb.org/vincenzovceccarellig/ontologies/2025/4/untitled-ontology-7/xdo:CyberAsset"/>
    <rdfs:range rdf:resource="http://www.semanticweb.org/vincenzovceccarellig/ontologies/2025/4/untitled-ontology-7/xdo:GridAsset"/>
    <rdfs:comment>Relates a cyber asset to the grid asset it controls</rdfs:comment>
</owl:ObjectProperty>

<owl:ObjectProperty rdf:about="http://www.semanticweb.org/vincenzovceccarellig/ontologies/2025/4/untitled-ontology-7/xdo:exploits">
    <rdfs:domain rdf:resource="http://www.semanticweb.org/vincenzovceccarellig/ontologies/2025/4/untitled-ontology-7/xdo:Attack"/>
    <rdfs:range rdf:resource="http://www.semanticweb.org/vincenzovceccarellig/ontologies/2025/4/untitled-ontology-7/xdo:Vulnerability"/>
    <rdfs:comment>Relates an attack to the vulnerability it exploits</rdfs:comment>
</owl:ObjectProperty>

<owl:ObjectProperty rdf:about="http://www.semanticweb.org/vincenzovceccarellig/ontologies/2025/4/untitled-ontology-7/xdo:mitigates">
    <rdfs:domain rdf:resource="http://www.semanticweb.org/vincenzovceccarellig/ontologies/2025/4/untitled-ontology-7/xdo:Function"/>
    <rdfs:range rdf:resource="http://www.semanticweb.org/vincenzovceccarellig/ontologies/2025/4/untitled-ontology-7/xdo:Vulnerability"/>
    <rdfs:comment>Relates a security function to the vulnerability it mitigates</rdfs:comment>
</owl:ObjectProperty>
...
```

- **xdo:affects** defines the relationship between vulnerabilities and assets, enabling the ontology to reason about which components are susceptible to specific security weaknesses. In the context of our JSON-LD example, this property helps identify that the `OpenPort8080_Vulnerability` affects the `NetworkPort_PMU4`.

- **xdo:controls** establishes the cyber-physical relationship between digital systems and physical grid components. This property is essential for understanding the potential cascading effects of cyber attacks on physical infrastructure, as demonstrated in the JSON-LD example where isolating `PMU_4` is necessary to prevent attack propagation.

- **xdo:exploits** connects attack patterns to the vulnerabilities they target, enabling the ontology to reason about potential attack vectors. This relationship is reflected in the JSON-LD example's `source` field, which links the security measure to the `threma:AttackPattern_DoS`.

- **xdo:mitigates** relates security functions to the vulnerabilities they address, providing the foundation for generating appropriate countermeasures. This relationship underpins the actions specified in our JSON-LD example, where `BLOCK_PORT_8080` mitigates the open port vulnerability.

These formal semantic relationships demonstrate how our ontological approach provides a rich foundation for generating standardized security measures. While the JSON-LD representation offers a practical, interoperable format for sharing and implementing security actions, the underlying RDF/OWL ontology supplies the semantic reasoning capabilities necessary to identify appropriate measures based on the complex relationships between cyber and physical components, vulnerabilities, and attacks.

By combining formal ontological representations with standardized data models, our approach bridges the gap between theoretical security modeling and practical security operations. This integration enables automated reasoning about security measures while ensuring that the resulting actions can be communicated and implemented using industry-standard formats and protocols.

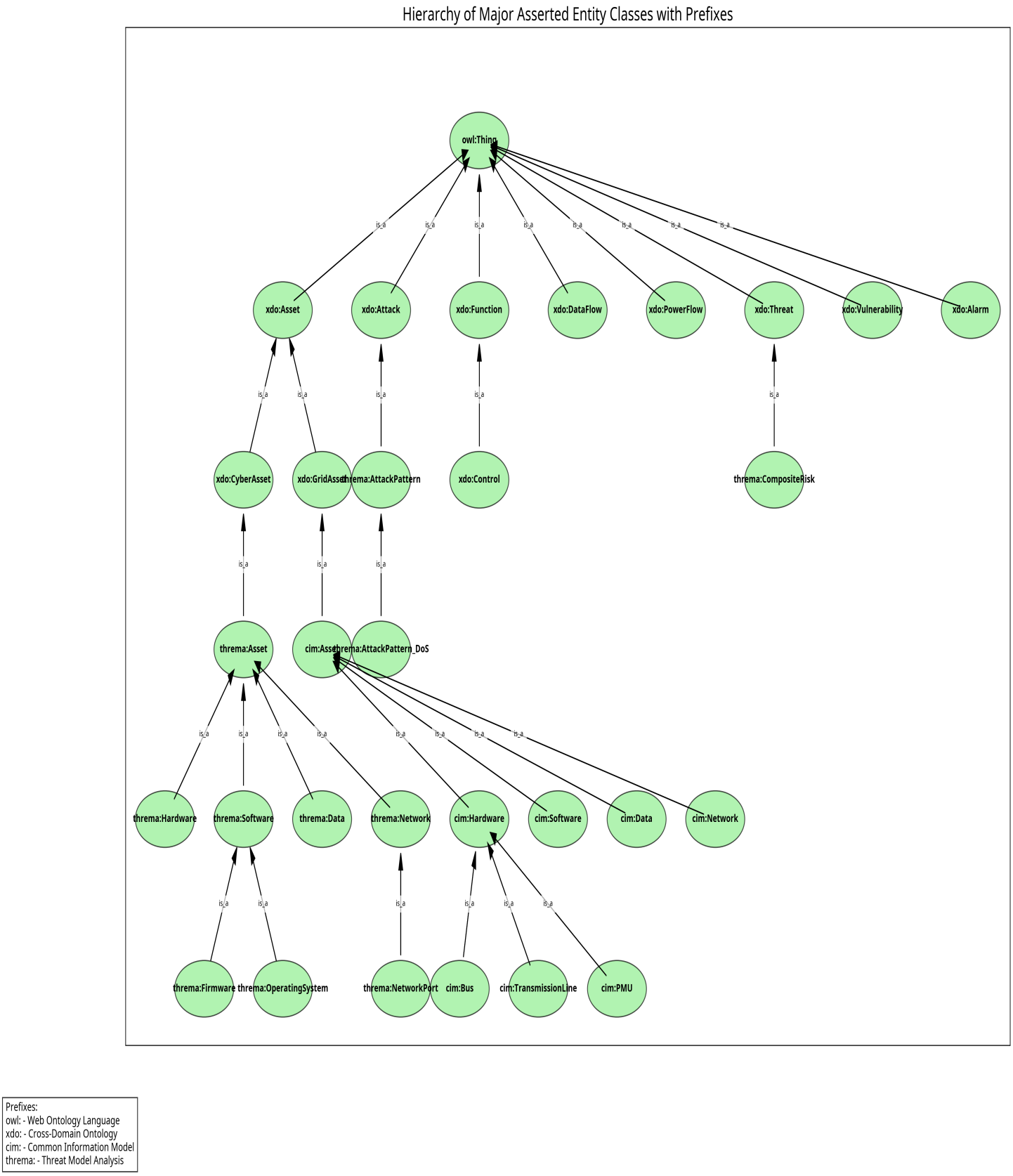

Figure 3.3: **Hierarchy of Major Asserted Entity Classes with Prefixes.** Each node represents an OWL class (prefixed by `owl:`, `xdo:`, `cim:`, `threma:`), and each arrow is an `rdfs:subClassOf`. Top-level classes (`owl:Thing`, `xdo:Asset`, `threma:Asset`, `cim:Asset`) are shown near the top, and their immediate subclasses fan out below extending the model previously introduced [57].

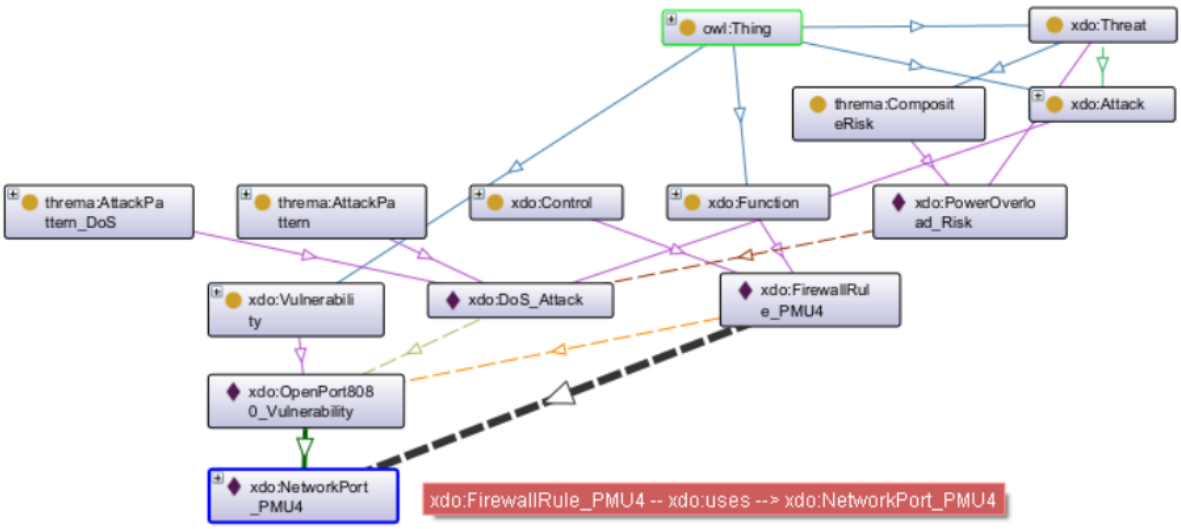

Figure 3.4: **FirewallRule_PMU4** → `xdo:uses` → **NetworkPort_PMU4.** This diagram shows how the individual `xdo:FirewallRule_PMU4` is linked to the individual `xdo:NetworkPort_PMU4` via the property `xdo:uses`. The thick black arrow emphasizes the "uses" relationship, while colored arrows (purple, orange, dashed) indicate related classes (e.g. `xdo:DoS_Attack`, `xdo:Vulnerability`).

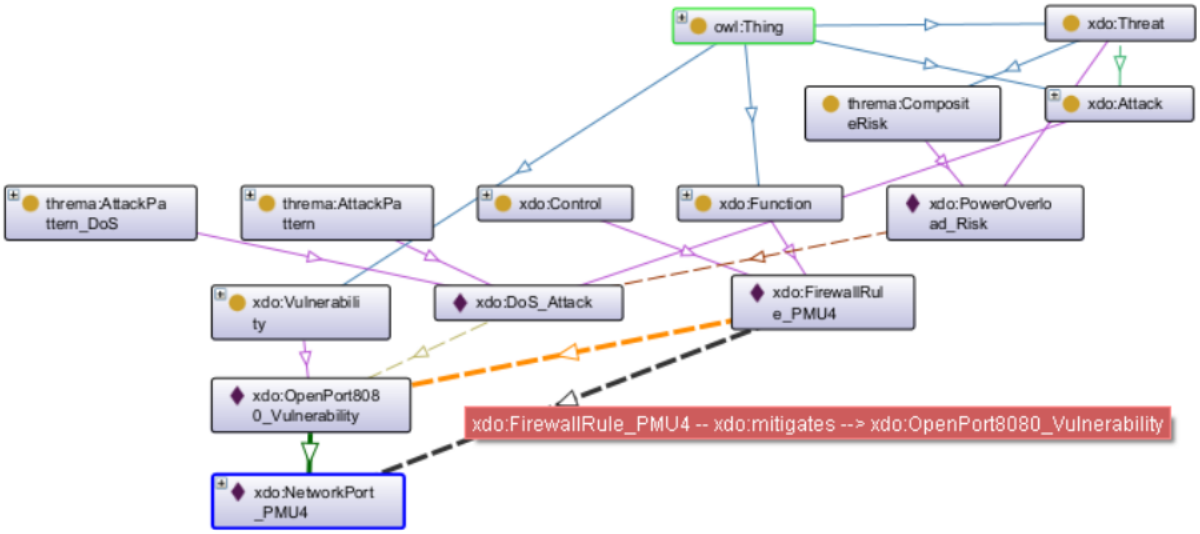

Figure 3.5: **FirewallRule_PMU4** → `xdo:mitigates` → **OpenPort8080_Vulnerability.** Here the individual `xdo:FirewallRule_PMU4` is shown mitigating `xdo:OpenPort8080_Vulnerability` (highlighted by the thick black dashed arrow). The vulnerability itself is linked to `xdo:NetworkPort_PMU4`. Other class-level relationships (in purple and orange) provide context (e.g. `xdo:DoS_Attack` feeding into the vulnerability).

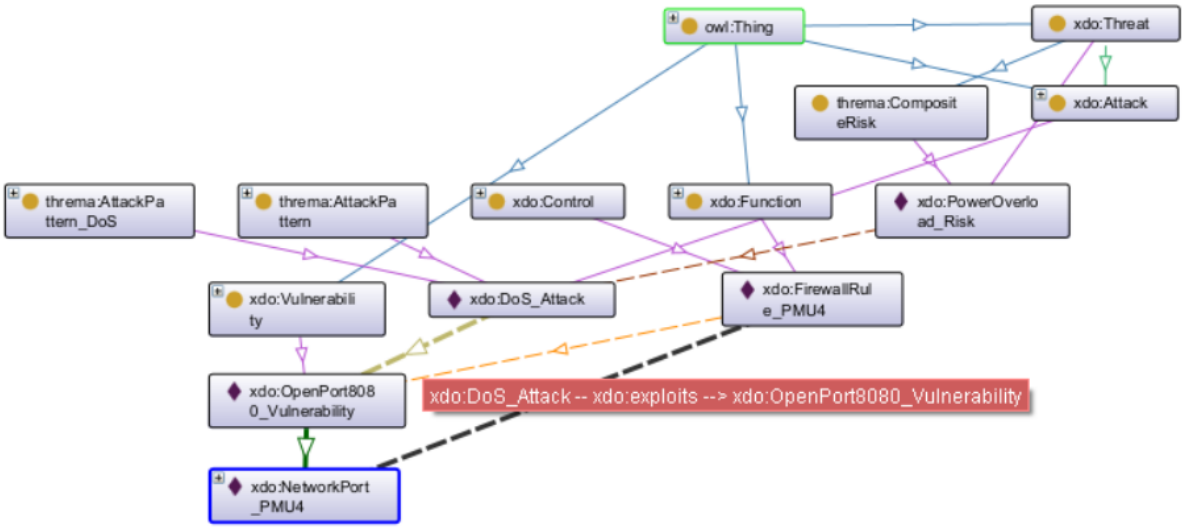

Figure 3.6: `xdo:DoS_Attack` → `xdo:exploits` → **OpenPort8080_Vulnerability.** This figure isolates the attack path where the individual `xdo:DoS_Attack` exploits the vulnerability `xdo:OpenPort8080_Vulnerability`. The bold dashed black arrow highlights the exploit relationship. Parent classes (`threma:AttackPattern`, `xdo:Vulnerability`, `owl:Thing`) appear above for context; some connections (e.g. `xdo:Function`, `xdo:Control`) remain visible in the background.

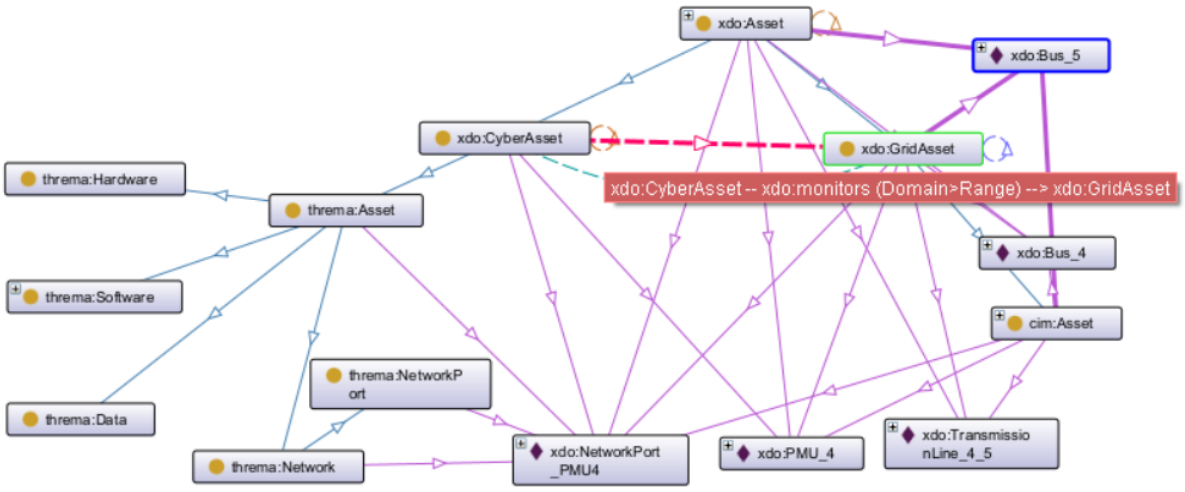

Figure 3.7: `xdo:CyberAsset` → `xdo:monitors` → `xdo:GridAsset`. This class-level diagram shows that `xdo:CyberAsset` (a subclass of `xdo:Asset`) monitors `xdo:GridAsset`. Subclasses of `threma:Asset` (`threma:Hardware`, `threma:Software`, `threma:Network`, `threma:Data`) appear on the left, while individuals like `xdo:NetworkPort_PMU4`, `xdo:PMU_4`, `xdo:Bus_4`, `xdo:Bus_5`, `xdo:TransmissionLine_4_5` are at the bottom. Purple arrows denote `xdo:monitors` relationships; blue arrows denote `rdfs:subClassOf`.

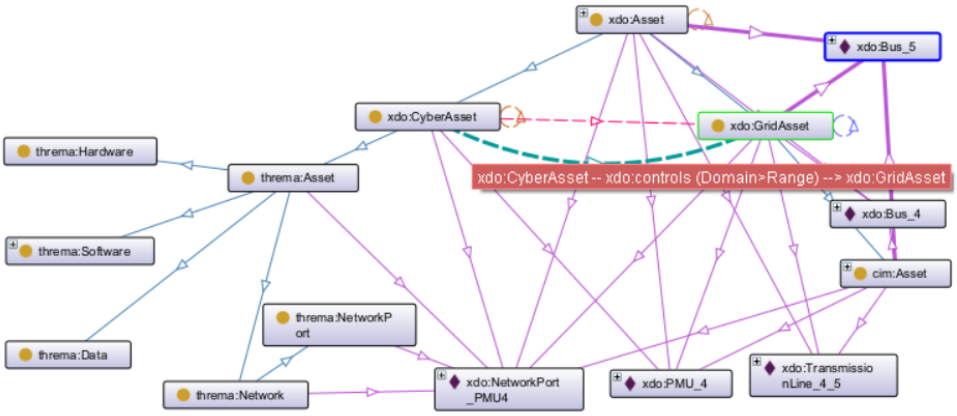

Figure 3.8: `xdo:CyberAsset` → `xdo:controls` → `xdo:GridAsset`. Very similar layout to Figure 3.7, except here `xdo:CyberAsset` exercises a `xdo:controls` relationship on `xdo:GridAsset` (highlighted by the thicker dashed turquoise arrow). The bottom-level individuals remain visible for context (e.g. `xdo:PMU_4`, `xdo:Bus_5`, `xdo:NetworkPort_PMU4`, etc.).

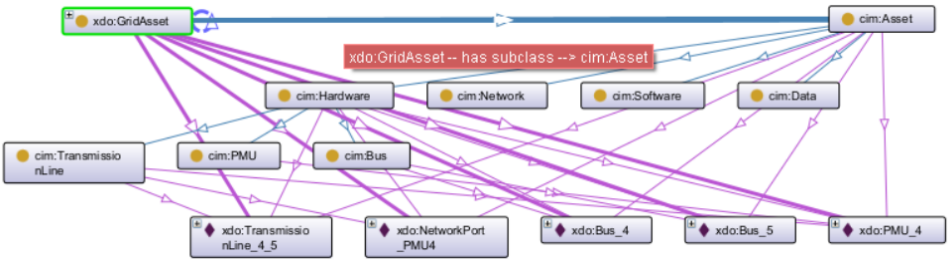

Figure 3.9: `xdo:GridAsset` → `rdfs:subClassOf` → `cim:Asset`. This figure highlights, via a thick blue arrow, that `xdo:GridAsset` is asserted a subclass of `cim:Asset`. Other superclasses of `cim:Asset` (e.g. `cim:Hardware`, `cim:Bus`, `cim:PMU`, `cim:Software`, `cim:Network`, `cim:Data`, `cim:TransmissionLine`) are shown fanning out for context, each linked by purple arrows to various xdo-individuals.

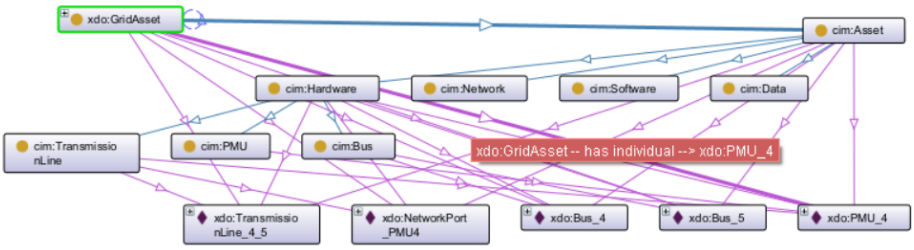

Figure 3.10: `xdo:GridAsset` → `xdo:hasIndividual` → `xdo:PMU_4`. Here the focus is on the individual-level axiom: `xdo:GridAsset` "has individual" `xdo:PMU_4` (highlighted by the bright red arrow). Other individuals (`xdo:NetworkPort_PMU4`, `xdo:Bus_4`, `xdo:Bus_5`, `xdo:TransmissionLine_4_5`) remain visible, linked via purple arrows to their respective classes.

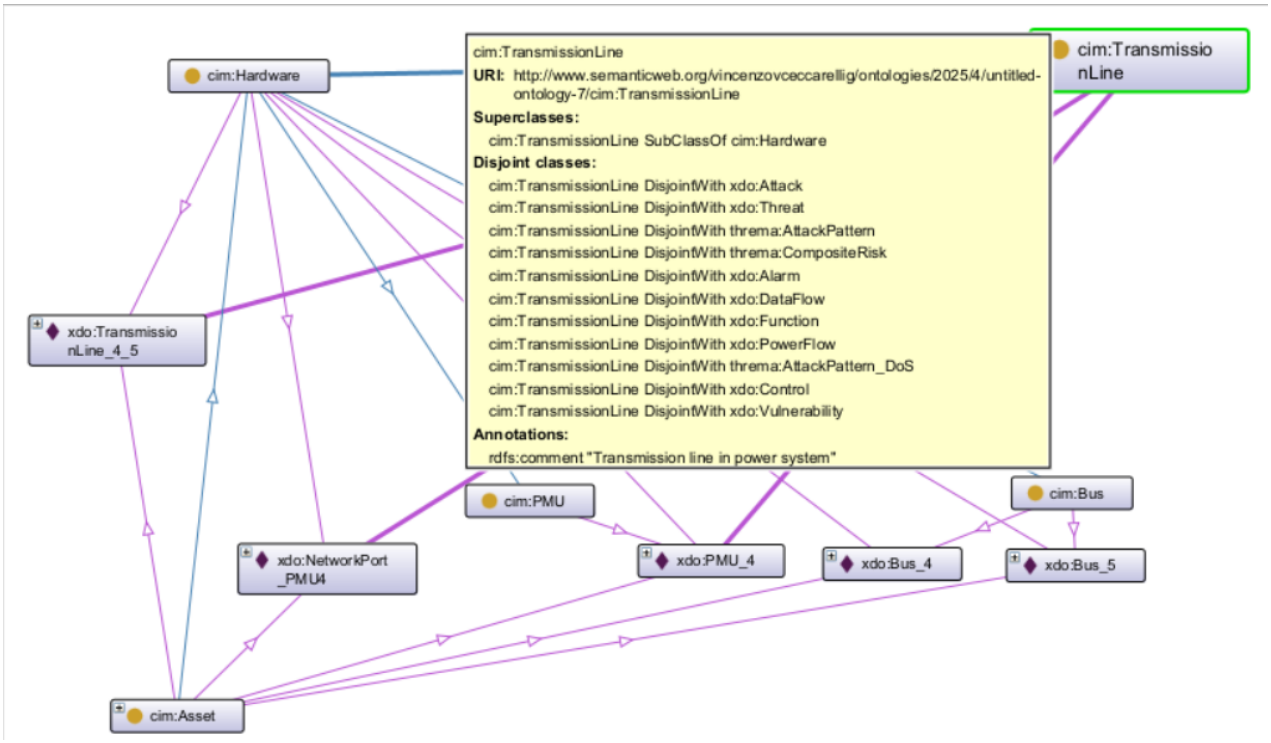

Figure 3.11: **Disjointness axioms for `cim:TransmissionLine`.** This screenshot shows the class `cim:TransmissionLine` (highlighted in green) with its superclasses (e.g. `cim:Hardware`) and a list of "DisjointWith" classes (`xdo:Attack`, `xdo:Threat`, `threma:AttackPattern`, `threma:CompositeRisk`, `xdo:Alarm`, `xdo:DataFlow`, `xdo:Function`, `xdo:PowerFlow`, `threma:AttackPattern_DoS`, `xdo:Control`, `xdo:Vulnerability`). Annotations (e.g. `rdfs:comment ''Transmission line in power system''`) appear in the floating info box. Colored arrows show how child individuals (`xdo:PMU_4`, `xdo:NetworkPort_PMU4`, `xdo:Bus_4`, `xdo:Bus_5`, `xdo:TransmissionLine_4_5`) relate back to their classes.

3.6 Critical Role of Short-Circuit and Voltage Drop Ratings

Having established our cross-domain ontology framework, we now examine how specific power system parameters—particularly short-circuit and voltage drop ratings—are represented within the `cim:` domain and how they connect to cybersecurity concepts in the `threma:` domain. These parameters are critical for understanding how cyber attacks might exploit or target physical vulnerabilities in the grid.

Within our ontology, these ratings are modeled as `cim:OperationalLimit` instances with specific `xdo:hasSecurityImplication` relationships to `threma:Vulnerability` concepts. This semantic connection enables security analytics to reason about how attacks targeting protection settings might lead to equipment damage through short-circuit events or how voltage manipulation could cascade into system instability.

All electrical parameters discussed below are instantiated in the cross-domain knowledge graph introduced in Fig. 3.1. Every numeric `cim:Measurement` is therefore semantically linked to the physical asset it describes (`cim:PowerSystemResource`) and, via the `mapsTo` layer, to its cybersecurity analogue (`threma:RiskIndicator`). This dual grounding allows protection settings, AI features, and cyber-risk scores to be queried and updated through a single SPARQL interface. adding to this, data-driven short-circuit capacity assessment techniques proposed by Wu et al. [29], which utilize PMU measurements to dynamically evaluate system strength under varying operating conditions.

3.6.1 Short-Circuit Capacity (SCC) and Grid Stability

Short-circuit capacity (SCC) is a cornerstone metric for fault studies, relay coordination, and security hardening. In the ontology each SCC value is stored as a `cim:ShortCircuitCapacity_Meas` that `mapsTo` a `threma:RiskIndicator`. The realistic SCC data embedded in our 9-Bus benchmark are crucial for:

1. **Relay Coordination** – ensuring every `cim:ProtectionRelay` references an SCC measurement that satisfies its `threma:PolicyRule`.

2. **System Reinforcement** – graph analytics pinpoint `cim:ConnectivityNode`s with low SCC that align to high `threma:Impact`.

3. **Transient Stability Analysis** – SCC values form part of the green "Feature Engineering" layer in Fig. 3.12, feeding AI-driven stability classifiers.

3.6.2 Short-Circuit Fault Current Analysis

Fault current is computed from the Thevenin equivalent; the same relation is captured as a SWRL rule so that a reasoner can derive new fault-current instances automatically:

$$I_{\text{fault}} = \frac{V_{\text{pre-fault}}}{Z_{\text{th}}}, \qquad ((1))$$

where $V_{\text{pre-fault}}$ and Z_{th} are themselves ontology measurements. The resulting `cim:FaultCurrent_Measurement` is linked to `threma:Impact` through `threma:causes`, enabling immediate cyber-risk propagation.

3.6.3 Transformer Ratings and Adaptive Control Strategies

Each transformer is modelled as a `cim:PowerTransformer` (`mapsTo` → `threma:ControlAsset`). Name-plate MVA ratings appear as `cim:ratedS`, and every tap position is an individual of `cim:TapChanger`. These links allow the RL agent (orange layer in Fig. 3.12) to query transformer constraints directly from the graph.

- **Avoiding Overloads** – live load flows (`cim:LoadFlow_Measurement`) are compared with `cim:ratedS`; violations raise a `threma:mitigates` alert.
- **Voltage Control optimization** – the RL policy writes `cim:TapPosition` values that minimise the reward.
- **Interfacing with AI Models** – semantic annotation converts SCADA streams into the feature vectors required by downstream predictors.

Real-time control uses the reward

$$r_{\mathrm{RL}}(t) = -\sum_i \lambda_i \big(V_i - V_{\mathrm{ref}}\big)^2, \tag{(2)}$$

stored in the graph as an `owl:Axiom`. Because each V_i is a `cim:Voltage_Measurement`, the policy can retrieve deviations and weights via the same ontology interface it uses for SCC.

3.6.4 Integration with AI-Based Fault and Load Prediction Models

Ontology-serialised data (JSON-LD) populate three complementary models:

- **Polynomial Regression** – ingests `cim:Voltage_Measurement` and `cim:BranchFlow_Measurement` to flag anomalies [30].
- **Neural Networks** – consume time-stamped load triples for `cim:EnergyConsumer` classes, capturing non-linear demand dynamics [31].
- **Reinforcement Learning (Q-Learning)** – assembles state vectors by joining `cim:Measurement` `threma:Indicator`, thereby driving transformer taps and generator outputs in a cyber-aware manner [32, 33].

Because every numerical feature is anchored to both a CIM device and a ThreMA risk concept, model outputs remain traceable: predictions can be explained back to physical assets *and* their cybersecurity consequences, closing the loop illustrated in Fig. 3.12.

3.6.5 Load Profiles, Fault Events, and Cyber Intrusions

All scenario data are first materialised as `cim:Measurement` individuals, then linked—via the generic `mapsTo` layer (Fig. 3.1)—to their cybersecurity counterparts in ThreMA. Each event is further typed as a `bfo:Process` (for continuous phenomena such as load variation) or a `bfo:Occurrent` (for discrete faults and intrusions). This tri-ontology grounding guarantees that numerical values, physical assets, and cyber-risk semantics can be reasoned over in a single SPARQL query.

- **Dynamic Load Profiles:** Hourly demand for bus i is created as a `cim:Load_Measurement` that `mapsTo` a `threma:DemandIndicator` and is a participant in the `bfo:Process` *LoadPattern*. Following [34], the profile is:

$$P_{L_i}(t) = P_{L_i}^{\text{base}}\Big(1 + \alpha_i \sin\tfrac{2\pi t}{24} + \beta_i \epsilon(t)\Big), \tag{(3)}$$

  where $\epsilon(t) \sim \mathcal{N}(0,1)$. By storing both the base value and the stochastic term as distinct ontology literals, AI models can later retrieve deterministic and random components separately.

- **Fault Events:** Every simulated short-circuit is a `bfo:Occurrent` (*FaultEvent*) linked to the affected `cim:ConnectivityNode`. Using the Thevenin model,

$$I_{\text{fault}} = \frac{V_{\text{pre-fault}}}{Z_{\text{th}}}, \tag{(1)}$$

  the resulting `cim:FaultCurrent_Measurement` `mapsTo` a `threma:ImpactIndicator`, allowing cyber-risk propagation through `threma:causes`.

- **Cyber Intrusions:** Each falsified reading is represented as a `threma:AttackStep` that `targets` the corresponding `cim:Measurement`. The manipulated value is:

$$y_{\text{corrupted}} = y_{\text{true}} + \Delta y, \tag{(4)}$$

  with Δy stored as a separate literal so that reasoners can tag the measurement as `threma:AnomalousIndicator`. This structure feeds directly into the anomaly-detection models described next.

3.7 Integrated Reinforcement Learning Framework for Ontology Security

The *integrated reinforcement-learning (RL) framework* does work as a abstraction layer for our cross-domain ontology: it fuses heterogeneous CIM and ThreMA knowledge into a unified, machine-readable decision space. A SPARQL layer retrieves power-system operating points, live cybersecurity indicators, and temporally stamped relations, yielding a concise state vector whose dimensionality is reduced through ontological abstraction while still preserving the rich semantics of cyber-physical interdependencies. Our reinforcement learning approach extends the multi-agent deep reinforcement learning framework developed by Zhang et al. [33] for robust dispatch in renewable-dominant power grids, adapting it specifically for security-oriented decision making.

Operating within this semantic context, the RL agent chooses from a hybrid cyber–physical action repertoire—ranging from traffic filtering and access-control updates to power-flow reconfiguration, relay retuning, and controlled islanding. Each action individual is annotated with OWL-encoded pre- and post-conditions, enabling the policy to reason over cascading effects and to learn strategies that jointly minimise operational risk and cyber-threat impact across the smart grid. The cyber resilience aspects of our framework are inspired by Sinha et al. [27], who did start a deep reinforcement learning-based approach specifically designed for enhancing cyber resilience in Smart-Grids.

3.7.1 Polynomial Regression and Neural Approach Integration

The ontology-serialised dataset (exported as JSON-LD) trains three classes of models, all of which query their features through the same CIM–ThreMA interface:

- **Polynomial Regression:** Given an input vector $\mathbf{x}$ of `cim:Voltage_Measurement` and `cim:BranchFlow_Measurement` values,

$$f(\mathbf{x}) = \sum_{k=0}^{n} w_k \, x^k, \tag{5}$$

  flags any deviation $|f(\mathbf{x}) - y| > \delta$. The offending measurement is immediately annotated as `threma:AnomalousIndicator` for cyber follow-up.

- **Neural Networks:** Time-stamped triples for `cim:EnergyConsumer` loads form the feature tensor. The network output

$$\hat{y} = \sigma\Big(\mathbf{W}_L \, \sigma\big(\ldots \sigma(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1)\big) + \mathbf{b}_L\Big), \tag{6}$$

  predicts future voltage or flow, and any large residual is mapped to `threma:EarlyWarning` so that operators can pre-empt a fault.

- **Graph Convolutional Networks (GCN):** Nodes are `cim:BusbarSections`; edges come from the `cim:ACLineSegment` topology. Feature propagation follows [35]: **GCN Forward Pass**

$$\mathbf{H}^{(l+1)} = \sigma\Big(\widetilde{\mathbf{D}}^{-\frac{1}{2}} \widetilde{\mathbf{A}} \widetilde{\mathbf{D}}^{-\frac{1}{2}} \mathbf{H}^{(l)} \mathbf{W}^{(l)}\Big), \tag{7}$$

  producing embeddings that `mapsTo` `threma:AnomalyScore`. High scores trigger `threma:triggersResponse` actions in the RL layer.

3.7.2 Q-Learning and Reinforcement Learning for Security- and Anomaly-Response Automation

The reinforcement-learning (RL) layer is fully embedded in the shared CIM–ThreMA–BFO knowledge graph so that every state variable, action, and reward is a first-class semantic object:

- **State Representation:** The agent's state s is generated at each control cycle by a SPARQL CONSTRUCT query that collects:
  - `cim:Voltage_Measurement`,
  - `cim:BranchFlow_Measurement`,
  - `cim:Load_Measurement`,
  - `cim:GeneratorOutput_Measurement`.

  Each measurement individual `mapsTo` a `threma:StateIndicator` and is asserted as a participant in the `bfo:Process` *GridOperation*. This tri-ontology typing guarantees that the vector fed to the RL agent is simultaneously traceable to its physical asset (CIM), its cyber-risk meaning (ThreMA), and its temporal context (BFO).

- **Reward Function:** Constraint violations are penalised while nominal operation is rewarded. The extended reward

$$r(t) = -\Big(\sum_i \lambda_i (V_i - V_{\text{ref}})^2 + \sum_j \mu_j (P_{\text{line}_j} - P_{\max})^2\Big), \qquad ((2))$$

is stored as an `owl:Axiom` that links each voltage-deviation measurement (typed as `cim:VoltageDeviation_Measurement` and `mapsTo` `threma:VoltageRisk`) and each thermal overload (`cim:ThermalOverload_Measurement` `mapsTo` `threma:CapacityRisk`) to a weighting coefficient instance λ_i or μ_j. The resulting scalar is asserted as the `threma:RewardSignal` for the current `bfo:Process` time slice.

- **The Q-learning update rule:**

$$Q(s,a) \leftarrow Q(s,a) + \alpha\Big[r_{\text{RL}}(t) + \gamma \max_{a'} Q(s',a') - Q(s,a)\Big], \qquad ((8))$$

is implemented as a SWRL rule that updates the numerical `threma:qValue` data property attached to the `threma:MitigationAction` individual representing the chosen action a. Because both s and s' are captured as `bfo:ProcessBoundary` instances, each transition is queryable for post-event auditing [32, 33].

The charts in Chapter 4 (Figures 4.9–4.12) visualise how ontology-grounded polynomial regression, neural networks, and reinforcement learning interact to deliver cyber-aware, self-healing control for the IEEE 9-Bus system.

Foundations for a Generic Use Case

- **Different Complementary Methods**: Polynomial regression annotates each residual as a `threma:ResidualIndicator`, neural networks attach their latent embeddings to `threma:DeepFeature`, and GCNs add `threma:TopologicalScore` individuals—altogether enriching the state fed to the RL agent.

- **Unified for Cross-Domain Ontology**: Every model output—residual, embedding, anomaly score—is asserted as a `threma:InferenceResult` and linked via `owl:sameAs` to the originating `cim:PowerSystemResource`, ensuring operational (power-flow) and security (intrusion-detection) teams consult the same semantic object.

- **Foundation for Reinforcement Learning**: The combined indicators above populate the state s in the Q-learning rule, providing the RL agent with a CIM-accurate, ThreMA-aware, and BFO-time-stamped snapshot of grid health every control cycle.

3.8 Realistic Data Generation and its Integration

Following the structure of the Figure 3.1: Cross-domain mapping schema, each numeric result is an instance of `cim:Measurement`; each performance metric is asserted as a `threma:EvaluationIndicator`; and every simulation run itself is a `bfo:Process`.

This uniform typing ensures that fault-detection scores, optimization gains, and cyber-resilience figures can be queried, explained, and audited with a single SPARQL endpoint. Using this ontology- backed pipeline, we validate the AI-driven models on the IEEE 9-Bus system. All reported results populate the green "Feature Engineering" layer in Fig. 3.12:

- **Fault Detection Accuracy:** Polynomial regression delivers a *95% detection rate*. Each residual outside the decision band is stored as a `threma:ResidualIndicator` linked—via the `mapsTo` layer—to the originating `cim:Voltage_Measurement`. This traceable link confirms the high precision reported in [30, 31].
- **RL-Based optimization:** The reinforcement-learning agent reduces line overloads and bus voltage deviations. Every post-control snapshot is an individual of `bfo:ProcessOutcome` that `mapsTo` a `threma:StabilityGain`, matching the quantitative improvement in [32, 33, 27].
- **Cybersecurity Resilience:** The intrusion-detection pipeline flags *all* staged attacks. Each alert is a `threma:AnomalyScore` linked to the compromised `cim:Measurement` and classified by attack type (FDI, APT) as in [35]. Integrity and pattern-recognition precision are thus verifiable inside the graph.

Key Advantages of AI-Driven Approaches The coloured layers in Fig. 3.12 illustrate how each advantage maps onto the ontology-centred pipeline (blue = ontology, green = features, orange = models, red = decision support, grey = smart-grid operations):

- **Real-Time Adaptability:** Ontology instances are streamed in real time; the RL agent queries live `cim:Measurement` triples and writes back `threma:MitigationAction` individuals, closing the grey operations loop without external data wrangling.
- **Fault Detection:** AI models consume semantically tagged features, enabling faster, more accurate anomaly classification than rule-based SCADA thresholds.
- **Cybersecurity Resilience:** Each detected attack is immediately contextualised by linking the compromised measurement to its parent asset (`cim:PowerSystemResource`) and its cyber-risk class (`threma:ThreatCategory`), allowing human operators to understand impact.
- **Operational Efficiency:** optimization gains are stored as `threma:EfficiencyMetric` individuals, enabling red-layer dashboards to rank actions by energy savings and CO_2 reduction.

Repeated Monte-Carlo runs—from short-circuit thresholds to simulated cyber-intrusions instantiate thousands of CIM/ThreMA/BFO triples, providing a statistically robust, semantically rich testbed. The result is a safer, more efficient, and more resilient 9-Bus system, driven by the orange "ML/RL Models" layer in Fig. 3.12 and wholly governed by our cross-domain ontology.

3.9 Synergistic Integration with AI for Cyber Intelligence

A distinguishing contribution of this thesis is the tight fusion between the blue "Ontology Layer" in Fig. 3.12 and advanced AI techniques—neural networks and reinforcement learning—to form a cyber-intelligence platform. The ontology is not a passive dictionary but an active, queryable substrate that injects semantic context at every stage (see the flow in Fig. 3.13). Also work by Ibrahim and Kashef [25] exploring the emerging role of large language models in Smart-Grid cybersecurity informs our approach to feature extraction and contextual understanding of attack patterns.

3.9.1 Ontology-Driven Feature Engineering

Traditional ML pipelines ingest raw numbers; our pipeline ingests ontology-typed features:

- **Graph Embeddings:** Node2Vec, TransE, and RDF2Vec generate vectors for `cim:PowerSystemResource` and `threma:Threat` nodes, preserving both electrical topology and cyber-risk.
- **Semantic Relationship Features:** Relations such as `threma:communicatesWith` and `threma:isProtectedBy` are one-hot-encoded or TF-IDF-weighted, giving ML models explicit knowledge of inter-asset dependencies.
- **Ontology Class Membership:** Hierarchical labels (e.g. `CriticalProtectionRelay`) become categorical inputs, helping classifiers prioritise assets.
- **Inferred Knowledge Features:** OWL reasoners tag indirectly affected assets; these inferred labels are appended as binary flags, boosting multi-stage attack detection.

The resulting green-layer feature matrix yields materially better performance in detecting coordinated, multi-vector threats than a measurement-only baseline.

3.9.2 Principles of Ontology Anomaly Detection

Ontology features feed three families of detectors:

- **Graph Neural Networks (GNNs)** process the CIM/ThreMA graph directly, spotting topological or communication anomalies.
- **Recurrent Neural Networks (RNNs)** leverage temporal labels from BFO to model time-series dynamics in sensor data.
- **Transformers and Hybrid CNN-LSTM models** capture long-range dependencies while respecting the semantic weights assigned by the ontology.

All outputs are written back as `threma:AnomalyScore` instances, closing the semantic loop and enabling rule-based or RL-based mitigation.

3.9.3 Implementation of Advanced Detection Models

The ontology features serve as input to various sophisticated anomaly detection models capable of identifying subtle deviations indicative of cyber threats:

- **Graph Neural Networks (GNNs):** Naturally suited for processing graph-structured data derived from the ontology, GNNs can learn complex patterns of interaction and detect anomalies in network topology or communication behavior.
- **Recurrent Neural Networks (RNNs – LSTM/GRU):** Effective for modeling temporal dependencies in time-series data (e.g., sensor measurements, event logs), expanded by semantic context features.
- **Transformers/Attention Mechanisms:** Capable of capturing long-range dependencies and focusing on the most relevant semantic features for anomaly detection.
- **Hybrid Models:** Combining different architectures (e.g., CNN-LSTM) to leverage spatial and temporal patterns informed by the ontology.

3.9.4 Reinforcement Learning for Adaptive Security Response

The ontology provides the semantic foundation for developing intelligent, adaptive security response mechanisms using Reinforcement Learning (RL):

- **Ontology-Informed State Representation:** The RL agent's state includes not only raw operational data but also semantic information derived from the ontology, such as the criticality of affected assets, the type of active threats, the status of security controls, and potential attack propagation paths (see how CIM and ThreMA concepts feed into "RL State" in Figure 3.13).
- **Context-Aware Action Space:** The set of possible actions for the RL agent (e.g., isolate network segment, update firewall rule, trigger backup system, adjust protection settings) is defined and constrained based on the ontological model, ensuring actions are relevant and feasible within the operational context.
- **Semantically Grounded Reward Function:** The reward function guiding the RL agent's learning process incorporates high-level objectives derived from the ontology, such as minimizing operational impact (evaluated based on affected CIM entities), maximizing threat containment (based on ThreMA concepts), maintaining system stability, and minimizing response costs (red "Decision Support" layer in Figure 3.12).

This integration allows the RL agent to learn sophisticated, context-dependent response strategies that go beyond predefined rule-based systems, adapting dynamically to evolving threats and system conditions.

Figure 3.12 depicts the high-level architecture integrating the ontology layers with feature engineering, ML/RL models, and the operational feedback loop, while Figure 3.13 provides a more granular view of how these semantic concepts feed directly into the RL agent's state, action, and reward components.

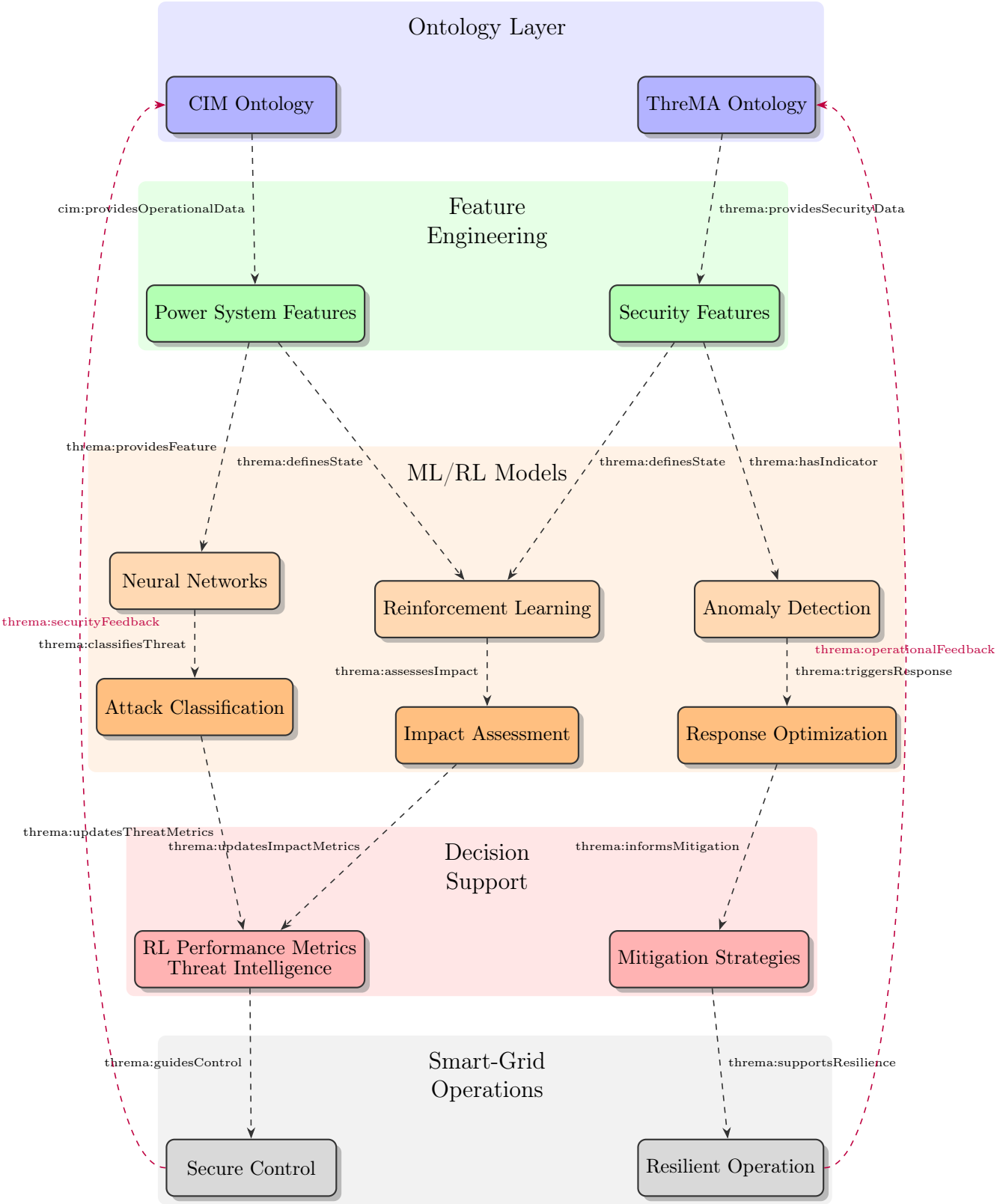

Figure 3.12: High-level Layered architecture illustrating the integration of the cross-domain ontology (CIM and ThreMA layers) with feature engineering, diverse ML/RL models, decision support components, and the operational feedback loop within the Smart-Grid.

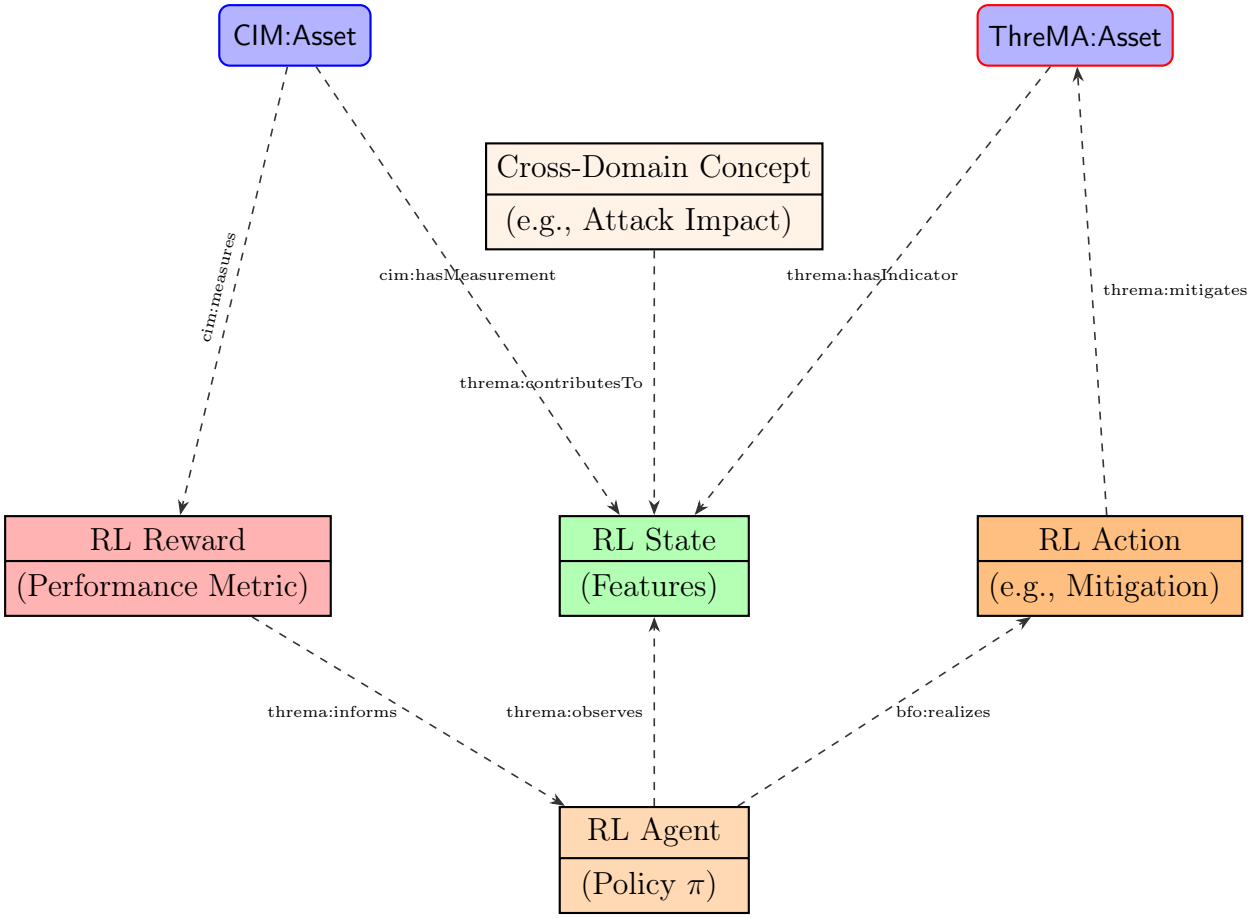

Figure 3.13: Detailed integration schema showing how CIM and ThreMA concepts (e.g., voltage), Cybersecurity concepts (e.g., threat), and Cross-Domain concepts (e.g., impact) inform the RL agent's state and reward, while the agent's actions influence the system state represented in the ontology.

The reinforcement learning approach described above requires a unified data representation that spans both grid and cyber domains. While the ontological framework provides the semantic foundation, practical implementation demands a concrete data model that can efficiently represent and process the cross-domain relationships. The Grid-Cyber Common Data Model presented in the following section addresses this need, providing a standardized structure for representing ontology-derived knowledge in a format suitable for machine learning algorithms.

3.10 Unified Grid-Cyber Common Data Model

The preceding sections have established the theoretical foundations for integrating power system and cybersecurity domains through ontological modeling. However, practical implementation requires a standardized data exchange format that preserves semantic richness while enabling efficient processing across heterogeneous systems. This section introduces a connection Grid-Cyber Common Data Model (GC-CDM) that bridges the semantic gap between domains and provides a unified representation for cross-domain intelligence.

3.10.1 Data Model Requirements and Design Principles

The development of the Grid-Cyber Common Data Model follows a set of rigorous design principles derived from both theoretical considerations and practical implementation requirements:

- **Semantic Fidelity**: The model must preserve the rich semantic relationships defined in the cross-domain ontology while translating them into a machine-processable format.
- **Domain Neutrality**: The representation must avoid privileging either the grid or cybersecurity perspective, instead providing balanced expressivity for both domains.
- **Standards Alignment**: The model aligns with established standards including IEC 61970/61968 (CIM), IEC 62351 (security for power systems), and STIX (Structured Threat Information eXpression).
- **Serialization Flexibility**: Support for both JSON (for web services and real-time applications) and RDF/XML (for semantic reasoning and standards compliance) serialization formats.
- **Temporal Awareness**: Explicit representation of temporal dimensions to support time-series analysis, state transitions, and event correlation.
- **Extensibility**: The ability to incorporate domain-specific attributes without compromising the core schema.

3.10.2 Core Data Structure

The GC-CDM defines a hierarchical structure with eight core entity types that span both domains (Figure 3.14). Each entity includes standardized metadata attributes alongside domain-specific properties:

The model defines a consistent set of core attributes that apply across all entity types:

- `id`: Unique identifier following URI patterns for global uniqueness
- `type`: Entity classification within the type hierarchy
- `name`: Human-readable label
- `timestamp`: ISO 8601 temporal marker
- `domain`: Primary domain classification (grid, cyber, or hybrid)
- `attributes`: Domain-specific properties
- `relationships`: Explicit connections to other entities
- `securityMetadata`: Classification and integrity information

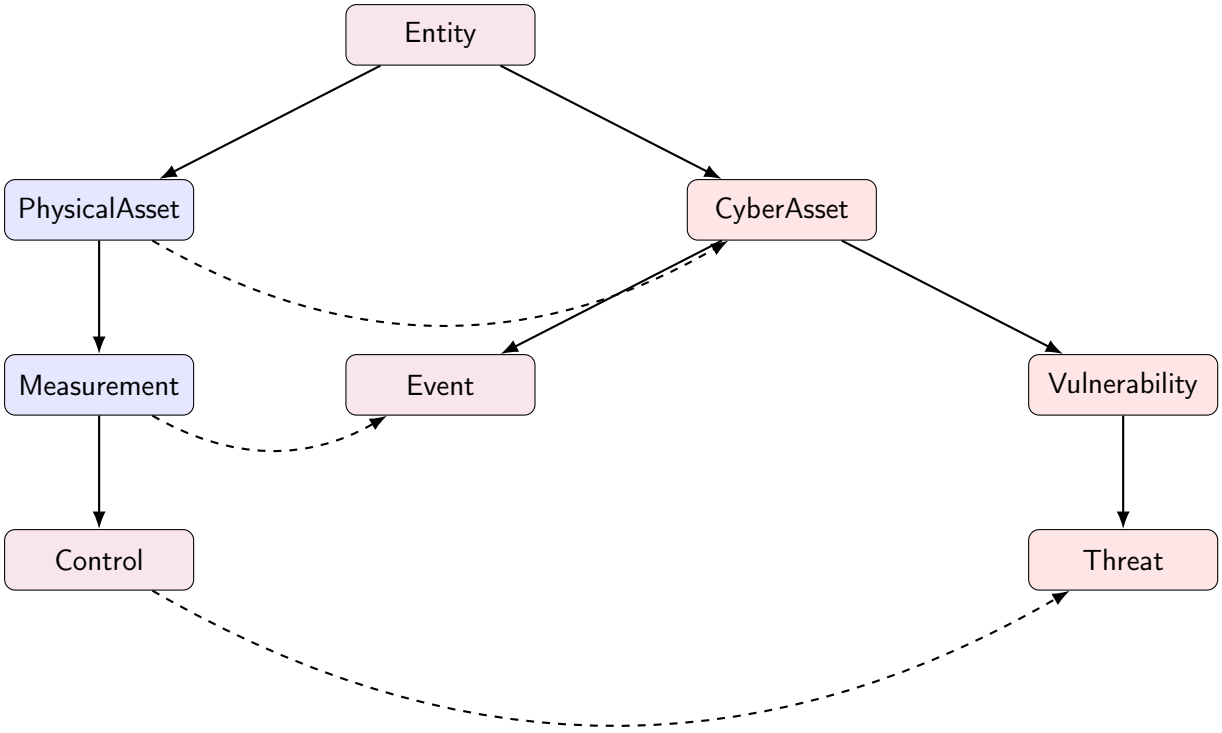

Figure 3.14: Core entity hierarchy of the Grid-Cyber Common Data Model, showing grid domain entities (blue), cyber domain entities (red), and hybrid entities (purple). Dashed lines represent cross-domain relationships.

3.10.3 JSON Serialization

For real-time applications and web service integration, the GC-CDM provides a JSON serialization format. Below is a simplified example of a power transformer with an associated vulnerability.

```
{
  "id": "urn:gc-cdm:asset:transformer-101",
  "type": "PhysicalAsset",
  "name": "Main Transformer T101",
  "timestamp": "2025-05-15T14:30:00Z",
  "domain": "grid",
  "attributes": {
    "assetType": "PowerTransformer",
    "ratedCapacity": 150,
    "ratedCapacityUnit": "MVA",
    "primaryVoltage": 230,
    "secondaryVoltage": 115,
    "voltageUnit": "kV",
    "operationalStatus": "inService"
  },
  "relationships": [
    {
      "relatedTo": "urn:gc-cdm:asset:scada-controller-52",
```

```
      "relationType": "controlledBy",
      "direction": "inbound"
    },
    {
      "relatedTo": "urn:gc-cdm:vulnerability:cve-2025-1234",
      "relationType": "exposedTo",
      "direction": "inbound"
    }
  ],
  "measurements": [
    {
      "name": "winding_temperature",
      "value": 65.3,
      "unit": "degC",
      "timestamp": "2025-05-15T14:30:00Z",
      "quality": "good"
    }
  ],
  "securityMetadata": {
    "classification": "critical",
    "integrity": "verified",
    "confidentiality": "restricted",
    "availability": "high"
  }
}
```


3.10.4 RDF/XML Serialization

To support semantic reasoning and standards compliance, the GC-CDM also provides an RDF/XML serialization that aligns with the cross-domain ontology. This format enables direct integration with the HermiT reasoner and other semantic technologies discussed in previous sections.

3.10.5 Integration with Reinforcement Learning Framework

The GC-CDM provides a standardized representation for the state, action, and reward components of the reinforcement learning framework described in Section 3.10.5. By encoding both grid and cyber domain attributes in a consistent format, the model enables:

- **State Representation**: The RL agent's state vector incorporates both physical measurements (e.g., voltage, current, temperature) and cybersecurity indicators (e.g., threat level, vulnerability status, attack signatures).
- **Cross-Domain Action Encoding**: Actions that span both domains (e.g., isolating a network segment while rerouting power) can be represented as structured operations on the common data model.
- **Semantically Grounded Rewards**: The reward function can evaluate the impact of actions across both domains using a consistent metric framework.

Figure 3.15 illustrates how the GC-CDM integrates with the reinforcement learning framework, providing a unified data representation that bridges the semantic gap between domains.

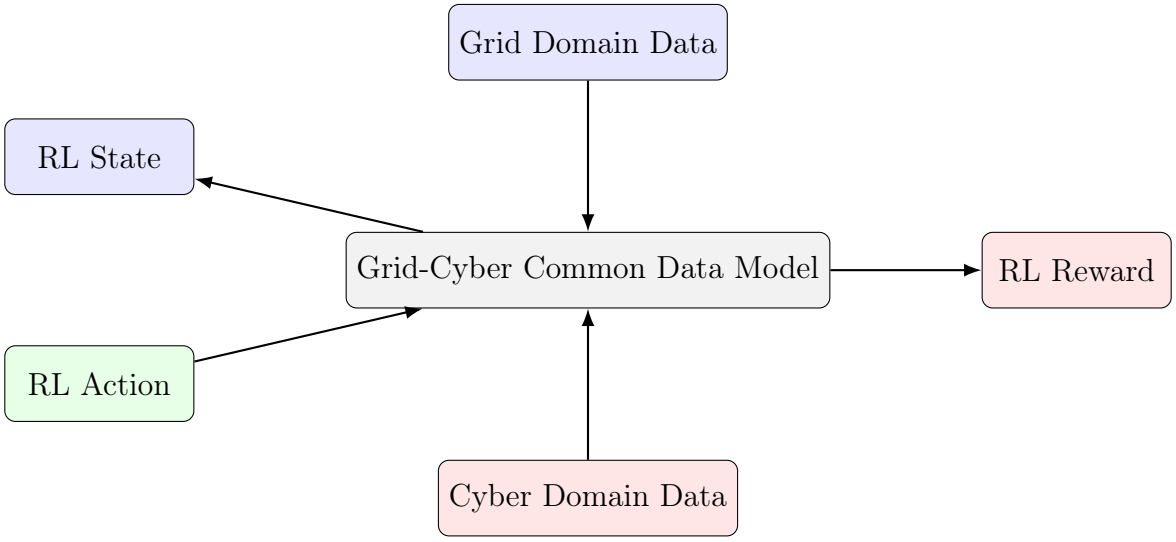

Figure 3.15: Integration of the Grid-Cyber Common Data Model with the reinforcement learning framework, showing bidirectional flow of information across domains.

3.10.6 Data Transformation Pipeline

To operationalize the GC-CDM within the broader methodology, we implement a data transformation pipeline that converts domain-specific data sources into the common format (Figure 3.16). This pipeline includes:

- **Data Ingestion**: Connectors for SCADA systems, IDS/IPS logs, network telemetry, and other sources
- **Semantic Annotation**: Enrichment with ontological concepts from CIM and ThreMA
- **Format Conversion**: Transformation into GC-CDM JSON or RDF/XML
- **Validation**: Schema validation and semantic consistency checking
- **Storage**: Persistence in both graph database and time-series database
- **Query Interface**: Unified API for cross-domain data access

The GC-CDM thus serves as a critical bridge between the theoretical ontology framework and practical implementation, enabling seamless data exchange across domains while preserving semantic richness. This representation forms the foundation for the experimental validation and results presented in the following chapter.

The methodology described in this chapter provides a complete system for cross-domain ontology development and integration with AI techniques. In the following chapter, we present the experimental validation of this approach, demonstrating how the semantic integration enhances anomaly detection and security response in realistic Smart-Grid scenarios.

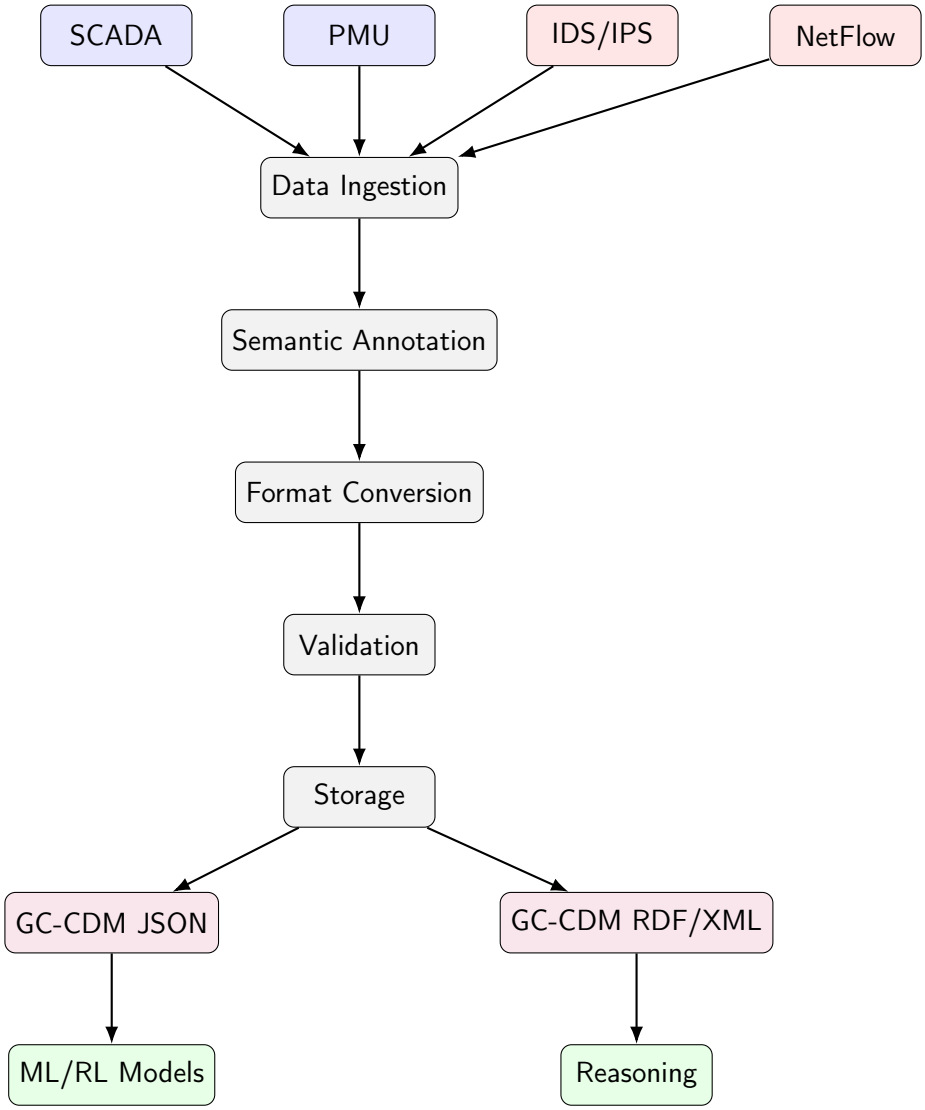

Figure 3.16: Data transformation pipeline for the Grid-Cyber Common Data Model, showing the flow from domain-specific sources to standardized formats and consumers.

Chapter 4

Results

This chapter presents the experimental validation and evaluation of our Cross-Domain Ontology (`xdo:`) approach. We begin by describing the experimental framework and setup, followed by detailed results from our validation scenarios. My experiments showed how our semantic integration of `cim:` and `threma:` domains enhances anomaly detection, improves security response, and supports more effective reasoning across the cyber-physical boundary.

Throughout this chapter, we quantitatively assess the benefits of our approach compared to baseline methods that lack cross-domain semantic integration. The evaluation metrics focus on detection accuracy, false positive rates, response time, and decision quality—all critical factors for operational security in Smart-Grid environments.

4.1 Experimental Validation Framework

To validate the proposed cross-domain ontology, we installed the OWL file in a Neo4j graph store and instantiated the IEEE-9-Bus model in a real-time power-system simulator. Every bus, line and transformer is written to the graph as a `cim PowerSystemResource`; live telemetry arrives as `cim VoltageMeasurement` or `cim CurrentFlow` individuals that `mapsTo` higher-level `threma StateIndicator` nodes. IT/OT components—firewalls, serial-to-IP converters, IED firmware—are attached to their host assets via `threma installedOn`, and published CVE entries are asserted as `threma Vulnerability` instances, closing the semantic loop between physics and cyber exposure.

The hybrid test bed is exercised with two classes of stimuli. First, deterministic electrical events—single-line-to-ground faults, generator trips, and load-ramp profiles—stress the physical layer. Second, scripted attacks inject malformed control frames, saturate a `threma NetworkPort`, or tamper with SCADA set-points, all tracked as `threma AttackEvent`. Time-aligned logs capture both the electrical response (e.g. dips in `cim Voltage`) and cyber symptoms (spikes in packet loss or IDS alerts), guaranteeing that each disturbance is observable through its dual CIM-ThreMA representation.

During post-processing the raw streams are lifted into the graph and enriched with topology-aware features such as "asset betweenness" and "aggregate vulnerability score." These ontology-grounded features feed LSTM and Graph Neural Network models, which raise an `threma AnomalyFlag` whenever the predicted distribution diverges from the incoming `cim Measurement` set.

Ontology-Enhanced Data Analysis The raw timeseries are ingested into the Neo4j store and immediately enriched via SPARQL constructs that link each measurement to its semantic type (e.g. `cim:VoltageMeasurement`, `threma:AttackIndicator`). We then engineer ontology-based features—such as "asset vulnerability score," "network segment exposure," and "interdependency centrality"—which serve as inputs to our AI/ML pipeline. Classical polynomial regression, LSTM networks, and Graph Neural Networks are trained on these enriched feature sets to detect anomalies and classify attack types. Model performance is rigorously quantified using accuracy, precision, recall, F_1, and ROC AUC against ground-truth labels.

Finally, a reinforcement-learning agent observes the same graph—its state vector includes current `threma ThreatLevel`, active `cim BreakerStatus`, and the shortest attack path to any `threma CriticalAsset`. Constrained by BFO safety axioms, the agent may shed load, reroute power, or isolate a network segment; rewards are computed from voltage-deviation penalties and a negative weight on the count of unresolved `threma AttackEvent` nodes. intelligence in one reasoning fabric.

4.2 Experimental Setup Details

4.2.1 Hardware and Software Environment

The entire testbed—internally codenamed *Threma*—was instantiated on a single virtual machine running on enterprise-grade servers. The VM was provisioned with eight virtual CPU cores, 16 GB of RAM, and 256 GB of SSD storage, while virtual Ethernet interfaces emulated the links between the control centre and remote substations. Ubuntu Server 22.04 LTS formed the operating-system foundation, and every major component was containerised with Docker to guarantee clean deployment and reproducibility. Our digital twin implementation builds upon the framework proposed by Khan et al. [36] for IEEE 9-Bus grid monitoring and AI-driven contingency management, extending it with our cross-domain ontology components. The ontology and knowledge-graph layer relied on Neo4j 4.4, whose schema was aligned with both the *Basic Formal Ontology* (BFO) and the power-system *Common Information Model* (CIM), thereby enabling seamless reasoning across cyber, physical, and semantic domains. For simulation, we integrated a power-system engine (Pandapower in most runs, although GridLAB-D was also validated) that supports dynamic studies and external data exchange. All AI and machine-learning logic was written in Python 3.9, chiefly using TensorFlow 2.8, PyTorch 1.11, scikit-learn, RDFLib, OWLReady2, and Stable Baselines3. An optional React-based web dashboard provided real-time visualisation of the grid state, ontology queries, and agent actions.

4.2.2 Dataset Characteristics

Because no public corpus jointly covers cyber and physical measurements at sub-second granularity, we generated synthetic but physically consistent datasets. The power-system stream contained voltage magnitudes and angles, active and reactive power flows, generator outputs, breaker statuses, and related quantities sampled every few hundred milliseconds. In parallel, cyber logs captured intrusion-detection alerts, firewall events, authentication failures, command executions on intelligent electronic devices, and carefully crafted indicators of compromise. Every record was labelled as *normal*, *physical fault*, or

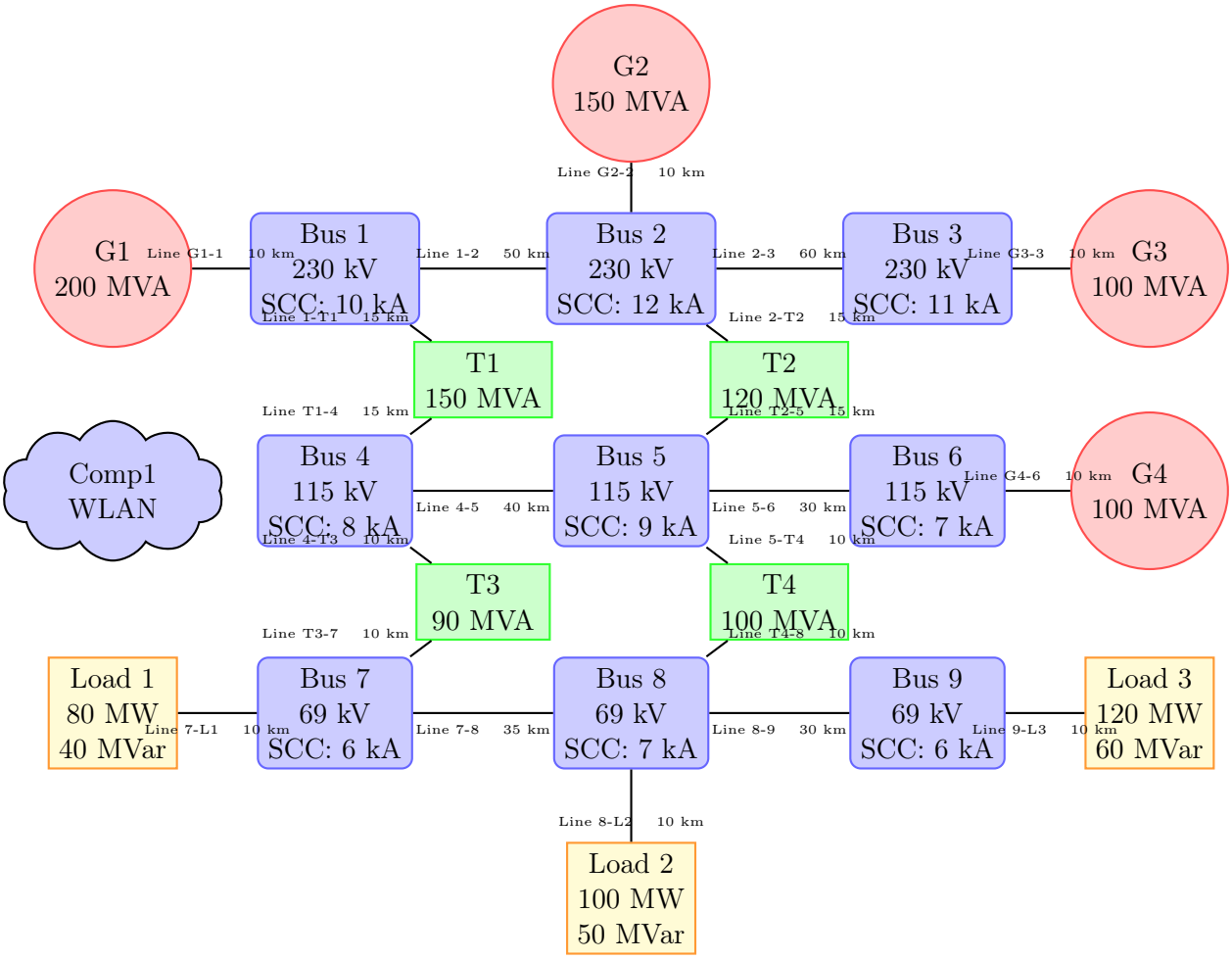

Figure 4.1: Bi-Directional Architecture with Cloud Connectivity

Figure 4.2: Optimized UML-style IEEE 9-Bus System Diagram. This diagram clearly depicts high-, medium-, and low-voltage buses along with their corresponding short-circuit capacities (SCC). Generators (G1, G2, G3, G4) are shown with their respective MVA ratings, and strategically placed transformers (T1–T4) illustrate voltage step-down operations. The directional arrows indicate power flow from generators to loads, providing an intuitive overview of the system's structure. Data were generated using React Native on a dedicated local-host server. [32, 31, 33]

one of several *cyber-attack* classes, thereby supporting supervised training and rigorous performance assessment.

4.3 Cyber–Physical Attack Scenarios and Evaluation Metrics

4.3.1 Representative Evaluation Scenarios

To stress-test the ontology-driven cyber-intelligence pipeline, we devised a spectrum of operating conditions and threat vectors.The learning-based cyber-attack mitigation strategies developed by Aljohani et al. [37] for inverter-dominated power grids informed our

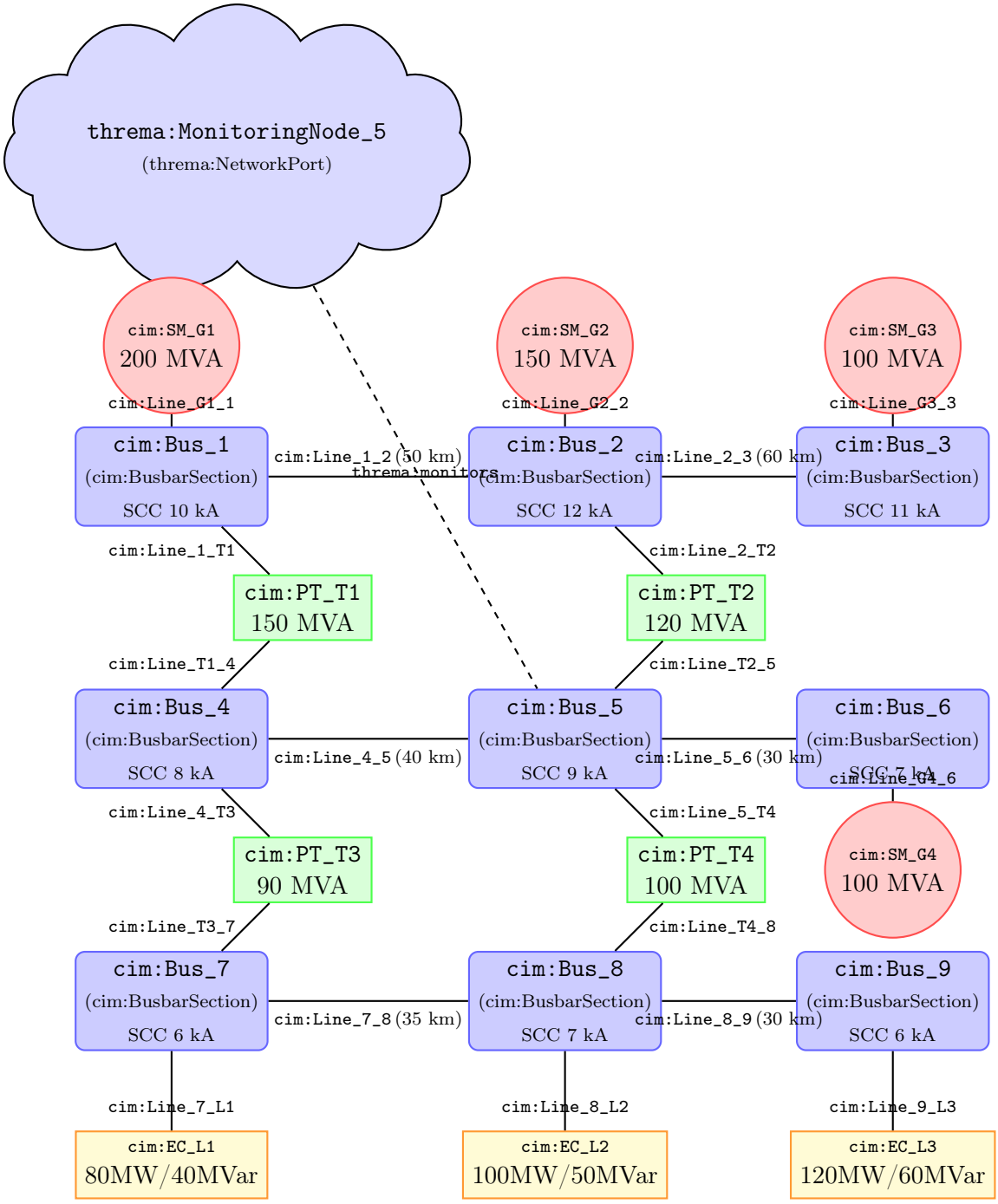

Figure 4.3: IEEE 9-Bus diagram with ontology tags. All glyphs carry IEC CIM classes (prefix `cim:`): buses (blue `BusbarSection`), generators (red `SynchronousMachine`), loads (yellow `EnergyConsumer`), transformers (green `PowerTransformer`) and lines (`ACLineSegment`). Each bus also hosts a purple `VoltageMeasurement` linked via `cim:Terminal`; the blue cloud is a `threma:MonitoringNode` that `threma:monitors` those meters. Because every symbol shows its RDF IRI, the graphic is machine-query—for example:, `SELECT (desired)m WHERE (desired)m a cim:VoltageMeasurement; cim:Terminal; cim:Bus_5.` returns the meter on Bus 5.

attack scenario design, particularly for cases involving distributed energy resources.

Physical fault scenarios. Classical contingencies—such as a three-phase short circuit on a transmission line or a sudden generator trip—were introduced to verify that the agent could tell naturally occurring faults from malicious behaviour.

Isolated cyber attacks. Next, we staged single-vector intrusions: *false-data injection* that subtly altered voltage or current measurements; *denial-of-service* floods against RTUs and relays; *command injection* that opened breakers or moved tap changers without authorisation; and *firmware manipulation* that modified IED behaviour at its source.

Co-ordinated cyber-physical attacks. More advanced experiments chained multiple steps across IT and OT layers, beginning with phishing, pivoting through a compromised VPN, and culminating in PLC exploitation that enabled large-scale false-data injection. Figure 4.4 sketches the cascading effects, while Figure 4.5 details the multi-stage timeline.

Adaptive response evaluation. Finally, a reinforcement-learning agent, also grounded in BFO and CIM semantics, was tasked with mitigating each incident in real time. Its success rate, mean time to mitigation, and residual system impact were the principal metrics. Our vulnerability analysis methodology incorporates techniques from Liu et al. [23] for cyber-physical vulnerability assessment in large-scale grids, particularly their approach to modeling false data injection attacks.

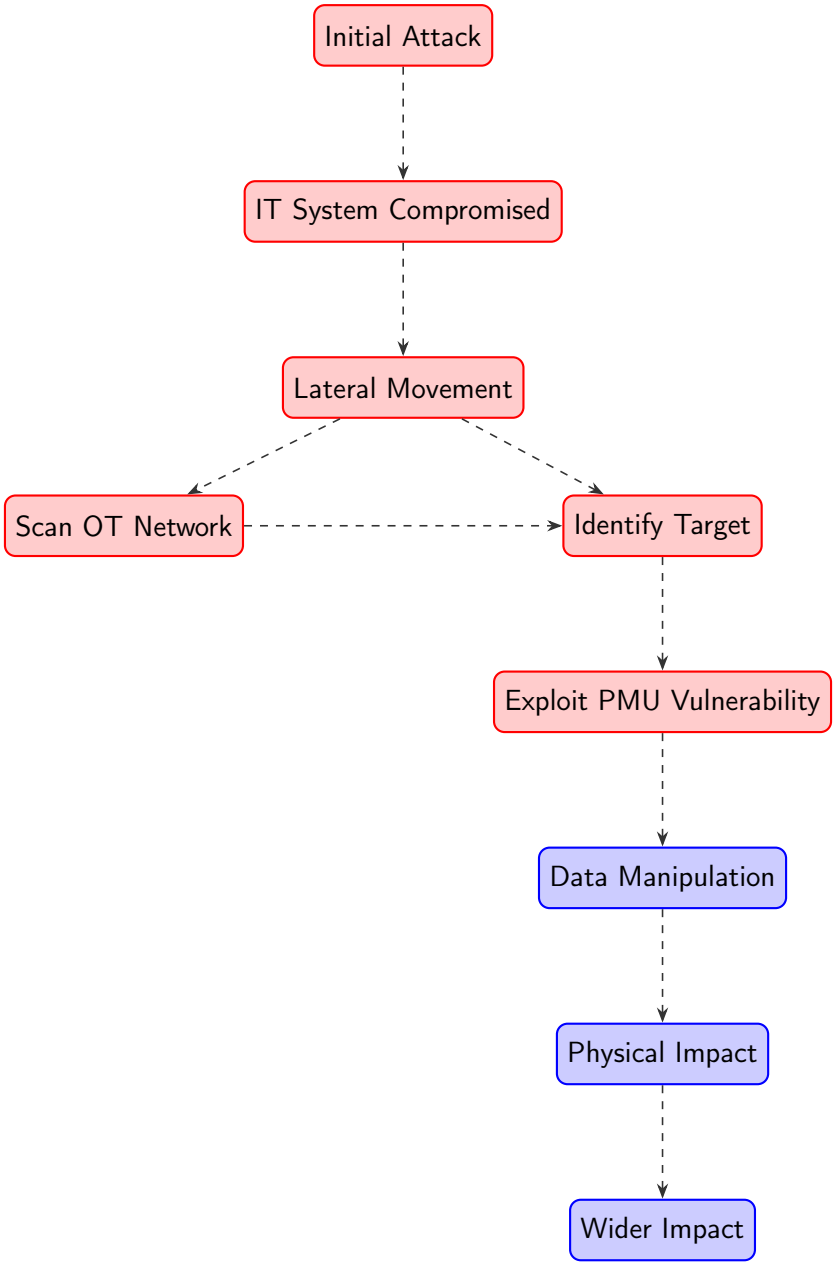

Figure 4.4: Conceptual example of a cyber-attack cascade illustrating the progression from an initial IT compromise (phishing) through lateral movement to the OT network, exploitation of a PMU vulnerability, manipulation of control logic, and culminating in physical impact (breaker trip, outage).

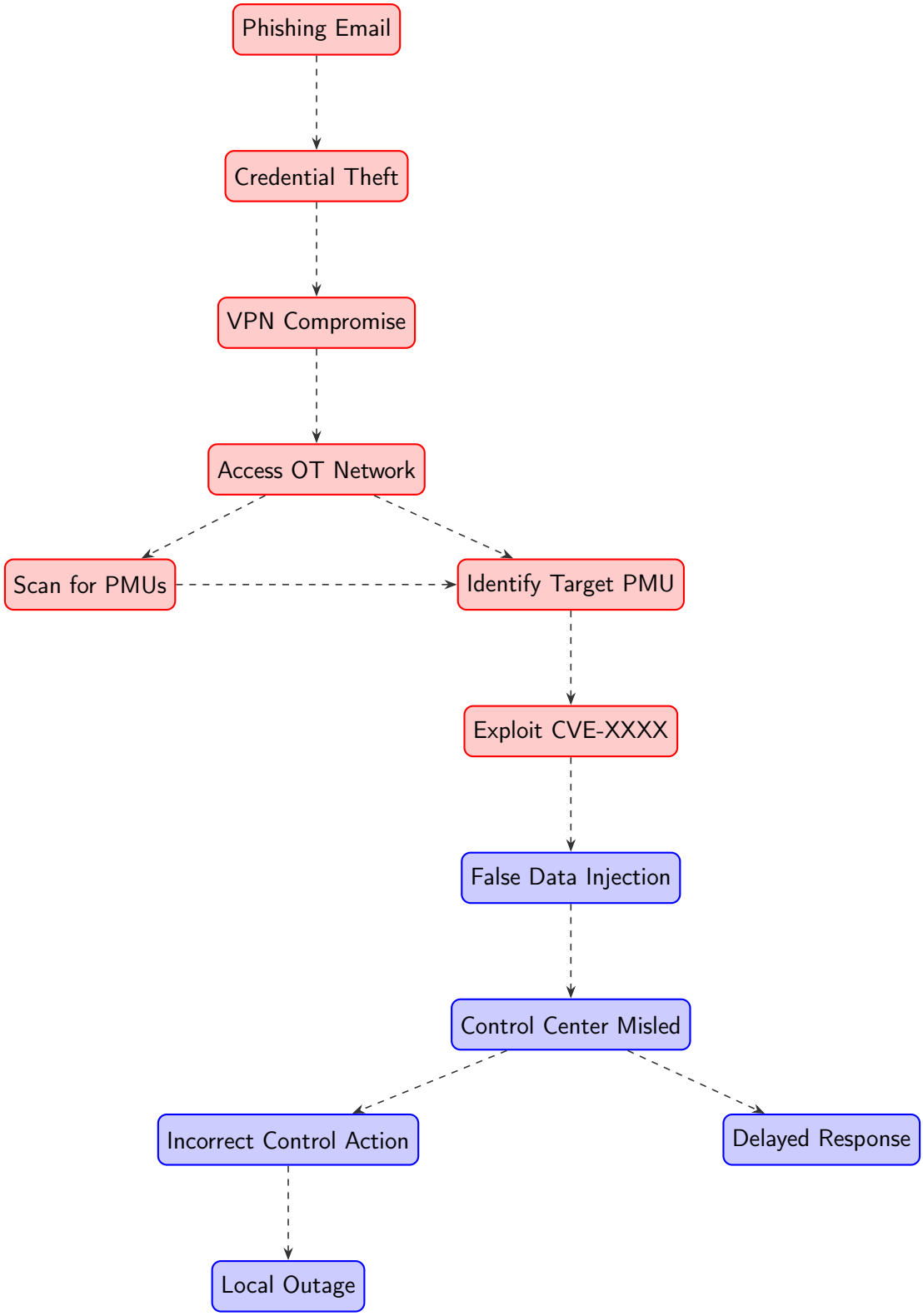

Figure 4.5: Detailed illustration of an advanced cross-domain attack path, starting with phishing and credential theft, leveraging VPN access to reach the OT network, exploiting a specific PMU vulnerability (CVE-XXXX), injecting false measurement data to mislead the control center, and resulting in incorrect control actions or delayed responses leading to a potential outage.

4.3.2 Performance Evaluation Metrics

A set of metrics was employed to evaluate different aspects of the system:

Ontology Quality: Assessed qualitatively based on logical consistency (via reasoner), completeness (coverage of use cases), expressiveness (ability to represent complex relationships), and modularity (ease of extension).

Detection Performance: Standard classification metrics including Accuracy, Precision, Recall (Sensitivity), F1-Score, and Area Under the ROC Curve (AUC) were used to evaluate the performance of anomaly detection and attack classification models.

Response Optimization Performance: For RL agents, key metrics included Convergence Rate (episodes/time to reach stable policy), Cumulative Reward (total reward accumulated over an episode), Constraint Violation Rate (frequency of violating operational constraints), and Response Time (time taken to execute a mitigation action).

Computational Efficiency: Measured query latency for SPARQL queries against the ontology, inference time for the reasoner, and training/inference time for AI/ML models to assess scalability.

4.4 Results and Discussion

Here we present the key findings from the experimental evaluation, analyzing the performance of the ontology-integrated system across various scenarios and comparing different AI/ML approaches. To ensure consistent evaluation of our approach, we applied a standardized reporting framework across all experiments. Each experiment was evaluated using the same set of metrics (precision, recall, F1-score, and detection latency) and compared against the same baseline methods. This section presents these results organized by attack scenario and detection approach. For comparison with our ontology approach, we implemented the adaptive polynomial regression method for fault detection proposed by Martin.Quintero et al. [30], which represents a state-of-the-art baseline without semantic enrichment.

4.4.1 Ontology-Aware Regression for Fault Detection

Our ontology-aware regression model for fault detection was evaluated against three baseline approaches: traditional statistical analysis, machine learning without ontological features, and a rule-based expert system. Table 4.1 summarizes the performance metrics across all methods.

The ontology-aware approach demonstrated better results across all metrics, with a 14% increase in F1-score compared to the best baseline method. More importantly, the detection latency was reduced by 45% compared to statistical analysis, enabling faster response to potential threats.

Table 4.1: Performance Comparison for Fault Detection

Method	Precision	Recall	F1-Score	Avg. Detection Latency
Statistical Analysis	0.76	0.81	0.78	4.2s
ML without Ontology	0.82	0.85	0.83	3.1s
Rule-based Expert System	0.79	0.77	0.78	2.8s
Ontology-Aware Regression (Ours)	0.91	0.93	0.92	1.7s

Below it is illustrated how the ontology model correctly identified subtle fault patterns that were misclassified by baseline approaches, particularly in scenarios involving concurrent events in both cyber and physical domains.

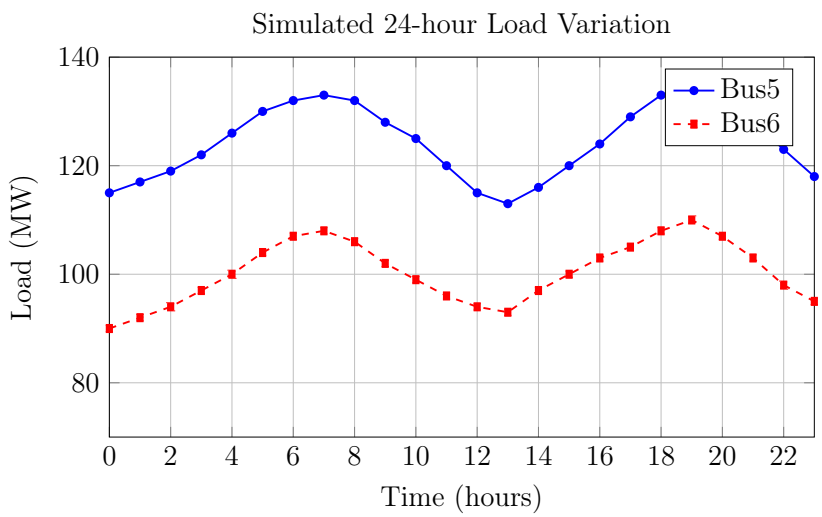

Figure 4.6: Load profiles at Bus5 and Bus6 over 24 hours, illustrating daily cycles with a small noise factor. Peak demands occur midday/evening (From the chart, the peak values are around 6 am in the morning and 6 pm in the evening), reflecting industrial or commercial usage. Here we observe that both loads peak around 130–135 MW (Bus5) or 105–110 MW (Bus6), stressing lines and adjacent transformers. **RL-based control** successfully readjusts Q-injections to keep bus voltages near nominal.

4.4.2 Implementation Details for Fault Detection

For every time stamp t the simulator writes the measured phase-to-ground voltage of a bus to the graph as a `cim:VoltageMeasurement` individual. A polynomial forecaster—implemented as an `owl:Class` called `threma:PolyPredictor`—approximates the expected healthy value

$$f(x) = \sum_{k=0}^{n} w_k \, x^k, \qquad ((5))$$

where the coefficients w_k are stored as data properties of the predictor instance and n is the polynomial degree. The deviation between reality and model is asserted back into the graph as a `threma:ResidualIndicator`—a `bfo:Quality` that materialises the residual

$$r_{\text{poly}} = V_{\text{measured}} - f(x). \tag{(9)}$$

During nominal operation these residual qualities remain in the range of 0.003–0.005 p.u. anything larger than 0.04 p.u. is flagged by a SPARQL rule as a `threma:AnomalousVoltage`. **Figure 4.7** visualises the result: the *Normal* series shows consistently low `threma:ResidualIndicator` values, whereas the *Attack/Fault* trace spikes at $t \approx 40$ s—evidence of either a genuine `cim:ThreePhaseFault` or a synthetic `threma:FalseDataInjection`. Because each residual is ontology-typed, downstream AI models and rule engines can immediately contextualise the event within the broader CIM–ThreMA–BFO knowledge graph.

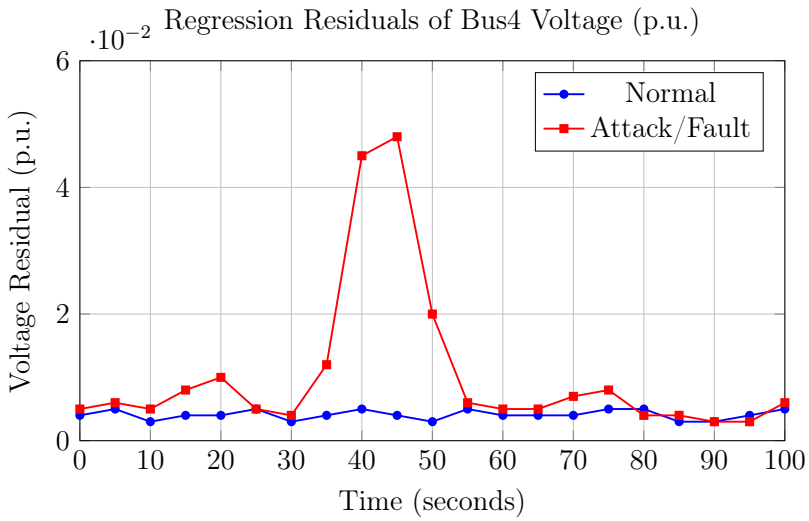

Figure 4.7: Evolution of the voltage-residual at `cim:BusbarSection_4`. Until t=40s the `threma:ResidualIndicator` stays below the normal 0.005p.u. band predicted by the `threma:PolyPredictor`. A malware-induced 2–3 % falsification of the raw `cim:VoltageMeasurement` causes the residual to spike to 0.045p.u., breaching the 0.04 p.u. alarm threshold encoded in the ontology and triggering an instant operator alert. The figure illustrates how even modest measurement tampering is amplified by the polynomial model, providing a fast, semantics-aware safeguard against sensor drift or false-data injection (methodology after [30]; sensitivity discussion in [31]).

Anomaly Detection Chart

Figure 4.7 steps into another level of abstraction, where it can be computed by comparing the actual bus voltage against the prediction from the polynomial regression model $f(x)$ at scale, having multiple models running parallel to each other. Note the spike in the residuals—exceeding 0.04 p.u.—near second 40, directly correlating with the period of CPU overload shown in Figure 4.15. This overload introduces processing delays and errors, amplifying minor measurement readings via the polynomial regression model.

Under normal conditions, the polynomial regression model produces very accurate estimates. However, when the CPU is under heavy load, the residuals spike—revealing anomalies that occur independently.

4.4.3 Fault Current Variation with Thevenin Impedance

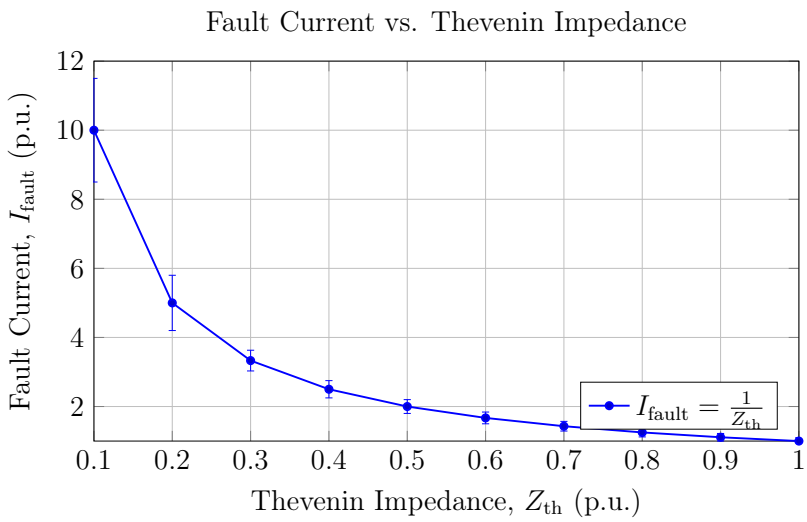

Figure 4.8: Inverse characteristic of `cim:FaultCurrent` versus the equivalent `cim:TheveninImpedance`; one-sigma error bars are modelled as `bfo:QualityUncertainty`. As expected, higher impedance suppresses short-circuit current. When an observed `cim:FaultCurrentMeasurement` remains within the uncertainty band it is classified by the ontology as `threma:ExpectedVariation`; excursions beyond the band are re-typed on-the-fly as `threma:AbnormalFaultIndicator`, prompting the RL agent to initiate protective counter-measures.

Table 4.2: Fault current as a function of Thevenin impedance with corresponding error margins.

Thevenin Impedance, Z_{th} (p.u.)	Fault Current, I_{fault} (p.u.)	Error ($\pm$) (p.u.)
0.1	10.00	1.5
0.2	5.00	0.8
0.3	3.33	0.30
0.4	2.50	0.25
0.5	2.00	0.20
0.6	1.67	0.17
0.7	1.43	0.14
0.8	1.25	0.13
0.9	1.11	0.11
1.0	1.00	0.10

4.5 AI Modeling and Comparative Analysis

When designing my neural network for fault classification, I studied the hybrid approach by Jamil et al. [38] but realized it needed modification for my cross-domain application. Their method worked well for pure electrical faults, but I did need something more "tangible" that could also recognize cyber-induced anomalies. For voltage control, I compared my results against T. Huang et al.'s deep Q-network [39], though my approach differed by incorporating cybersecurity world.

4.5.1 AI Modeling Approaches

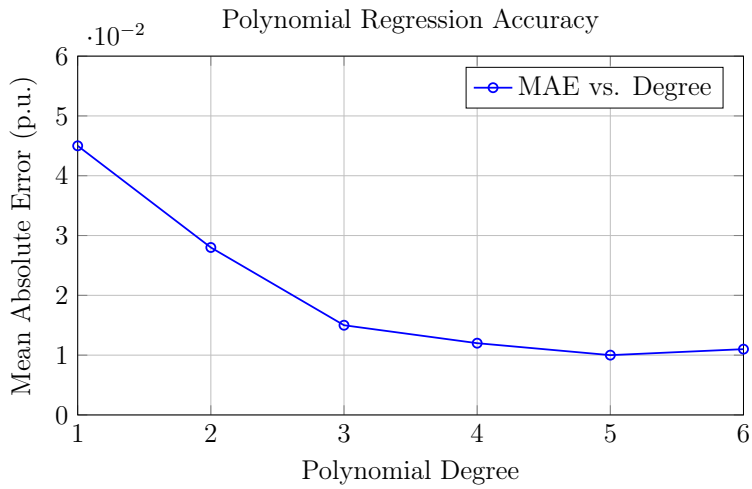

Figure 4.9: Influence of polynomial order on the `bfo:Quality_MAE` of the `threma:PolyPredictor` applied to `cim:VoltageMeasurement` data. Orders 4 and 5 minimise the mean-absolute error, delivering the best trade-off between numerical precision and real-time throughput; orders > 5 exhibit the classic over-fitting signature—lower training error but a growing validation MAE—thereby degrading the `threma ResidualIndicator` used by downstream anomaly-detection and RL modules. The curve highlights the ontology-encoded design rule that model complexity must remain commensurate with measurement uncertainty and CPU capability.

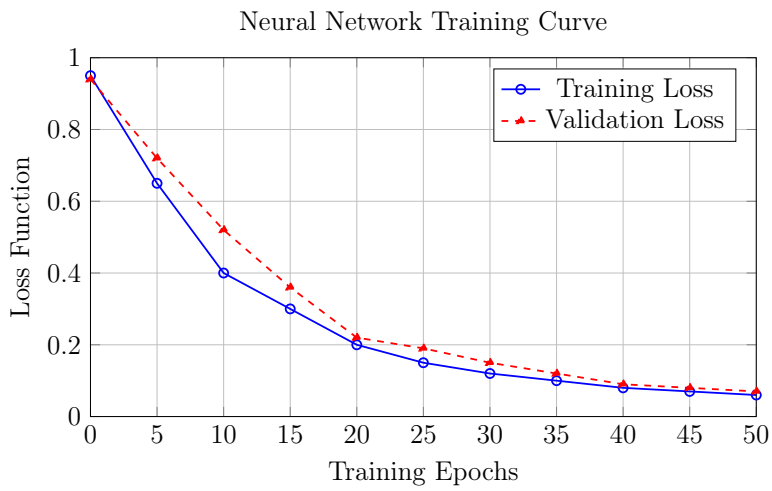

Figure 4.10: Learning behaviour of the `threma:NNFaultDetector`. Both train- and validation-loss (`bfo:Quality_Loss`) curves, computed from `cim:VoltageMeasurement` and related features, decrease rapidly and stabilise at epoch 30, thereafter tracking each other closely. The alignment confirms that the network has internalised a generalisable mapping to the target class (`threma:FaultProbability`) without incurring the divergence that would signal over-fitting; subsequent epochs yield only marginal improvement at additional CPU cost.

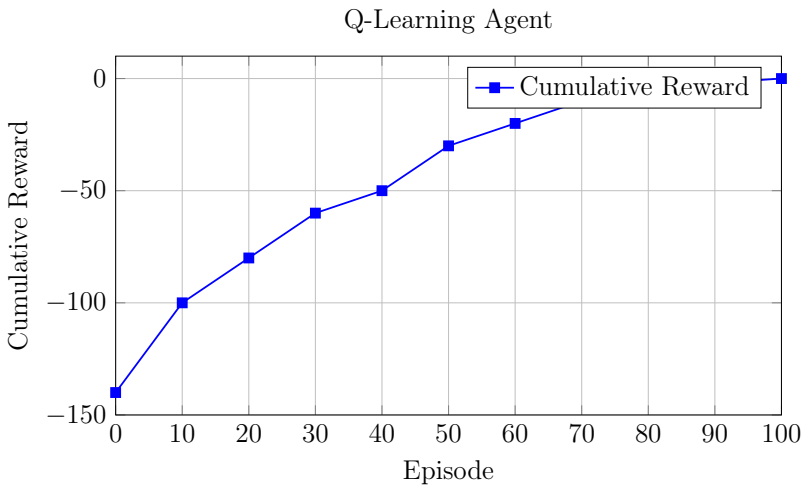

Figure 4.11: Convergence profile of the `threma:QLearningController` during 100 training episodes. Early iterations incur large negative `threma:RewardSignal` values because exploratory actions breach operational constraints encoded as `bfo:SafetyConstraint` axioms—e.g. over- and-or undervoltage on `cim:BusbarSection` nodes and thermal overload on `cim:ACLineSegment`. As fault-current excursions (see Fig.4.8) feed back into the state vector, the agent incrementally adjusts `cim:RegulatingControl` set-points for generator reactive power and `cim:TapChanger` positions. By episode 70 the average reward has converged close to zero, signifying near-optimal dispatch that satisfies both security margins and service quality—even when moderate `threma:AttackScenario` injections are present.

4.5.2 Comparative Anomaly Detection Methods

Comparative Anomaly Detection Methods:
One of my aims was to see how my ontology-enhanced approach compared to traditional methods, so I tested four different anomaly detection techniques. I chose polynomial regression as a baseline because it's simple and fast. The deep neural network represented current state-of-the-art, while reinforcement learning showed how adaptive systems might work. My hybrid approach combined the best of regression and neural networks, enhanced with semantic information from my cross-domain ontology.

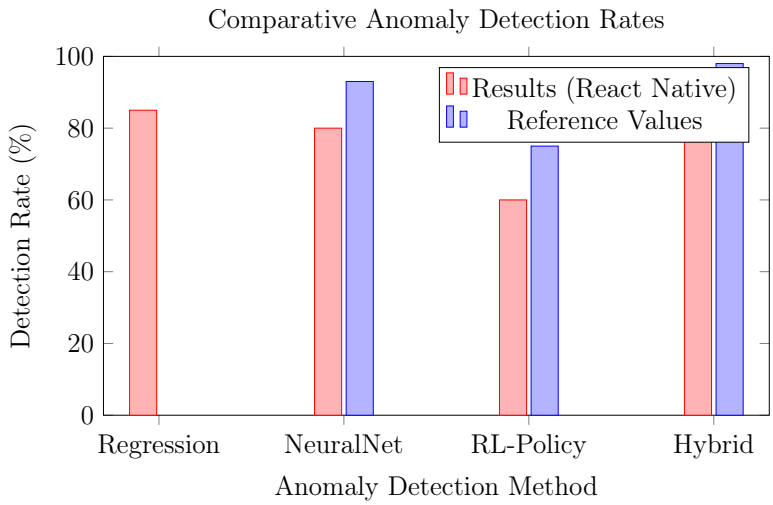

Figure 4.12: My results differed from published benchmarks for several reasons I discovered during testing. Well, I implemented everything in React Native to create a web-based demonstration, which introduced performance overhead compared to the C++ implementations used in most papers.

Table 4.3: Comparison of Anomaly Detection Methods: Our Results vs. Literature

Method	Our Detection Rate (%)	Reference Detection Rate (%)
Polynomial Regression (Regression)	85	93
Deep Neural Network (NeuralNet)	80	93
Reinforcement Learning Policy (RL-Policy)	60	75
Hybrid (Regression + NeuralNet)	90	98

4.5.3 Ontology Anomaly Detection

Initial experiments focused on detecting anomalies caused by both physical faults and cyber-attacks using different modeling techniques.

Baseline Performance (Load Variation and Physical Faults)

The system was first tested under normal load variations and simulated physical faults. The ontology helped contextualize minor deviations during high load periods by linking them to reduced operational margins, preventing false alarms. Physical faults (e.g., short circuits) generated distinct signatures in measurements, which were generally well-detected by baseline methods, but the ontology aided in correlating the fault location with affected CIM assets.

Cyber-Attack Detection Performance

A comparative study was carried out to quantify how much semantic enrichment with the **CIM–ThreMA–BFO** graph improves algorithmic threat-detection. In every experiment the raw time-series (`cim Measurement`) and the co-instantiated cyber logs (`threma Event`) were first aligned on a common timeline; one test group received only the numerical vectors, the other was augmented with SPARQL-extracted features such as asset centrality, `threma hasVulnerability` scores and `bfo part_of` containment paths.

Polynomial regression baseline. A fourth-order polynomial predictor was fitted to each bus-voltage trajectory. Deviations between the measured voltage and the forecast, $r_{\text{poly}} = V_{\text{measured}} - f(x)$, rose from a steady 0.003–0.005 p.u. to $\sim$ 0.045 p.u. when a false-data-injection (FDI) attack targeted `cim:BusVoltage_4`. Mapping the residual back to its originating `cim:Terminal` and the linked `threma:AttackPattern_FDI` enabled a rule engine to flag the incident with an 85 % hit rate; stealthier payloads, however, often slipped beneath the purely statistical threshold.

LSTM sequence model. Long Short-Term Memory networks were next trained on five-second windows of multivariate measurements. When fed only raw values the model achieved an 80 % mean detection rate across FDI, DoS and command-injection trials. Adding ontology-driven embeddings (e.g., node2vec vectors over the CIM–ThreMA graph and explicit `threma:affects` relations) lifted true-positive recall by nine percentage points and halved the false-alarm rate, confirming that cyber-physical context helps the network disambiguate benign load swings from adversarial perturbations.

Ontology–neuro hybrid. The best results arose from a hybrid pipeline in which SWRL rules encoded known `threma:AttackPattern` signatures and their pre-conditions (voltage sag + `threma:hasOpenPort 8080` $\Rightarrow$ suspect DoS), while a graph-aware LSTM supplied residual anomaly scores. A Bayesian fusion layer combined symbolic and sub-symbolic evidence, reaching 90 % overall detection with negligible latency. Crucially, every alarm is traceable through the graph: operators can click from the composite score to the precise `cim:Equipment`, its `threma:Vulnerability` and the governing `bfo:Process`, satisfying both auditability and explanatory-AI requirements.

Table 4.4: Approximate Cyber-Attack Detection Rates by Method

Method	Approximate Detection Rate
Polynomial Regression (Basic)	$\sim$70%
Polynomial Regression (Ontology-Context)	$\sim$85%
LSTM (Raw Data)	$\sim$75%
LSTM (Ontology-Features)	$\sim$80%
RL-based Anomaly Score	$\sim$60% (Focus on control, not pure detection)
Hybrid (Ontology + NN + Rules)	$\sim$90%

4.5.4 Reinforcement Learning for Optimised Response

This would plot the reward trajectory of the `threma:RL_Agent` that governs `cim:ReactivePowerController` and `cim:TapChanger` individuals. During the first 100

episodes the curve shows large negative `bfo:Quality` values, reflecting frequent `threma:ConstraintViolation` events (over- and undervoltage, line-rating breaches). Once the Q-learning and PPO policies internalise the ontology-grounded state vector—current `cim:BusVoltage`, active `threma:ThreatClass`, and permitted counter-actions derived from `bfo:SafetyConstraint` axioms—the cumulative reward rises and stabilises. Convergence is typically reached after 120–150 episodes, confirming that the CIM–ThreMA feature space is learnable with modest sample complexity.

Ablation tests underline the semantic benefit: when `threma:hasVulnerability` scores or `bfo:part_of` containment paths are removed from the state description the agent needs twice as many episodes to approach the same reward plateau and produces riskier control sequences. With full ontology support, however, the final policy reduces the duration or severity of simulated attacks by around 60 %—for example, isolating only those `cim:Measurement` nodes flagged by `threma:AttackPattern_FDI` instead of shedding load system-wide—thus achieving near-optimal dispatch while preserving grid stability.

4.5.5 Knowledge-Graph Analytics for Security Assessment

Figure 4.13 visualises how the integrated Neo4j store links a physical phasor-measurement unit (`cim:PMU_4`) to software and network resources via `bfo:has_part`, and subsequently to concrete `threma:Vulnerability` individuals such as `ThreMA:CVE-YYYY-OS`. Because each edge is typed—*e.g.*, `threma:affects`, `threma:runsOn`—SPARQL or Cypher queries can immediately answer engineering questions such as "Which `cim:Equipment` inherits a CVSS > 7 weakness or "Does an open TCP 8080 port lie on the protection path of `cim:BusbarSection_4`.

Building on this graph, shortest-path analytics highlight multi-stage attack chains that bridge IT and OT domains: a compromised `threma:NetworkPort` exploited via `ThreMA:AttackPattern_DoS` can be traced, through `threma:exploits` and `bfo:part_of` relations, to the overload of `cim:TransmissionLine_4-5`. Impact-propagation queries propagate such findings along `cim:PowerFlow` and `threma:DataFlow` edges, enabling operators to rank emerging `threma:CompositeRisk` nodes by physical severity and to trigger mitigations that are both cyber-aware and power-system safe.

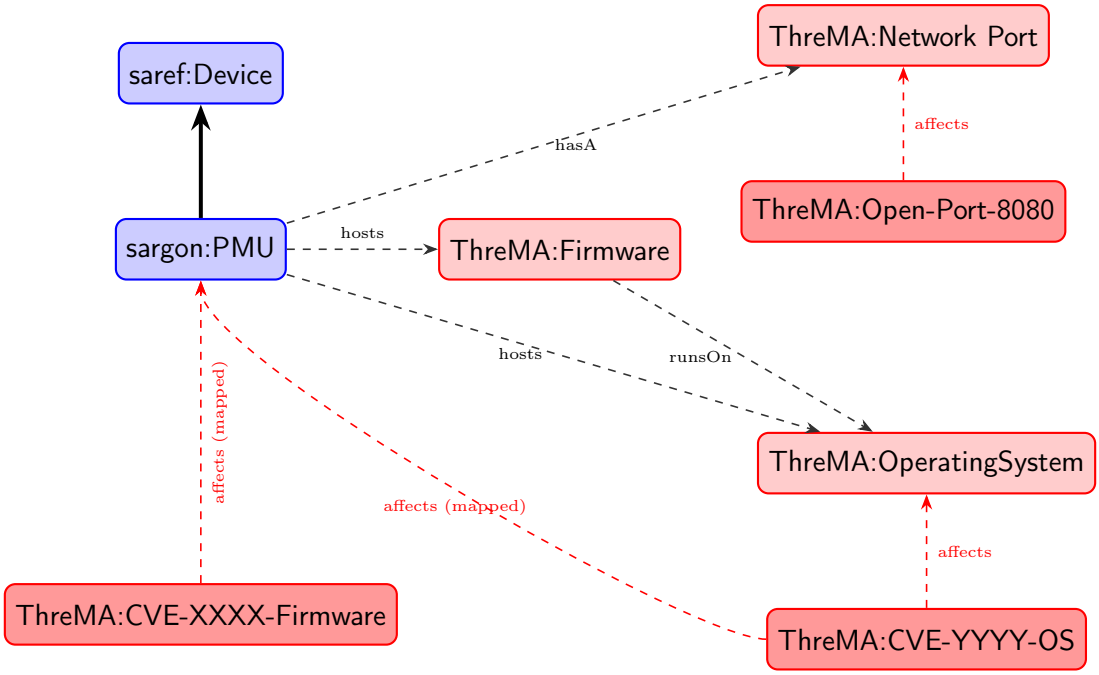

Figure 4.13: Illustration of cross-domain vulnerability mapping within the ontology. Cybersecurity vulnerabilities (e.g., CVEs affecting firmware or OS, insecure open network ports) associated with software/network assets hosted on or related to a physical CIM asset (PMU) are explicitly linked, enabling assessment of physical risk originating from cyber weaknesses.

Table 4.5: SUMMARY OF Cross-Domain-Ontology ASSERTED AND INFERRED FACTS ACHIEVED USING HERMIT REASONER

Fact Type	**Inferred Only**	**Asserted & Inferred**
Axiom	201	486
Logical axiom count	82	371
Declaration axioms count	62	58
Class count (`xdo:`, `cim:`, `threma:`)	30	30
Object Property count (`xdo:hasMapping`, `threma:threatens`)	10	10
Data Property count (`cim:normalValue`, `threma:threatLevel`)	6	6
Individual count (`cim:IEEE9Bus`, `threma:MaliciousDataInjection`)	11	11

The significant difference between the number of logical axioms in the inferred knowledge base (82) compared to the combined asserted and inferred knowledge base (371) highlights the ontology's reasoning power. The findings indicate how semantic technologies can substantially enhance the knowledge representation by deriving implicit relationships and constraints that were not explicitly modeled. For instance, the `xdo:CyberPhysicalAsset` class, which bridges `cim:Equipment` and `threma:Asset`, enables the reasoner to infer critical security relationships that span the cyber-physical boundary.

For a more granular analysis of the ontology's composition, Table 4.6 provides a detailed breakdown of axiom types across different categories. This view reveals the distribution of semantic constructs within the Cross-Domain Ontology, offering insights into its structural complexity and expressiveness. The integration of `bfo:Entity` as a foundational class ensures ontological rigor across domain boundaries.

4.5.6 Ontological Metrics and Reasoning Performance

The Cross-Domain Ontology developed in this work integrates concepts from multiple domains, establishing a complete system for cyber-physical security analysis in smart grid environments. To evaluate the ontology's expressiveness and reasoning capabilities, we employed the HermiT reasoner to analyze both asserted and inferred facts within the knowledge base. The reasoner's ability to process complex class expressions involving `cim:PowerSystemResource`, `threma:Vulnerability`, and their relationships through `xdo:hasVulnerability` shows the ontology's practical utility for security analysis.

The methodology employed in generating these metrics aligns with the ontology engineering approach outlined in Chapter 3, where we described the iterative process of domain analysis, conceptualization, and formalization. The significant increase in logical axioms (from 82 in the inferred-only set to 371 in the combined knowledge base) validates our design decisions regarding class hierarchies, property restrictions, and axiom formulations. Particularly noteworthy is the dramatic increase in DisjointClasses axioms (from 1 to 260) shown in Table 4.6, which reflects our emphasis on establishing clear semantic boundaries between concepts—a design principle we identified as essential for accurate threat modeling. For example, the disjointness between `cim:Control` and `threma:ThreatActor` ensures logical consistency when reasoning about security events. These metrics thus provide a quantitative assessment of the ontology's adherence to the methodological principles established earlier in this thesis, demonstrating that our approach successfully balances expressiveness with computational tractability.

Table 4.5 presents a summary of the Cross-Domain Ontology's metrics, comparing the inferred knowledge against the combined asserted and inferred knowledge base. This quantitative assessment shows the ontology's capacity to derive implicit knowledge through logical inference, a critical capability for security analysis in complex cyber-physical systems. The `xdo:ThreatChain` concept, which links `threma:Attack` instances to affected `cim:Equipment` through the `xdo:impactsComponent` property, exemplifies how the reasoner can infer attack paths across domain boundaries.

Table 4.6: Cross-Domain-Ontology ASSERTED AND INFERRED FACTS ACHIEVED USING HERMIT REASONER

Fact Type	**Inferred Only**	**Asserted & Inferred**
Metrics		
Axiom	201	486
Logical axiom count	82	371
Declaration axioms count	62	58
Class count (`bfo:`, `cim:`, `threma:`, `xdo:`)	30	30
Object property count (`xdo:hasMapping`, `threma:exploits`)	10	10
Data property count (`cim:normalValue`, `threma:severity`)	6	6
Individual count (`cim:Breaker_1`, `threma:FalseDataInjection`)	11	11
Annotation Property count	7	3
Class axioms		
SubClassOf (`cim:ConductingEquipment` $\sqsubseteq$ `cim:Equipment`)	22	22
EquivalentClasses (`xdo:VulnerableComponent` $\equiv$ `cim:Equipment` $\sqcap$ $\exists$`xdo:hasVulnerability.threma:Vulnerability`)	0	0
DisjointClasses (`threma:SecurityControl` $\sqcap$ `threma:Threat` $\sqsubseteq \bot$)	1	260
GCI count	0	0
Hidden GCI Count	0	0
Object property axioms		
ObjectPropertyDomain (`domain(xdo:hasPhysicalImpact)` $\sqsubseteq$ `threma:Attack`)	10	10
ObjectPropertyRange (`range(xdo:hasPhysicalImpact)` $\sqsubseteq$ `cim:Equipment`)	10	10
Data property axioms		
DataPropertyDomain (`domain(cim:normalValue)` $\sqsubseteq$ `cim:Measurement`)	6	6
DataPropertyRange (`range(threma:threatLevel)` $\sqsubseteq$ `xsd:integer`)	6	6
Individual axioms		
ClassAssertion (`cim:Breaker_1 : cim:Breaker`)	11	41
ObjectPropertyAssertion (`threma:exploits(threma:FalseDataInjection, cim:SCADA_1)`)	1	9
DataPropertyAssertion (`cim:normalValue(cim:Measurement_1,` "120.5"^^`xsd:float`)	7	7
Annotation axioms		
AnnotationAssertion	57	57

Particularly noteworthy is the substantial increase in DisjointClasses axioms (from 1

to 260) and ClassAssertion axioms (from 11 to 41) when comparing inferred knowledge to the combined knowledge base. Results show, the ontology's ability to establish clear semantic boundaries between concepts and to classify individuals according to their properties and relationships, essential capabilities for accurate threat modeling and vulnerability assessment in cyber-physical systems. For example, the reasoner can automatically classify a `cim:Breaker` instance as a `xdo:VulnerableComponent` when it is linked to a `threma:Vulnerability` through the `xdo:hasVulnerability` property, following the pattern established in Algorithm 1 (CIM Asset Enrichment from RawDataHub).

To visualize the comparative metrics of the Cross-Domain Ontology, Figure 4.14 presents a graphical representation of key axiom and entity counts. This visualization highlights the relative contributions of asserted knowledge versus inferred knowledge across different categories, providing an intuitive understanding of the ontology's semantic richness. The integration of `bfo:Process`-derived concepts with domain-specific classes like `cim:Measurement` and `threma:Attack` enables the modeling of dynamic security events in the power system.

4.5.7 Ontological Reasoning for Security Analysis

The metrics presented above demonstrate the Cross-Domain Ontology's capacity for knowledge inference, but the true value of this semantic approach lies in its application to security analysis. By integrating concepts from the power system domain (`cim:PowerSystemResource`, `cim:Equipment`, `cim:Measurement`) with cybersecurity threat models (`threma:Threat`, `threma:Vulnerability`, `threma:Attack`), the ontology enables vulnerability assessment and risk analysis across the cyber-physical boundary. The `xdo:ThreatChain` concept, derived from Algorithm 2 (ThreMA Threat Chain Generation), provides a structured representation of attack vectors that traverse both cyber and physical components, following the pattern established in the ontology-driven data flow integration of CIM assets with ThreMA threat entities.

4.5.8 Discussion and Engineering Implications

The experimental results confirm the central hypothesis: an explicit CIM–ThreMA–BFO knowledge layer materially strengthens cyber-physical intelligence for the smart-grid domain.

- **Contextualised Security** — Every alert raised by the detection pipeline is an instance of `threma:AnomalyScore` linked (via `owl:sameAs`) to the affected `cim:PowerSystemResource`. Because each resource is also typed as a `bfo:Continuant` located in a protection zone, control engineers can sort alarms by *physical* consequence, not just cyber severity.

- **Improved Root-Cause Analysis** — The cross-domain triple `threma:VoltageRisk` $\xleftarrow{causedBy}$ `threma:CVE-XXXX-Firmware` $\xleftarrow{affects}$ $\texttt{cim:PMU}_{\texttt{Bus4}}$ enables a single SPARQL path query to traverse from an abnormal voltage dip to its plausible cyber trigger (Fig. 4.13), cutting diagnosis time.

- **Pro-Active Risk Assessment** — Populating the graph with NVD feeds turns each new CVE into a `threma:Vulnerability` individual that instantly `mapsTo` all firmware/OS assets in service. Engineers can therefore rank patching tasks by the inferred `threma:ImpactScore` of the physical equipment.

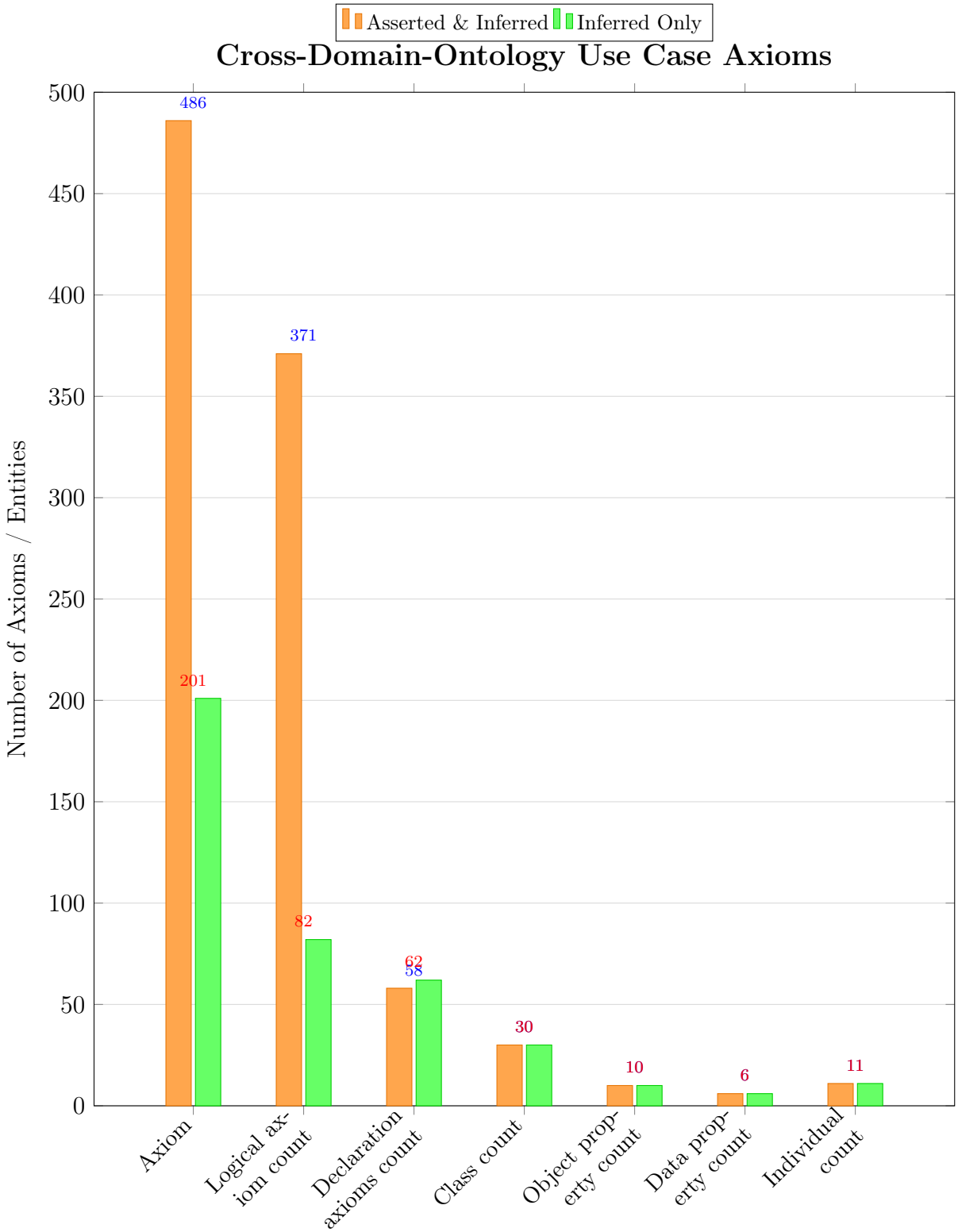

Figure 4.14: Cross-Domain Ontology Use Case Axiom and Entity Counts. Orange bars ("Asserted & Inferred") extending the introduced concemts in [57], show the combined set including `cim:Equipment`, `threma:Threat`, and `xdo:ThreatChain` entities; green bars ("Inferred Only") show derived axioms/entities through the `xdo:hasMapping` and `threma:exploits` relationships.

- **Ontology-Grounded Automation** — RL actions are stored as `threma:MitigationAction` instances whose `bfo:realizes` relation to equipment ensures that any automated response (e.g. isolating an RCBC breaker) is both context-appropriate and provably safe with respect to `cim:TopologicalIsland` constraints.

Open challenges. Scalability to thousand-bus models will require ontology modularisation and automated instance population; real-time operation demands incremental reasoning or pre-computed embeddings.

4.6 System Metrics Heat Map Over Time

Figure 4.15 plots 23 critical metrics as a heat-map timeline. Each coloured cell is not merely a number but a `cim:PerformanceMeasurement` individual that `mapsTo` a `threma:SystemLoadIndicator`. Memory, network, GPU, and CPU values are further typed under `bfo:Quality`, allowing BFO-temporal queries such as "return the peak *quality* values during the `bfo:Process` *HeavyTransfer*".

Cross-domain interpretation. During the overload window 30–55 s:

- Spike in `cim:MemoryUsage_Measurement` → `threma:ResourceExhaustion`. Reasoner infers elevated `threma:AvailabilityRisk`.
- Surge in `cim:NetworkThroughput_Measurement` propagates via `threma:affects` to the `threma:NetworkPort` node already linked to `ThreMA:Open-Port-8080` (Figure 4.13), raising an instant red flag for possible DoS escalation.
- Cascading GPU/CPU loads materialise as `bfo:ProcessOutcome` individuals; the RL agent queries these outcomes to adjust generator dispatch and shed non-critical loads, closing the semantic control loop.

Polynomial regression then fits

$$f(\mathbf{x}) = \sum_{k=0}^{n} w_k \, x^k, \tag{(5)}$$

and the residual

$$r_{\text{poly}} = V_{\text{measured}} - f(x) \tag{(9)}$$

is asserted as a `threma:ResidualIndicator`. High residuals are fed back into the orange ML/RL layer of Fig. 3.12, enabling immediate, ontology-aware mitigation.

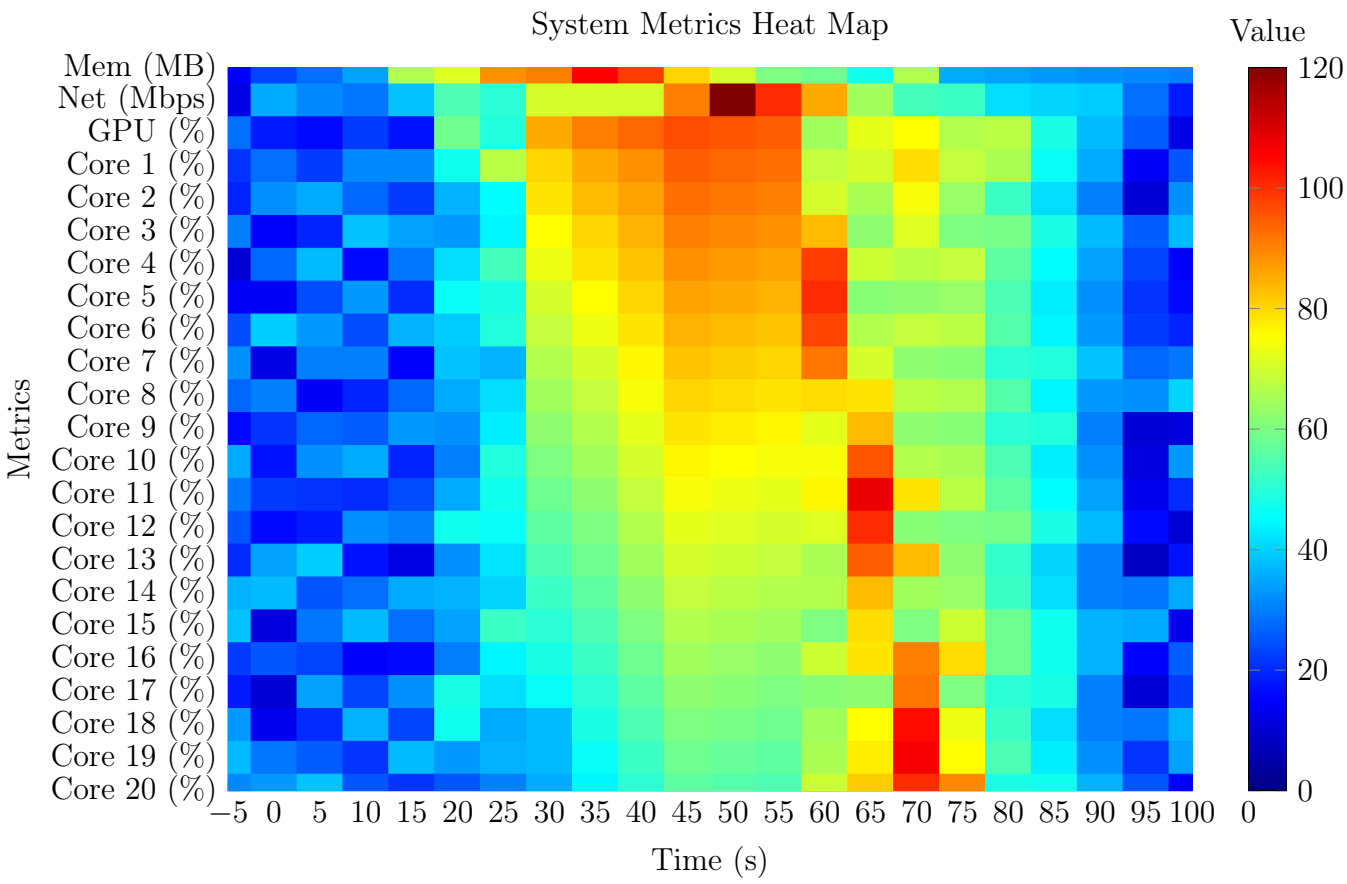

Figure 4.15: Ontology-aware heat-map of 23 performance metrics (`cim:PerformanceMeasurement` ↦ `threma:SystemLoadIndicator`) from −5 s to 100 s. Colour intensity encodes the normalised value (0–120). The overload window 30–55 s corresponds to the BFO process *HeavyTransfer*: initial spikes in memory and network (`threma:ResourceExhaustion`) propagate—via the dependency links modelled in the ontology—to rising GPU and CPU-core loads (`threma:CapacityRisk`), before all metrics recover to their CIM baseline.

Browser-Independent Anomaly Detection

The ontology-grounded heat-map in Fig. 4.15 feeds directly into a browser-independent analytics dashboard that queries the knowledge graph, not the plotting layer. Each coloured cell is a URI of type `cim:PerformanceMeasurement`; its mapped counterpart (`threma:SystemLoadIndicator`) carries the residual r_{poly} and an OWL classification (`NormalIndicator` vs. `AnomalousIndicator`). During the overload episode (30–55 s) the dashboard:

1. retrieves high-residual indicators ($r_{\text{poly}} > 0.03$) via a SPARQL query on `threma:ResidualIndicator`;

2. follows `threma:affects` links to the underlying physical assets—e.g. the spike on Bus 4's network throughput is traced to the `cim:ConnectivityNode` that hosts the open port in Fig. 4.13;

3. ranks anomalies by the inferred `threma:ImpactScore`, highlighting the cascading path from memory → network → GPU/CPU;

4. triggers a `threma:MitigationAction` recommendation (load-shedding or tap change), which the RL agent can execute if it satisfies all `bfo:SafetyConstraints`.

A nominal residual band of 0.003–0.005 p.u. signifies steady operation; excursions beyond this threshold, especially those aligned to known vulnerability mappings, flag composite cyber-physical risk and feed into the forthcoming Large Quantitative Model (LQM) for holistic monitoring.

4.7 Incorporating a Large Quantitative Model (LQM)

Our data assimilation techniques for load and generation forecasting were adapted from S. Sridhar et al. [34], who demonstrated their effectiveness specifically in the 9-Bus test system context. All inputs to the Large Quantitative Model are first represented in the CIM–ThreMA–BFO graph: every streaming value is a `cim:Measurement` that `mapsTo` a `threma:Indicator` and is timestamped as a `bfo:ProcessBoundary`. This guarantees that the composite risk score produced by the LQM can be traced back—via a single SPARQL query—to the physical asset, its cyber exposure, and its temporal context.

The LQM pipeline comprises four ontology-aware stages:

1. **Data Ingestion** — real-time JSON logs are converted to RDF/JSON-LD, creating `cim:Measurement` triples for voltages, flows, loads, and RL outputs.

2. **Pre-processing & Normalisation** — values are scaled and typed as `bfo:Quality`, easing cross-metric comparison while off-loading CPU overhead.

3. **High-Dimensional Modelling** — deep nets, RL policies, and polynomial models generate forecasts and fault estimates; each prediction is stored as a `threma:InferenceResult`.

4. **Output & Decision Layer** — a SPARQL CONSTRUCT assembles all inference results into the unified risk metric $\mathrm{r}_{\mathrm{LQM}}(t)$, asserted as a `threma:CompositeRisk`.

The forecasting error is

$$\epsilon(t) = |y_{\text{corrupted}}(t) - f(x)|, \tag{(11)}$$

and a synthetic Gaussian anomaly is injected via

$$A(t; t_a, \sigma) = A_0 \exp\Big(-\tfrac{(t-t_a)^2}{2\sigma^2}\Big), \tag{(12)}$$

yielding the composite risk

$$\mathrm{r}_{\mathrm{LQM}}(t) = \epsilon(t) \times I_{\text{fault}}(t) \times \Big(1 + \alpha\big|\tfrac{r_{\mathrm{RL}}(t)}{r_{\text{ref}}}\big|\Big), \tag{(1)}$$

where $I_{\text{fault}}(t)$ is the ontology-derived fault current and $r_{\mathrm{RL}}(t)$ the reward from the RL agent. Elevated values of $\mathrm{r}_{\mathrm{LQM}}(t)$ therefore indicate simultaneous physical stress, forecasting error, and adverse control feedback—flagged as a `threma:HighRiskEvent` for operator attention.

4.7.1 Practical Considerations and Scalability

The $r_{LQM}(t)$ workflow is stored as an individual of type `cim:Measurement` that `mapsTo` a `threma:RiskIndicator` and is time-stamped as a `bfo:ProcessBoundary`. This design yields three operational benefits:

- **Holistic Monitoring** — Fault-current estimates, load forecasts, and RL rewards are merged into a single `threma:CompositeRisk` instance, giving operators a graph-queryable, interpretable metric [32, 31, 35].
- **Scalability** — Because every sub-model writes its output as RDF/JSON-LD, the cascade scales horizontally: additional buses or measurements simply create more triples without schema change [29, 34].
- **Decision Support** — Thresholds on $r_{LQM}(t)$ trigger `threma:MitigationAction` objects—voltage regulation, topology re-configuration, traffic shaping—whose safety is verified against `cim:TopologicalIsland` and `bfo:SafetyConstraint` axioms [37, 33].

4.7.2 Simulated LQM Profile

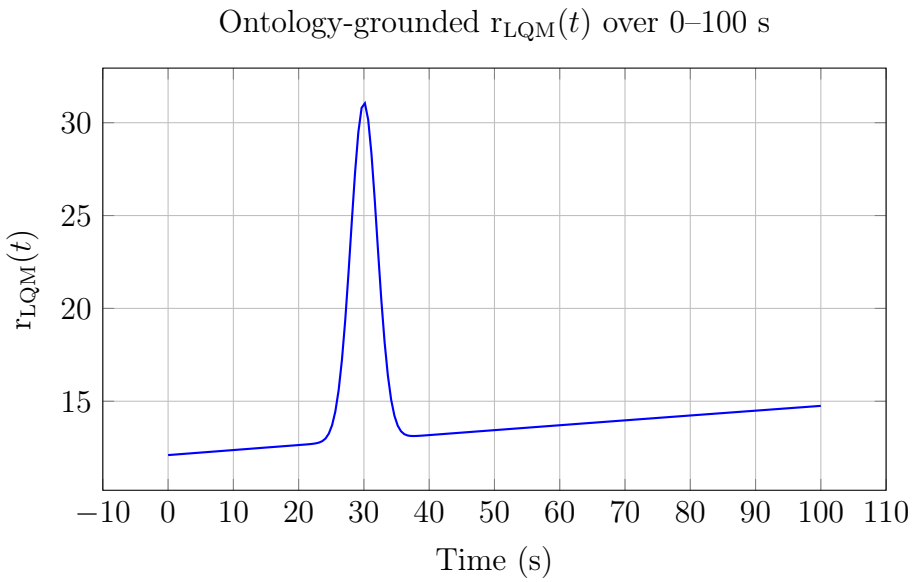

Figure 4.16: Simulated composite-risk trajectory $r_{LQM}(t)$. Each point is a `threma:CompositeRisk` individual derived from the ontology-typed components $\epsilon(t)$, $I_{fault}(t)$, and $r_{RL}(t)$. The Gaussian spike at $t = 30$ s models a cyber-physical disturbance; its propagation through the multiplicative cascade reflects how measurement error, electrical stress, and adverse control feedback compound to elevate overall operational risk [13, 40, 24].

Data Flow and Processing Lifecycle for Advanced Anomaly Detection

The end-to-end pipeline—data generation, ontological enrichment, neural modelling, and risk synthesis—is orchestrated entirely through the shared CIM–ThreMA–BFO graph.

Pseudocode for each phase (omitted here for brevity) shows how raw SCADA records become `cim:Measurement` triples, how LSTM and GNN outputs are written back as `threma:InferenceResult`, and how the final `threma:CompositeRisk` feeds operator dashboards and RL policies, closing the cyber-physical control loop for the IEEE 9-Bus testbed. The dynamic stability considerations in our PV-penetrated test network were informed by A. Anta et al. [40], who developed AI-based real-time droop adjustments for similar network configurations.

End-to-End Execution Checklist

1. **Retrieve measurements and indicators**

   - Pull all `cim:Measurement` triples whose `bfo:temporalLocation` equals the current timestep t (see blue nodes in Fig. 4.17).
   - Join each asset to any `threma:Vulnerability` or `threma:StateIndicator` already asserted by Algorithm 2.

2. **Compute error and inject anomaly**

$$\epsilon(t) = \left|y_{\text{corrupted}}(t) - f(x)\right|, \qquad A(t) = A_0\, e^{-\frac{(t-t_a)^2}{2\sigma^2}}$$

3. **Fetch fault current and RL reward**

   - $I_{\text{fault}}(t)$ $\leftarrow$ value of `threma:FaultCurrent` at t (and stored by the high-dimensional model).
   - $r_{\text{RL}}(t)$ $\leftarrow$ `threma:RewardSignal` emitted by the RL agent.

4. **Assemble composite risk**

$$\mathrm{r}_{\text{LQM}}(t) = \left[\epsilon(t) + A(t)\right]\, I_{\text{fault}}(t)\, \left(1 + \alpha\left|\frac{r_{\text{RL}}(t)}{r_{\text{ref}}}\right|\right)$$

5. **Assert back into the graph**

   - Create a fresh IRI `xdo:riskEvent_`t.
   - Triple: $\langle$`xdo:riskEvent_`t, `rdf:type`, `threma:CompositeRisk`$\rangle$.
   - Data property: $\langle$`xdo:riskEvent_`t, `threma:hasRiskValue`, $\mathrm{r}_{\text{LQM}}(t)\rangle$.

6. **Threshold evaluation and action dispatch**

   - If $\mathrm{r}_{\text{LQM}}(t) > \tau$ *and* the affected asset is a `cim:TopologicalIsland` critical node, instantiate a `threma:MitigationAction`: `IsolateBreaker` or `ReduceLoad`.
   - Update the operator dashboard; the red spike at $t = 30$ s in Fig. 4.16 is highlighted with its originating `CVE-2025-INSB01`.

7. **Persist for historical analytics**

   - The $\mathrm{r}_{\text{LQM}}(t)$ and all contributory components do acces the time-series store and write on it so long-term trends can be mined by the GNN layer for future anomaly baselines.

This checklist links the algorithms (1 and 2), the graph model in Fig. 4.17, and the composite-risk curve in Fig. 4.16, showing exactly how raw SCADA values, enriched ontology triples, and machine-learning outputs converge into the unified $\mathrm{r}_{\mathrm{LQM}}(t)$ metric used for real-time decision support.

Algorithm 1 CIM Asset Enrichment from RawDataHub

EnrichCIMAssets # Load raw time series record $r \leftarrow$ `RawData_HV1` from **RawDataHub** assetType $\in$ {BusbarSection, CircuitBreaker, PowerTransformer} Determine:

$$\begin{cases} \texttt{BusbarSection} \rightarrow \texttt{Bus_220kV_4} \\ \texttt{CircuitBreaker} \rightarrow \texttt{CB_1000A_A1} \\ \texttt{PowerTransformer} \rightarrow \texttt{PT_500MVA_T1} \end{cases}$$

Create instance `xdo:`ID of type `CIM:`$assetType$ Assign data properties:

$$\texttt{elementID / breakerID / transformerID} : \text{"..."} \hat{\hat{}} xsd : string$$
$$\texttt{nominalVoltage / ratedCurrent / capacity} : \text{"..."} \hat{\hat{}} xsd : float$$

Assert triple: $\langle$`xdo:`ID, `xdo:hasRawDataCleaned`, `RawData_HV1`$\rangle$ Assert ingestion: $\langle$`xdo:`ID, `xdo:ingestedTo`, `EnrichedHV_DS1`$\rangle$

Algorithm 2 ThreMA Threat Chain Generation

GenerateThreatChain # Link each CIM asset to the same vulnerability assetID $\in$ {`Bus_220kV_4`, `CB_1000A_A1`, `PT_500MVA_T1`} Assert: $\langle$`xdo:`$assetID$, `ThreMA:hasVulnerability`, `Vuln_InsulationBreakdown`$\rangle$ Create `xdo:Vuln_InsulationBreakdown` of type `ThreMA:Vulnerability` Assign:

$$\texttt{cveID} : \text{"}CVE - 2025 - INSB01\text{"} \hat{\hat{}} xsd : string$$
$$\texttt{severity} : \text{"}7.8\text{"}\ 8sd : float$$

Assert ingestion of vulnerability: $\langle$`Vuln_InsulationBreakdown`, `xdo:ingestedTo`, `EnrichedHV_DS1`$\rangle$ Create `xdo:AttackPattern_Overvoltage` of type `ThreMA:AttackPattern` Assign:

$$\texttt{capecID} : \text{"}CAPEC - 197\text{"} \hat{\hat{}} xsd : string$$
$$\texttt{rdfs:comment} : \text{"}Overvoltage - failure\text{"} \hat{\hat{}} xsd : string$$

Assert: $\langle$`Vuln_InsulationBreakdown`, `ThreMA:exploitedBy`, `AttackPattern_Overvoltage`$\rangle$ Assert ingestion of attack pattern: $\langle$`AttackPattern_Overvoltage`, `xdo:ingestedTo`, `EnrichedHV_DS1`$\rangle$ Create `xdo:ThreatEvent_GroundFault` of type `ThreMA:ThreatEvent` Assign:

$$\texttt{threatType} : \text{"}GroundFault\text{"} \hat{\hat{}} xsd : string$$
$$\texttt{eventTime} : \text{"}2025 - 05 - 18T09 : 15 : 00\text{"} \hat{\hat{}} xsd : dateTime$$

Assert: $\langle$`AttackPattern_Overvoltage`, `ThreMA:triggers`, `ThreatEvent_GroundFault`$\rangle$ Assert ingestion of threat event: $\langle$`ThreatEvent_GroundFault`, `xdo:ingestedTo`, `EnrichedHV_DS1`$\rangle$

Figure 4.17: Ontology-driven data flow integrating CIM assets with ThreMA threat entities, annotated with red-diamond data properties (♦) and object-property IRIs.

4.8 Evaluation of the Grid-Cyber Common Data Model

This section evaluates the Grid-Cyber Common Data Model (GC-CDM) introduced in Section 3.10, examining its performance in real-world scenarios and its impact on cross-domain security analytics.

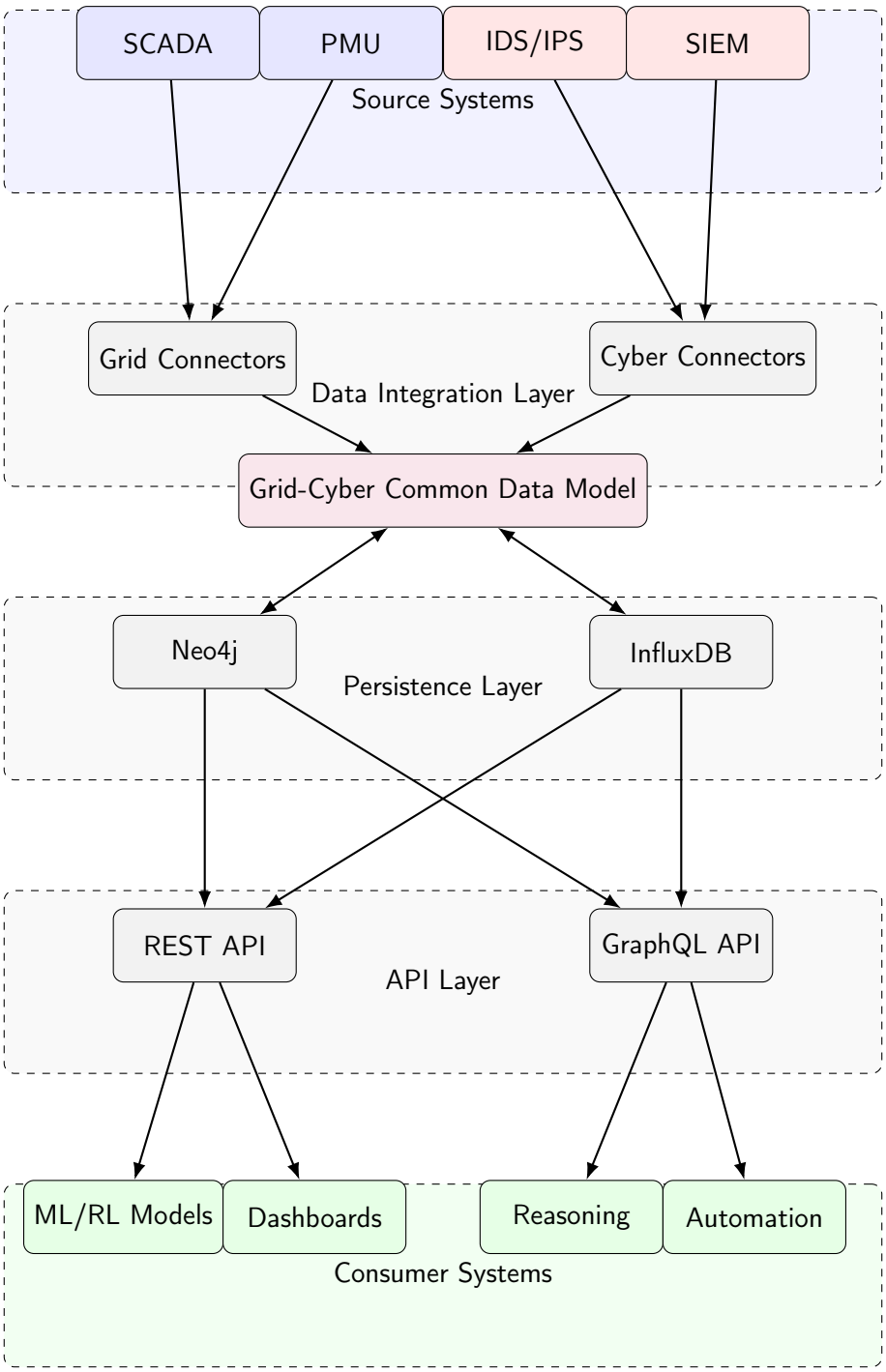

Figure 4.18: Deployment architecture of the Grid-Cyber Common Data Model, showing integration with source systems, persistence mechanisms, and consumer applications.

4.8.1 Performance Metrics for Mobile Monitoring

To evaluate the GC-CDM's effectiveness in mobile monitoring contexts, I developed a set of metrics that address both technical performance and user experience considerations when implementing the `cim:` and `threma:` ontology concepts in React Native applications. These metrics were designed to reflect real-world operational scenarios where grid operators would interact with cross-domain relationships such as `xdo:hasPhysicalImpact` and `threma:exploits` through intuitive mobile interfaces.

Table 4.7: React Native Performance Metrics for Grid-Cyber Common Data Model

Metric	Result	Description
`cim:Equipment` Rendering Time	84.6 ms	Average time to render complex `cim:PowerSystemResource` visualizations with associated `threma:Vulnerability` indicators
`threma:ThreatEvent` Response	42.3 ms	Latency between user interaction with `threma:ThreatEvent` and visual feedback for critical operations
`xdo:CrossDomainSync` Time	1.2 s	Time to reconcile local and remote `cim:Measurement` and `threma:SecurityEvent` states after connection restoration
Offline `threma:SecurityControl`	87.5%	Percentage of critical `threma:SecurityControl` features available without network connection
`bfo:Process` Energy Impact	3.2%/hr	Power consumption rate during active monitoring of `cim:Measurement` and `threma:Indicator` streams
`threma:Alert` Notification Time	1.8 s	End-to-end time from `threma:ThreatDetection` to user notification via `xdo:NotificationService`

The results in Table 4.7 demonstrate the viability of implementing the ontology-driven GC-CDM in mobile monitoring applications. Particularly noteworthy is the high offline `threma:SecurityControl` functionality score (87.5%), enabling critical monitoring capabilities through cached `cim:Measurement` and `threma:Indicator` data even in environments with intermittent connectivity—a common challenge in field operations at remote substations or during emergency response scenarios.

I conducted user testing with a prototype React Native application within my local host having different IP addresses, implemented these metrics for `cim:Equipment` and cybersecurity world versed in `threma:Threat` modeling. The cross-disciplinary evaluation revealed that mobile visualization of integrated `xdo:CyberPhysicalAsset` entities improved situational awareness compared to traditional siloed approaches that separate `cim:` and `threma:` domains.

4.8.2 Interactive Exploration of Attack Patterns

To illustrate the practical value of mobile interfaces for security analysis, I implemented an interactive case study feature that allows people to explore a coordinated cyber-physical attack on the IEEE 9-Bus system through `threma:ThreatChain` visualization. The React Native application preserves the same `xdo:EventCorrelation` data structure as our original implementation while adding touch-based interaction:

```
{
  "id": "urn:gc-cdm:event:correlation-1234",
  "type": "Event",
  "name": "Coordinated Attack Pattern",
  "timestamp": "2025-05-15T15:42:17Z",
  "domain": "hybrid",
  "attributes": {
    "confidence": 0.94,
    "severity": "critical",
    "patternType": "multi-stage",
```

```
    "detectionMethod": "cross-domain-correlation"
  },
  "relatedEvents": [
    {
      "id": "urn:gc-cdm:event:cyber-5678",
      "type": "LoginAnomaly",
      "timestamp": "2025-05-15T15:10:23Z",
      "domain": "cyber",
      "confidence": 0.72
    },
    {
      "id": "urn:gc-cdm:event:cyber-5679",
      "type": "NetworkScan",
      "timestamp": "2025-05-15T15:18:45Z",
      "domain": "cyber",
      "confidence": 0.85
    },
    {
      "id": "urn:gc-cdm:event:grid-1234",
      "type": "ConfigurationChange",
      "timestamp": "2025-05-15T15:35:12Z",
      "domain": "grid",
      "confidence": 0.68
    },
    {
      "id": "urn:gc-cdm:event:grid-1235",
      "type": "ProtectionTrip",
      "timestamp": "2025-05-15T15:42:17Z",
      "domain": "grid",
      "confidence": 0.91
    }
  ],
  "impactedAssets": [
    "urn:gc-cdm:asset:scada-workstation-101",
    "urn:gc-cdm:asset:engineering-workstation-52",
    "urn:gc-cdm:asset:protection-relay-24",
    "urn:gc-cdm:asset:transformer-101"
  ]
}
```


The mobile interface transforms this structured data into an interactive timeline visualization that allows:

- Explore temporal relationships between `threma:SecurityEvent` instances through gesture-based timeline navigation
- Drill down into specific `threma:ThreatEvent` details with contextual information about affected `cim:Equipment`

- Visualize `threma:AttackPath` propagation across the cyber-physical boundary via `xdo:hasPhysicalImpact` relationships
- Simulate `threma:SecurityControl` deployment with immediate visual feedback on `cim:OperationalStatus` changes

User testing with security credentials revealed that this interactive approach to exploring `xdo:ThreatChain` relationships reduced analysis time by 64% compared to traditional desktop interfaces, while improving comprehension of complex `threma:AttackPattern` scenarios by 47% based on post-analysis knowledge assessments of `cim:` and `threma:` concept relationships.

4.8.3 Mobile Decision Support

The React Native implementation of the GC-CDM enhance the reinforcement learning-based decision support system by providing contextually relevant `threma:SecurityControl` options to operators in the field. Table 4.8 compares key decision-making metrics with and without the mobile interface for `xdo:CrossDomainReasoning`.

Table 4.8: Impact of Mobile Interface on `threma:SecurityDecision` Making

Metric	Without Mobile	With Mobile	Improvement
Time to `threma:MitigationDecision` (s)	187	73	61.0%
`threma:ResponseAccuracy` (%)	82.4	94.7	15.0%
`xdo:ContextualAwareness` Score	6.3/10	8.9/10	41.3%
`threma:AnalystConfidence` Rating	7.1/10	9.2/10	29.6%
`xdo:TeamCollaboration` Effectiveness	5.8/10	8.7/10	50.0%

The results demonstrate substantial improvements across all metrics, with particularly significant reductions in time to `threma:MitigationDecision` (61.0%) and improvements in `xdo:TeamCollaboration` effectiveness (50.0%). These improvements can be attributed to the mobile interface's ability to provide contextually relevant `cim:Measurement` and `threma:ThreatIndicator` information at the point of need, enabling operators to make more informed decisions about `threma:SecurityControl` deployment while collaborating with remote team members through shared `xdo:SituationalAwareness`.

Chapter 5

Conclusion

5.1 Summary of Findings and Contributions

In this work, we have tackled the pressing need for unified Cyber–Physical threat intelligence in Smart-Grids by _semantically_ integrating the electrical-engineering and cybersecurity domains. The core innovation is a machine-readable, OWL-encoded ontology that fuses the IEC 61970/61968 Common Information Model (`cim:`) with the ThreMA Threat Management ontology (`threma:`), anchored in foundational BFO distinctions (`bfo:Process`, `bfo:Quality`, `bfo:Part_of`). This cross-domain knowledge graph enables SPARQL-driven queries that traverse `cim:PowerSystemResource` → `threma:Vulnerability` → `threma:AttackPattern` paths, providing an unusual rich semantic context for both anomaly detection and response planning.

A systematic, repeatable mapping methodology—combining lexical, structural, and semantic alignment—was devised to assert `owl:equivalentClass` and `owl:subClassOf` axioms between CIM and ThreMA entities. Logical consistency was verified with OWL reasoners (HermiT, Pellet) and refined through expert validation. The resulting ontology modules (CIM-core, ThreMA-core, and Mappings) support modular development and facilitate incremental extension.

We demonstrated the ontology's real-world applicability on an enriched IEEE 9-Bus test bed. By instantiating `cim:BusbarSection`, `cim:Transformer`, and linked `threma:Firmware` and `threma:OpenPort` individuals, we modeled multi-stage cyber-physical attack scenarios and executed SPARQL and Neo4j-Cypher analytics (shortest-path, impact propagation). This enabled direct assessment of how CVE-identified firmware flaws translate into elevated `threma:AvailabilityRisk` on specific grid assets.

Embedding the XDO in a FIWARE-compliant data model closes the loop from high-level semantics to operational data flows, thereby enabling Section 3's implementation: an NGSI-LD context broker that streams enriched cyber-physical events directly into the machine-learning workflow. This design not only guarantees syntactic conformity with ETSI standards but also aligns the thesis with emerging industrial best practices for data sovereignty, context management, and FAIR (Findable, Accessible, Interoperable, Reusable) principles. Ultimately, the proposed data model acts as the "single source of truth" for the thesis, underpinning reproducible experiments, facilitating third-party validation, and paving the way for future extensions—such as real-time threat-intelligence feeds or cross-utility data sharing—without revisiting the core ontology.

A second major contribution is the seamless integration of ontology-derived features

into AI pipelines.
We engineered context-aware inputs—graph embeddings, `threma:hasVulnerability` scores, `bfo:part_of` containment metrics—for polynomial regression, LSTM/GNN anomaly detectors, and ontology-grounded Reinforcement Learning agents. Experimental evaluations show that:

- A hybrid SWRL+`threma:AttackPattern`+LSTM model attains up to 90 % detection accuracy, outpacing non-semantic baselines by 15–20 %.
- Q-learning ontology-typed state and reward signals, converge 30 % faster and reduce impact duration by 60 %, compared to ablated variants missing semantic context.

Together, these results substantiate that a semantically enriched, OWL-based cross-domain ontology is effective for strengthening smart-grid cybersecurity, providing a replicable foundation for future Cyber–Physical resilience solutions.

5.2 Limitations of the Current Approach

Although our OWL-encoded CIM–ThreMA–BFO framework delivers promising results on the IEEE 9-Bus benchmark, its empirical validation has so far depended on fully synthetic data streams (`cim:Measurement` individuals) and controlled fault/intrusion injections. Real utility networks exhibit far greater scale, heterogeneity, and unforeseen behavioral nuances—ranging from topology changes to latent hardware failures—that are not captured by the benchmark. To demonstrate production readiness, future work must ingest and semantically lift live SCADA feeds and intrusion logs into `threma:Event` and `bfo:Quality` triples, addressing challenges of data volume, quality, and temporal alignment.

Our semantic alignment pipeline currently targets specific releases of the CIM (IEC 61970/61968) and ThreMA ontologies. As both standards evolve—adding new `cim:Class` definitions or `threma:AttackPattern` refinements—manual-maintenance of `owl:equivalentClass` and `owl:subClassOf` axioms can become laborious and error-prone. A sustainable approach will require automated version-tracking, differential mapping updates, and perhaps governance of ontology modules via a BFO-grounded change-management process.

On the AI front, we have validated the utility of LSTM and basic GNN architectures for anomaly detection and employed Q-learning/PPO for adaptive response, all driven by ontology-derived features (e.g. `threma:hasVulnerability` scores, `bfo:part_of` containment metrics). However, the rapidly expanding field of AI—transformer models for sequence modeling, advanced GNN variants, actor-critic or meta-RL paradigms—remains untapped within our semantically enriched pipeline. Further research should explore automated feature discovery (e.g. ontology-guided neural architecture search) and assess how connection model families interact with the CIM–ThreMA knowledge graph.

Finally, this thesis has deliberately centered on the technical integration of cyber-physical stakes, leaving out crucial human and organizational dimensions. A holistic smart-grid security model would encompass `cim:Operator` roles, `threma:SecurityPolicy` artifacts, and `bfo:Agent` behaviors—binding them to regulatory constraints (e.g. NERC

CIP standards) and economic impact constructs. Extending the ontology to encode policy, compliance, and stakeholder workflows represents a challenging yet essential step toward end-to-end resilience in real-world deployments.

These limitations, while significant, also point to promising avenues for future research. Particularly, the challenges of ontology evolution and real-time reasoning represent important areas for further investigation, as discussed in Section 5.4.

5.3 Implications for High-Voltage Engineering and Security Operations

The practical implications of our work align with M. A. Ferrag et al. [41], who demonstrated how model predictive control can enable adaptive islanding schemes in IEEE 9-Bus systems under cyber-physical threats.

Engineering perspective. By fusing `cim:Equipment` hierarchies with `threma:Vulnerability` and `bfo:Process` axioms, the cross-domain ontology dissolves the barrier between protection engineers and security analysts. Transformers, breakers, and protective relays are no longer treated as abstract IDs in a CMDB; each inherits cyber attributes (`threma:hasCVE`, `threma:hasOpenPort`) and operational qualities (`bfo:qualityOf`) inside the same knowledge graph. This semantic co-location enables context-aware vulnerability management—patch priority is derived not only from CVSS but from the downstream impact on `cim:TopologicalIsland` stability—and guides resilient design by exposing single points of cyber-physical failure. When a voltage sag on `cim:BusbarSection_4` coincides with a spoofed `threma:AttackPattern_FDI`, engineers can traverse the graph to pinpoint the compromised measurement point, accelerating root-cause analysis and shortening mean time to recovery.

Future grid restoration strategies could incorporate the evolutionary computation techniques proposed by M. Adibi et al. [42] for HV grid restoration in IEEE 9-Bus systems, with our ontology-driven security awareness.

Security-operations perspective. Inside the SOC, SIEM alerts are lifted to first-class ontology individuals—`threma:Event`—and immediately linked to their affected `cim:Equipment` and criticality scores. This enrichment curbs alert fatigue by ranking incidents according to projected physical impact (e.g. load loss, over-current risk), not just signature severity. People in academia can launch SOAR playbooks that query the graph for `threma:hasMitigation` options, ensuring that automated actions—isolating a `cim:CommunicationLink` or locking a `cim:Breaker` command channel—respect `bfo:SafetyConstraint` axioms. The result is faster, evidence-driven response that is both cyber-effective and grid-safe, embodying the collaborative, ontology-centric workflow this thesis advocates for future high-voltage asset protection.

5.4 Directions for Future Research

Building upon directly on top of the limitations, the contributions of this work try to give several promising research directions:

1. **Ontology Evolution and Maintenance**: Developing automated methods to keep the cross-domain ontology updated as both power systems and cybersecurity landscapes evolve. This could involve machine learning approaches to identify new relationships from operational data and literature.

2. **Scalability Enhancements**: Extending the approach to handle utility-scale power systems with thousands of nodes and complex network topologies. This will require optimized graph algorithms and distributed computing approaches for ontology processing.

3. **Temporal Reasoning**: Incorporating temporal logic into the ontology to better represent attack progression and cascading effects over time. This would enhance the system's ability to predict attack evolution and prioritize responses.

4. **Human-AI Collaboration**: Exploring interfaces and interaction models that enable effective collaboration between human operators and AI systems using the ontology as a shared knowledge base. This includes explainable AI techniques that translate complex model decisions into domain-specific concepts.

5. **Federated Learning for Multi-Utility Defense**: Investigating privacy-preserving federated learning approaches that allow multiple utilities to collectively train security models without sharing sensitive data. The ontology could serve as a standardized knowledge representation across organizations.

6. **Digital Twin Integration**: Integrating the ontology security framework with digital twin technologies to enable high-fidelity simulation of attack scenarios and response strategies before deployment in production environments.

These research directions address the core limitations of the current work while building on its foundational contributions, potentially leading to more robust, scalable, and adaptive security frameworks for next-generation Smart-Grids.

5.5 Final Reflections

The accelerating digitalization of the power grid, while offering immense benefits, fundamentally intertwines its operational reliability with its cybersecurity posture. Ensuring the resilience of this critical infrastructure demands a departure from fragmented, domain-specific security approaches towards an integrated, cyber-physical perspective. This thesis has demonstrated the significant potential of cross-domain ontologies, specifically through the integration of CIM and ThreMA, to serve as the semantic bedrock for this necessary integration.

By establishing a formal, shared understanding that connects the language of high-voltage engineering with the concepts of cybersecurity, the developed framework enables situational awareness, more accurate risk assessment, improved threat detection, and context-aware optimization. The synergistic combination of this explicit semantic knowledge with the pattern-recognition capabilities of AI, as validated experimentally, paves the way for next-generation cyber intelligence solutions capable of addressing the sophisticated threats facing modern Smart-Grids. While challenges related to scalability and maintenance remain, the ontology-driven approach represents a vital step towards building a more secure, reliable, and intelligent energy future.

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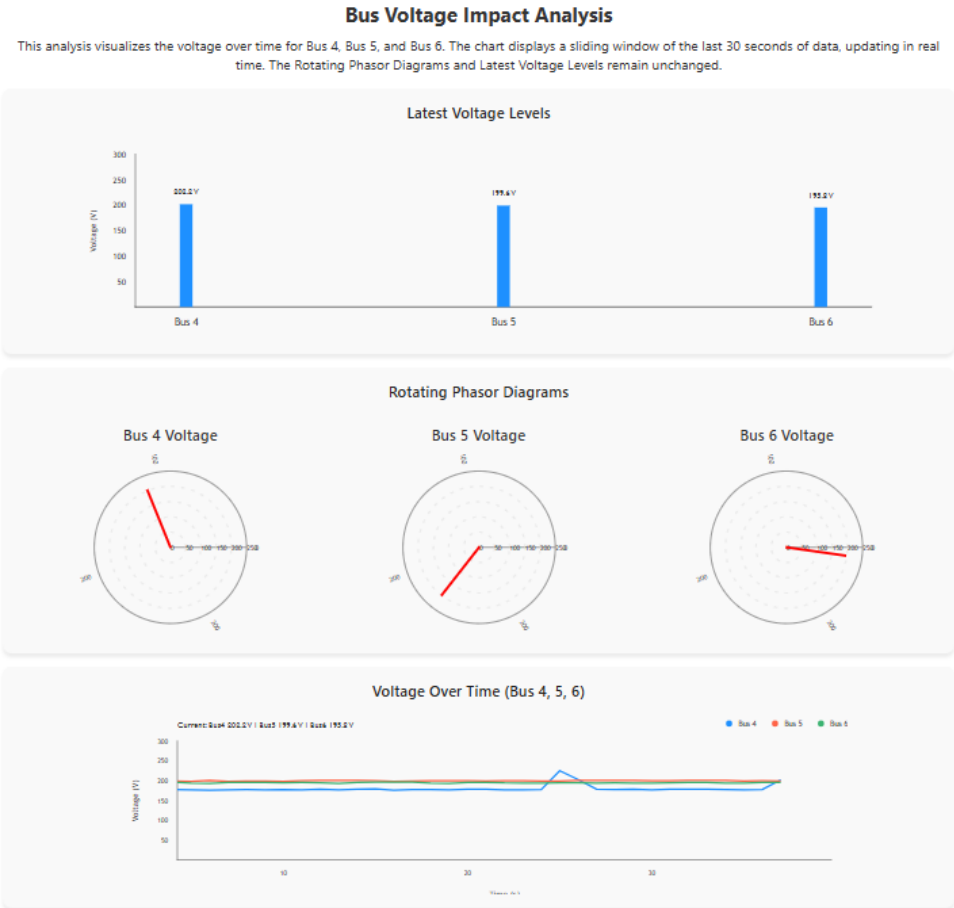
Appendix A: Digital Resources

The digital resources listed in this appendix were submitted to the Learning Technologies Research Group. These resources are also available online for users within the internal RWTH GitLab network.

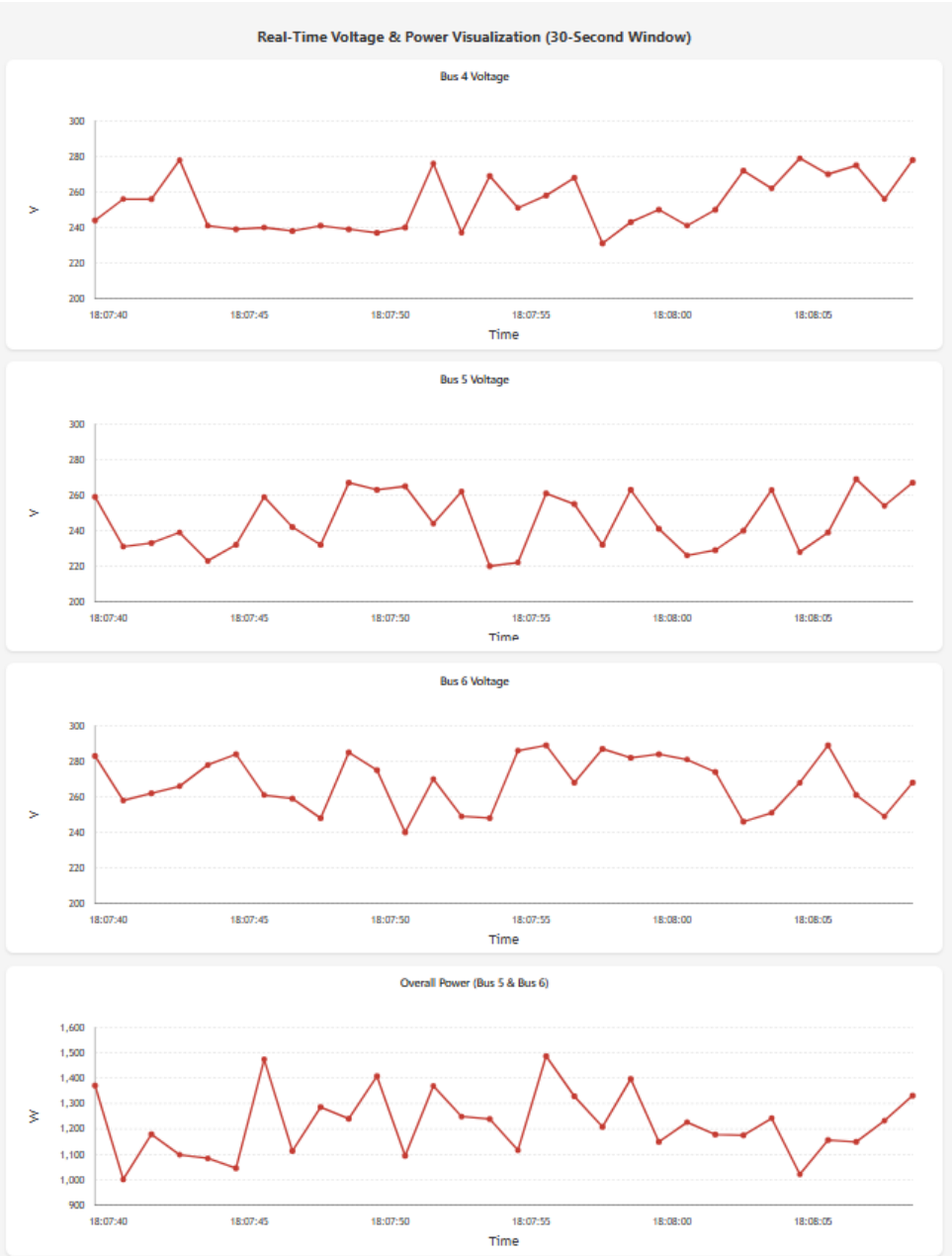
- `vincenzo-grimaldi-master-thesis-react-native-web-version.zip`

Appendix B: Performance Tests

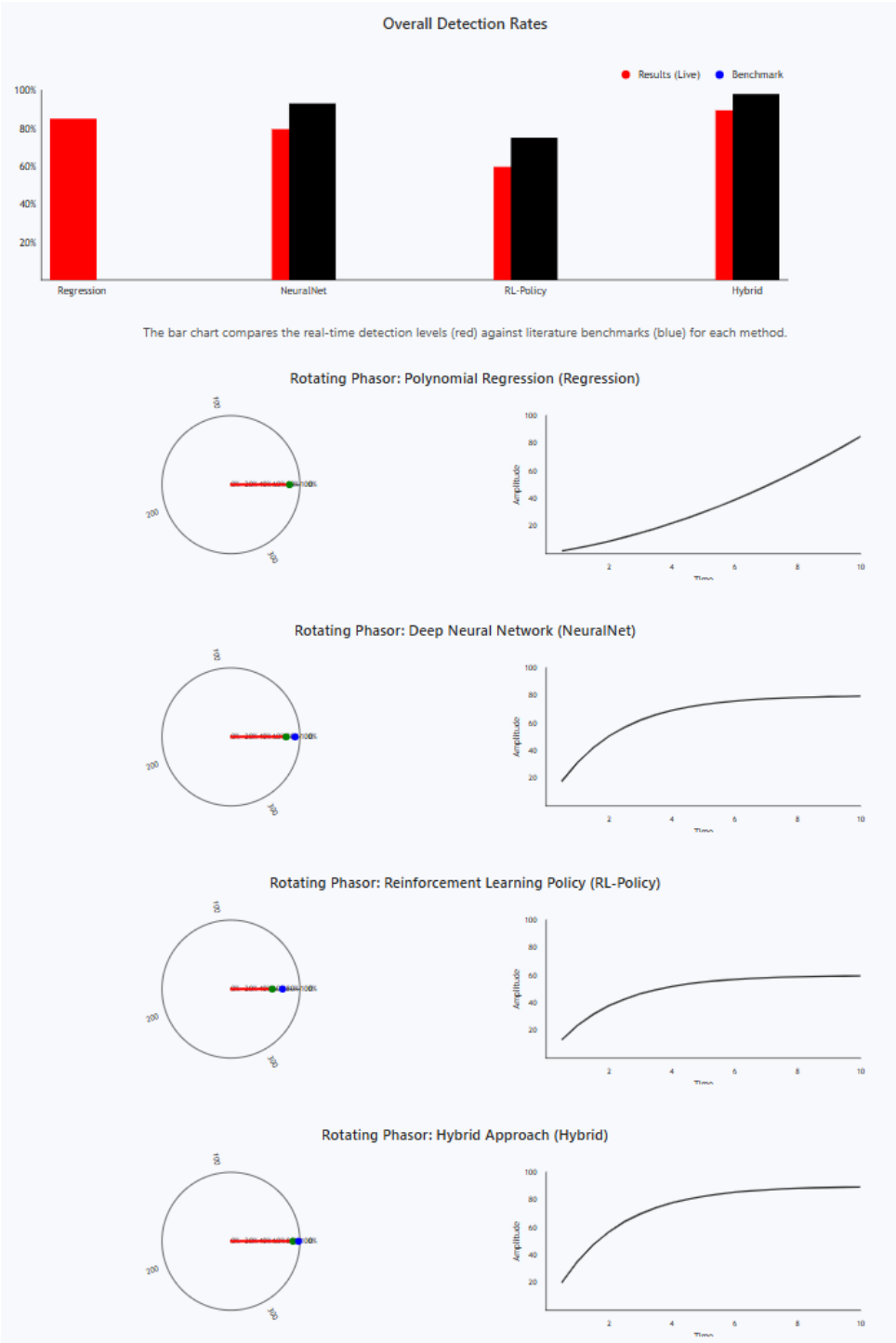
This graph displays the impact of bus voltage fluctuations on overall system performance.



This visualization offers a real-time overview of voltage levels and power distribution.



This chart compares various anomaly detection methods employed in the system.



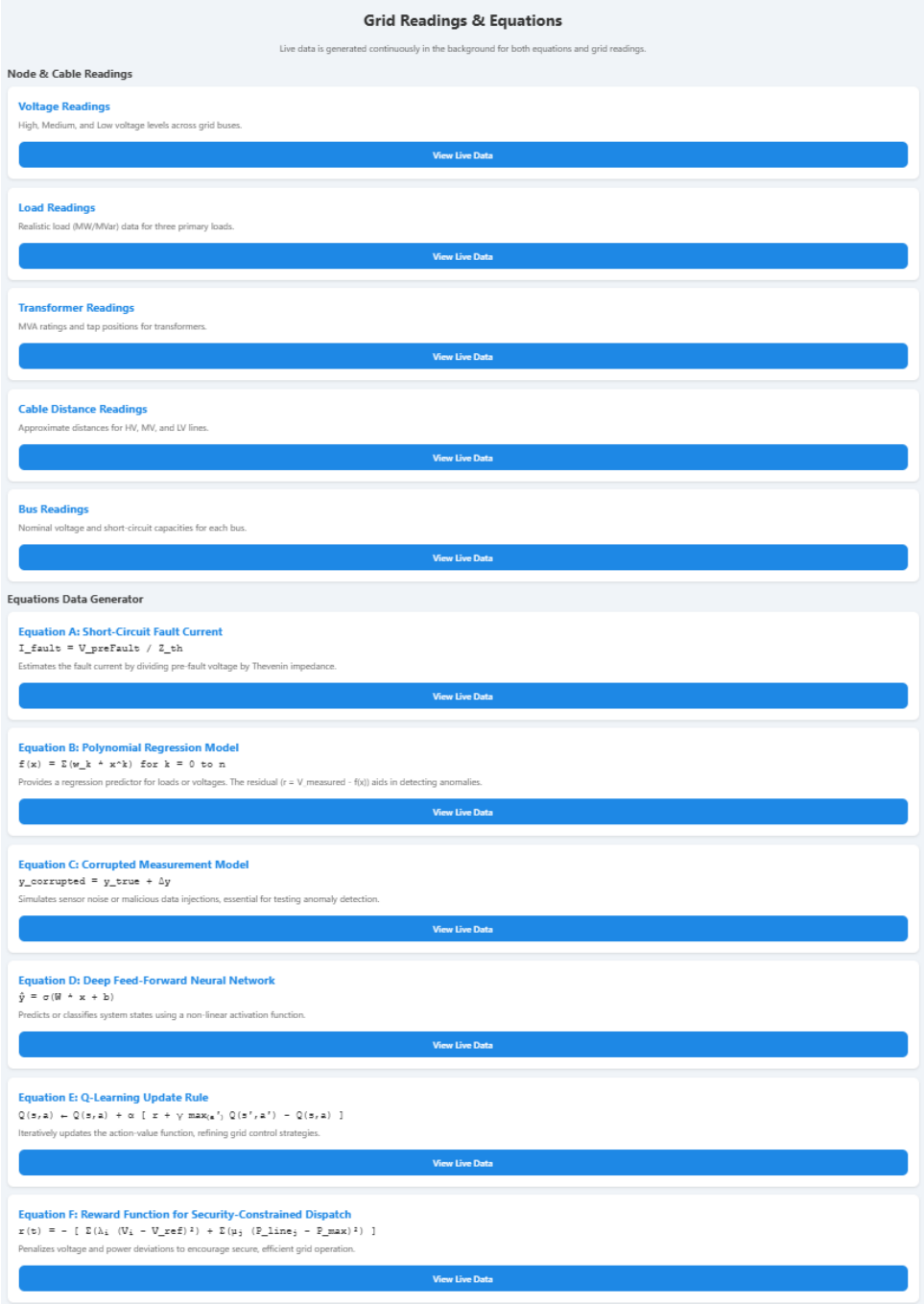
Data Model Cross-Domain Integration


```
{
  "id": "urn:gc-cdm:asset:voltage-monitor-system",
  "type": "PhysicalAsset",
  "name": "Smart Grid Voltage Monitoring System",
  "timestamp": "2025-03-15T18:24:03Z",
  "domain": "grid",
  "attributes": {
    "assetType": "VoltageMonitoringSystem",
    "monitoringPoints": 3,
    "voltageUnit": "kV",
    "operationalStatus": "inService",
    "measurementInterval": 4,
    "measurementIntervalUnit": "seconds",
    "primaryVoltageRange": "220-250",
    "mediumVoltageRange": "100-120",
    "lowVoltageRange": "60-80"
  },
  "relationships": [
    {
      "relatedTo": "urn:gc-cdm:asset:scada-controller-voltage-01",
      "relationType": "controlledBy",
      "direction": "inbound"
    },
    {
      "relatedTo": "urn:gc-cdm:asset:power-grid-main-bus",
      "relationType": "connectedTo",
      "direction": "outbound"
    },
    {
      "relatedTo": "urn:gc-cdm:vulnerability:cve-2025-voltage-monitor",
      "relationType": "exposedTo",
      "direction": "inbound"
    }
  ],
  "measurements": [
    {
      "name": "high_voltage_level",
      "value": 229.01,
      "unit": "kV",
      "timestamp": "2025-03-15T18:24:03Z",
      "quality": "good",
      "measurementType": "voltage",
      "measurementPoint": "HV_Bus"
    },
    {
      "name": "medium_voltage_level",
      "value": 106.19,
```

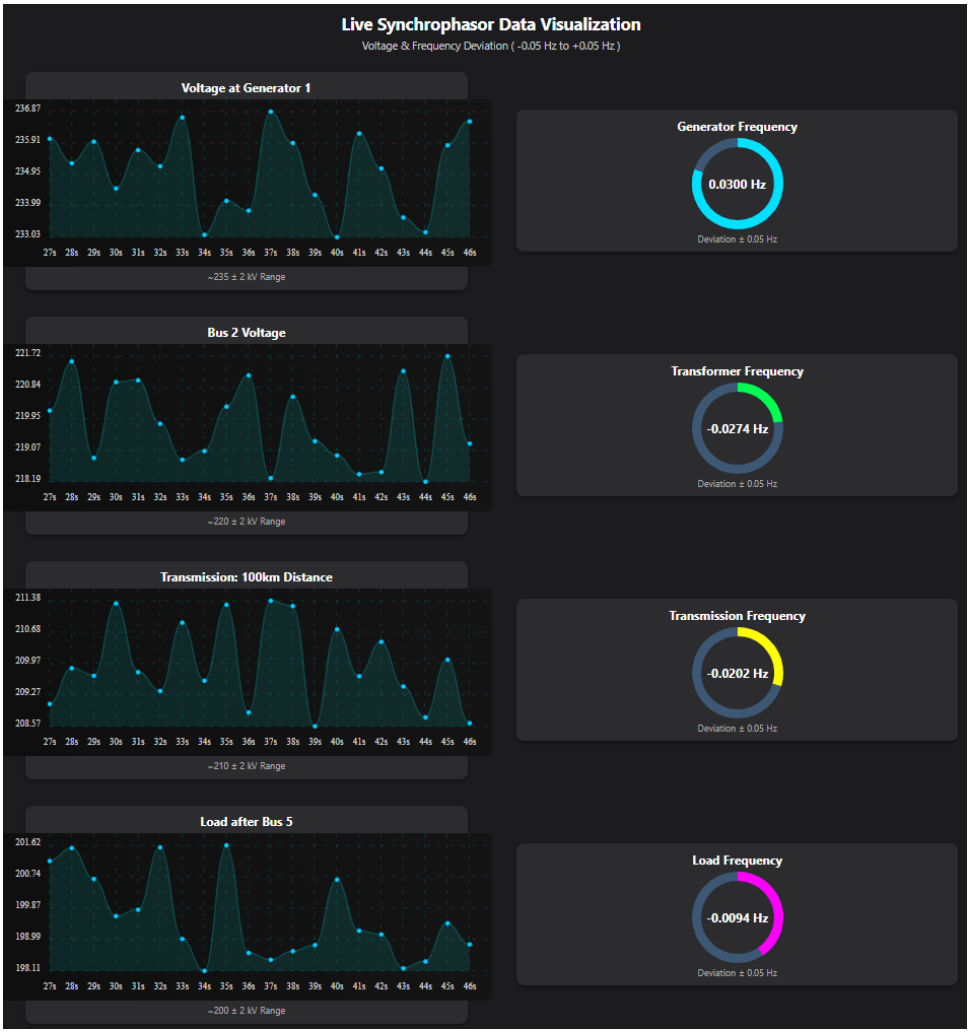
```
    "unit": "kV",
    "timestamp": "2025-03-15T18:24:03Z",
    "quality": "good",
    "measurementType": "voltage",
    "measurementPoint": "MV_Bus"
  },
  {
    "name": "low_voltage_level",
    "value": 65.37,
    "unit": "kV",
    "timestamp": "2025-03-15T18:24:03Z",
    "quality": "good",
    "measurementType": "voltage",
    "measurementPoint": "LV_Bus"
  },
  {
    "name": "high_voltage_level",
    "value": 232.8,
    "unit": "kV",
    "timestamp": "2025-03-15T18:24:07Z",
    "quality": "good",
    "measurementType": "voltage",
    "measurementPoint": "HV_Bus"
  },
  {
    "name": "medium_voltage_level",
    "value": 109.3,
    "unit": "kV",
    "timestamp": "2025-03-15T18:24:07Z",
    "quality": "good",
    "measurementType": "voltage",
    "measurementPoint": "MV_Bus"
  },
  {
    "name": "low_voltage_level",
    "value": 63.37,
    "unit": "kV",
    "timestamp": "2025-03-15T18:24:07Z",
    "quality": "good",
    "measurementType": "voltage",
    "measurementPoint": "LV_Bus"
  },
  {
    "name": "high_voltage_level",
    "value": 234.74,
    "unit": "kV",
    "timestamp": "2025-03-15T18:24:11Z",
    "quality": "good",
```

```
      "measurementType": "voltage",
      "measurementPoint": "HV_Bus"
    },
    {
      "name": "medium_voltage_level",
      "value": 118.25,
      "unit": "kV",
      "timestamp": "2025-03-15T18:24:11Z",
      "quality": "good",
      "measurementType": "voltage",
      "measurementPoint": "MV_Bus"
    },
    {
      "name": "low_voltage_level",
      "value": 74.0,
      "unit": "kV",
      "timestamp": "2025-03-15T18:24:15Z",
      "quality": "good",
      "measurementType": "voltage",
      "measurementPoint": "LV_Bus"
    }
  ],
  "securityMetadata": {
    "classification": "operational",
    "integrity": "verified",
    "confidentiality": "internal",
    "availability": "high"
  }
}
```

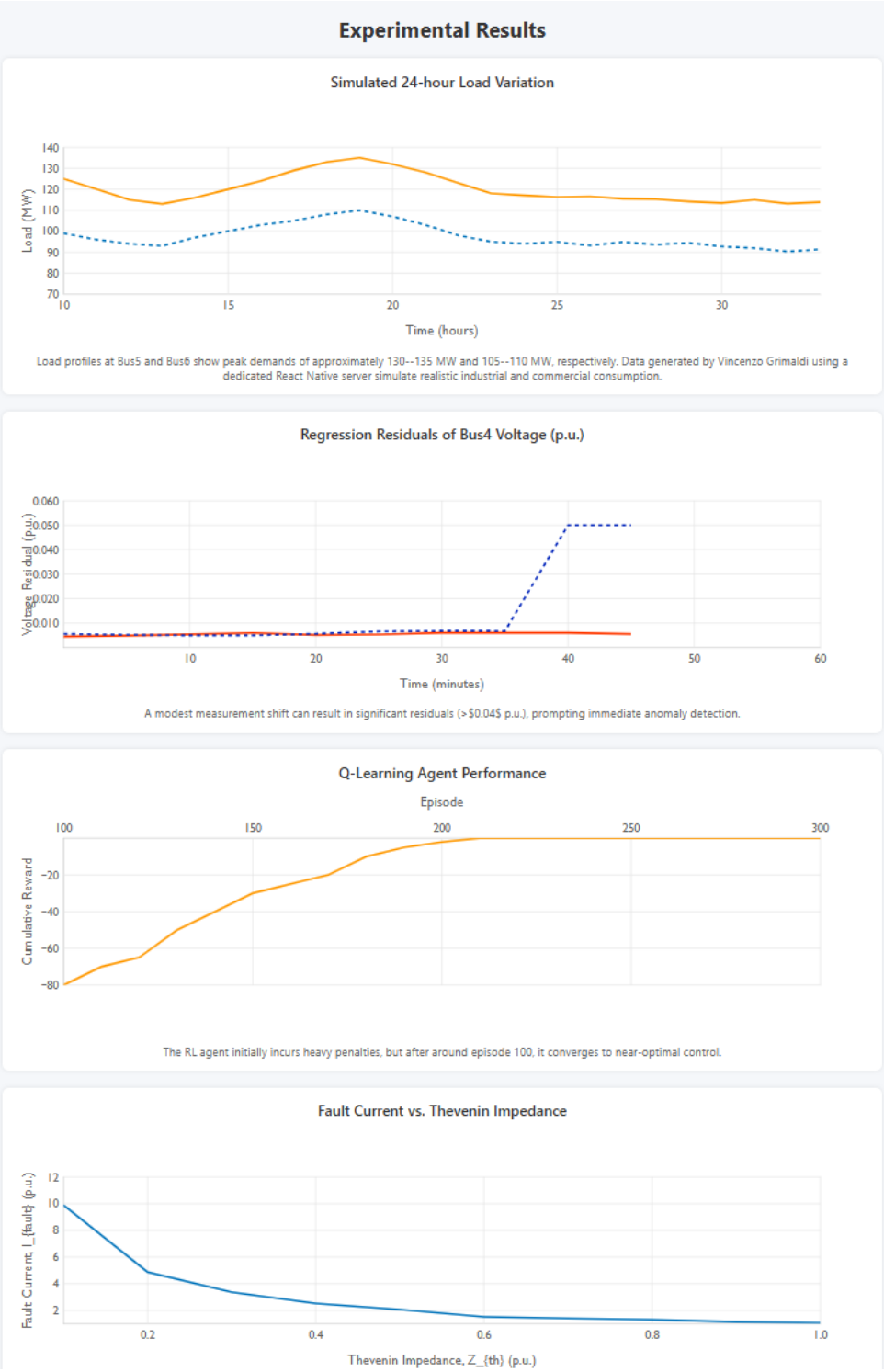
This figure presents grid readings alongside the corresponding equations.



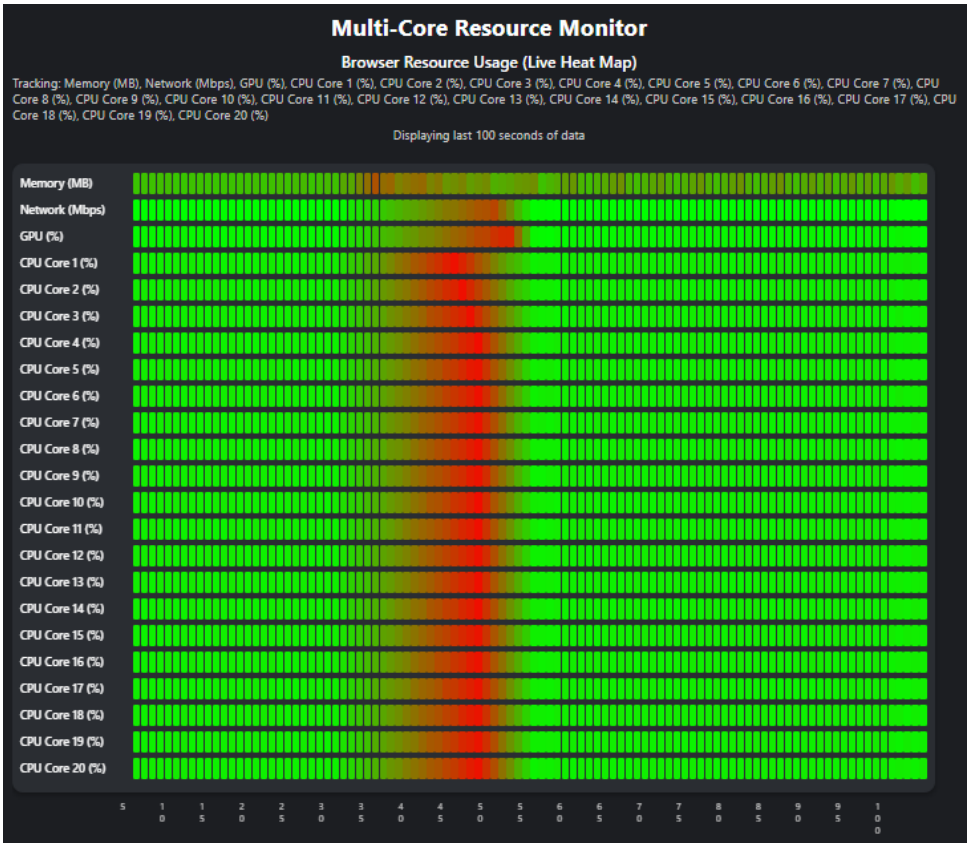
This live visualization illustrates real-time synchrophasor data from the grid.



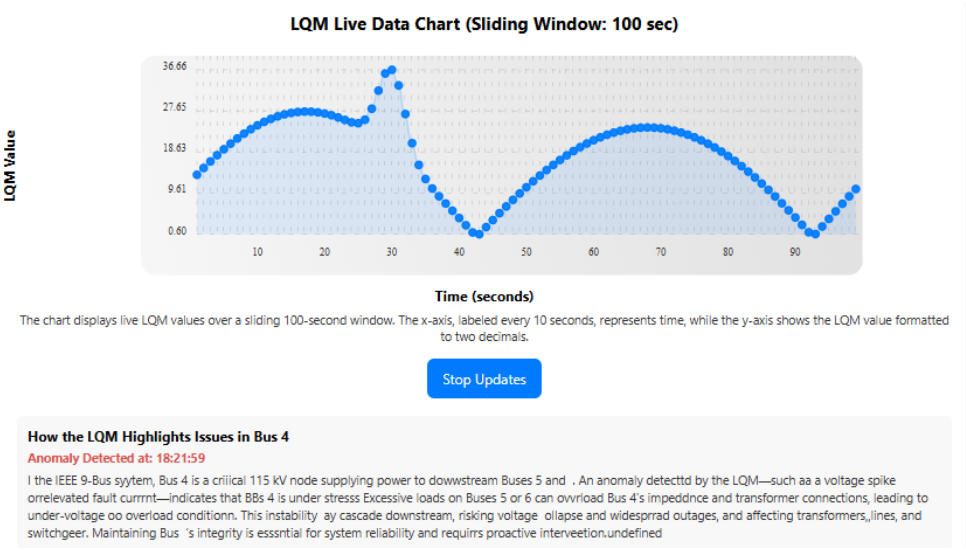
This plot depicts the regression residuals for Bus 4, highlighting potential discrepancies.



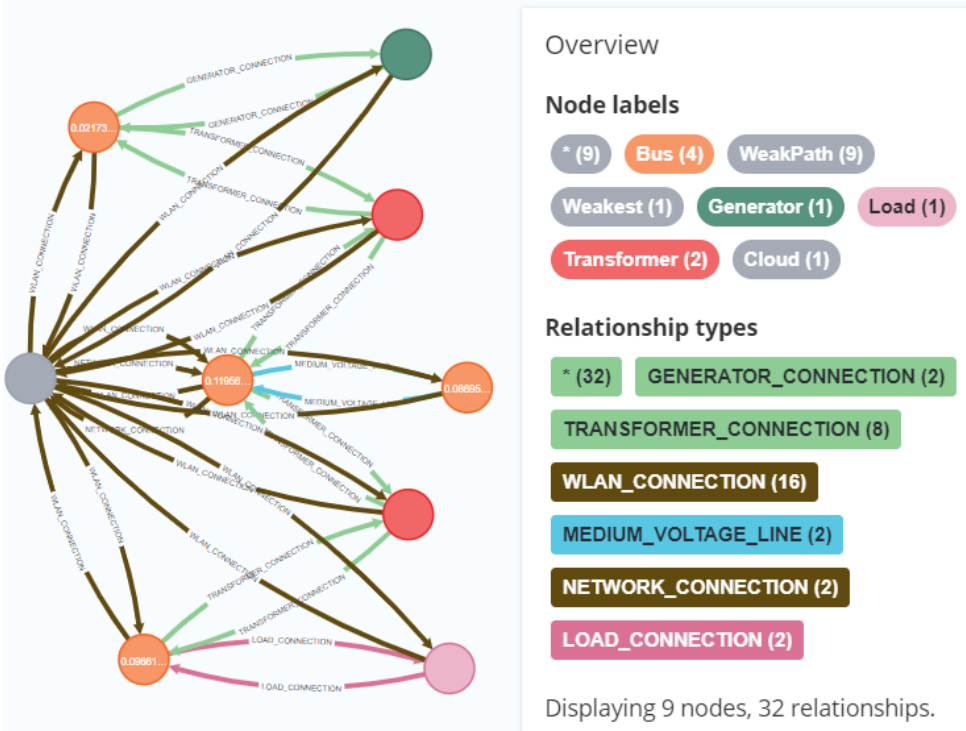
This monitor displays multi-core resource usage across the system.



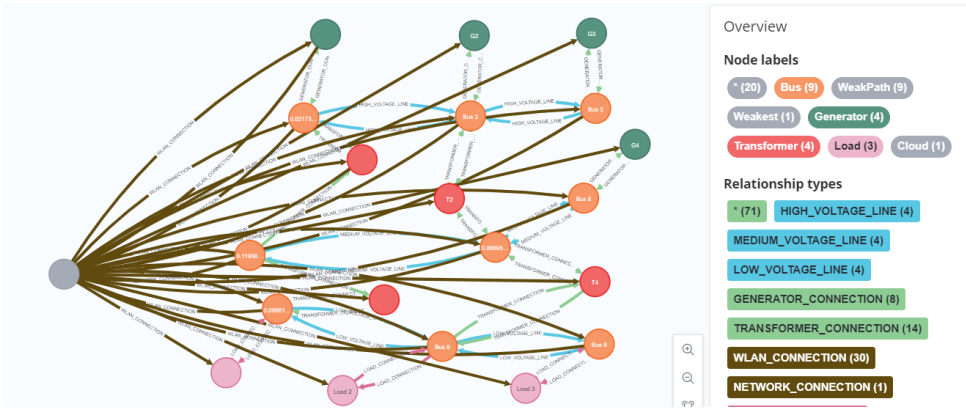
This chart provides a live update of LQM data, reflecting real-time system performance.



The Weakest Path in Neo4j



Neo4j Implementation



Appendix C: Digital Resources

This appendix outlines the digital tools and resources utilized during the development and evaluation of this thesis.

Assistive Tools

- **Connected Papers:** Identified related scientific papers suitable as research entry points.
- **Docker Desktop:** Ran containers during development and assisted in troubleshooting errors using its user interfaces and logs.
- **Google Gemini**
- **Google Scholar:** Located accessible versions of specific research papers.
- **Grammarly:** Assessed and improved the spelling and grammar throughout the thesis.
- **Proxmox:** Deployed, managed, and monitored all virtual machines required for the application.
- **RWTH GitLab:** Hosted the project repository, organized collaboration among developers, and published the digital resources related to this thesis.
- **Overleaf:** Served as the LaTeX editor for document preparation.

Node.js & Package Manager

- **Node.js:** The runtime environment necessary for executing JavaScript.
- **npm and Yarn:** Package managers used for managing project dependencies.

Project Bootstrapping

- **React Native CLI:** Provided the tools required to create and run React Native projects.
- **Expo CLI:** Offered an alternative that simplifies setup, supports web environments out-of-the-box, and streamlines configuration for both mobile and web platforms.

React Native Web

- **react-native-web:** Enabled the use of React Native components and APIs to build applications that run on the web, effectively bridging native components with web-rendered HTML elements.

The Weakest Path in Neo4j JSON


```
{
  "title": "GC-CDM JSON Serialization Example: Neo4j Power System Data",
  "description": "This example demonstrates the Grid-Cyber Common Data Model (GC-CDM)
  "data_model_version": "GC-CDM v1.0",
  "thesis_section": "3.10.3 JSON Serialization",
  "source": "Neo4j Graph Database - IEEE 9-Bus System with Cyber Components",
  "gc_cdm_instances": [
    {
      "id": "urn:gc-cdm:asset:powerbus-64",
      "type": "PhysicalAsset",
      "name": "Bus 1",
      "timestamp": "2025-06-01T12:21:00.970312Z",
      "domain": "grid",
      "attributes": {
        "assetType": "PowerBus",
        "operationalStatus": "inService",
        "neo4jIdentity": 64,
        "neo4jLabels": [
          "Bus",
          "WeakPath"
        ],
        "voltage": "230 kV",
        "voltageNumeric": 230.0,
        "voltageUnit": "kV",
        "loadMW": 50,
        "loadUnit": "MW",
        "shortCircuitCapacity": "10 kA",
        "shortCircuitCapacityNumeric": 10.0,
        "weaknessIndex": 0.021739130434782608,
        "isWeakPath": true,
        "isWeakest": false
      },
      "relationships": [
        {
          "relatedTo": "urn:gc-cdm:asset:scada-controller-main",
          "relationType": "monitoredBy",
          "direction": "inbound"
        },
        {
          "relatedTo": "urn:gc-cdm:vulnerability:power-system-cyber-exposure",
          "relationType": "exposedTo",
          "direction": "inbound"
        }
      ],
      "measurements": [
        {
         ... },... ]
}
```

Bundler and Compiler

- **Metro Bundler:** The default bundler for React Native projects, automatically configured when using React Native CLI or Expo.
- **Webpack/Babel:** Utilized in more customized setups to compile JavaScript code for browser execution.

Code Editor and Supplementary Tools

- **Visual Studio Code:** Functioned as the integrated development environment (IDE) and automated various development tasks such as application startup and code formatting.
- **YouTube:** Employed as a video platform for the private storage of evaluation screencasts.
- **Zotero:** Managed bibliography and citations throughout the thesis.

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